SAT Math — Teacher Copies

Explanations with worked examples · a labelled figure for every rule, every example and every question · Desmos questions · Grade 11 · Ellipse International School

Domain	Lessons	Worked examples	Questions	Figures
SAT Math 2 — Advanced Math	8	144	176	**303**
SAT Math 4 — Geometry & Trigonometry	6	108	132	**235**
All domains	**14**	**252**	**308**	**538**

SAT Math 2 — Advanced Math

1. **2.1** Equivalent Expressions
2. **2.2** Quadratic Equations
3. **2.3** Polynomials
4. **2.4** Rational & Radical Equations
5. **2.5** Exponential Functions
6. **2.6** Quadratic & Polynomial Graphs
7. **2.7** Function Notation & Transformations
8. **2.8** Nonlinear Systems

SAT Math 4 — Geometry & Trigonometry

1. **4.1** Lines, Angles & Triangles
2. **4.2** Similarity & Congruence
3. **4.3** Right Triangles & Trigonometry
4. **4.4** Circles
5. **4.5** Area & Volume
6. **4.6** Coordinate Geometry

Each domain is also supplied as its own file. Every numeric answer in this booklet is recomputed by the checkers in the source folder; every figure is drawn to the numbers in its own question.

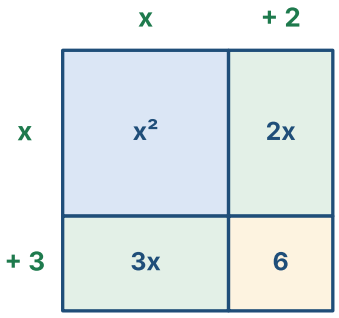
SAT Math 2.1 — Equivalent Expressions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Multiplying and collecting

multiply every term by every term | then collect only like terms

Like terms share the same variable to the same power: $3x^2$ and $5x^2$ combine, $3x^2$ and $5x$ do not. The area model keeps every product in view and is worth drawing whenever a minus sign is involved.

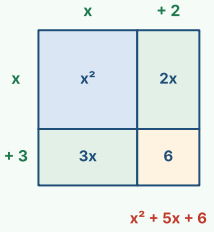


$$(x + 2)(x + 3) = x^2 + 5x + 6$$

The rectangle is (x + 2) wide and (x + 3) tall, so its area is the product. Each piece is one of the four multiplications, and the two green pieces are the like terms that collect into 5x.

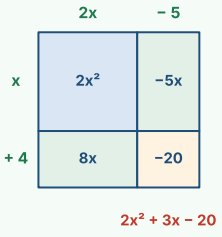
Example 1. **Expand (x + 2)(x + 3).**

Solution.
x·x + 3x + 2x + 6.
= **$x^2 + 5x + 6$**.



Example 2. **Expand (2x − 5)(x + 4).**

Solution.
$2x^2 + 8x - 5x - 20$.
= **$2x^2 + 3x - 20$**.



Example 3. **Simplify $(3x^2 + 2x - 1) - (x^2 - 4x + 6)$.**

Solution.
Distribute the minus to *every* term: $3x^2 + 2x - 1 - x^2 + 4x - 6$.
$= \mathbf{2x^2 + 6x - 7}$.

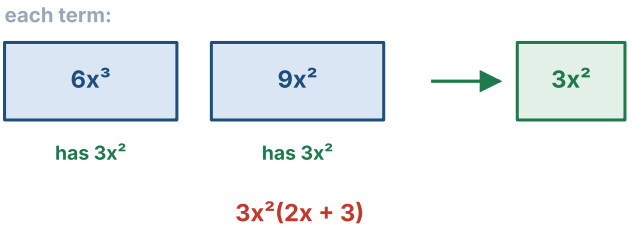
the minus belongs to every term in the bracket

$-(x^2 - 4x + 6)$	$= -x^2 + 4x - 6$
every sign inside flips	all three, not just the first

2 Common factor first, always

take out the greatest common factor before anything else

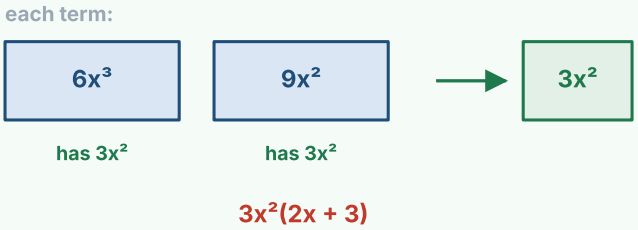
Pulling out the GCF turns an awkward expression into a familiar one and is the step most often skipped. Check the numbers and every variable: the GCF of $6x^3$ and $9x^2$ is $3x^2$, not 3.



Both terms carry three lots of x^2, so $3x^2$ comes out in front and what is left goes inside the bracket. Take the numbers *and* the letters — 3 alone would not be greatest.

Example 1. **Factor $6x^3 + 9x^2$ completely.**

Solution.
GCF is $3x^2$.
$= \mathbf{3x^2(2x + 3)}$.

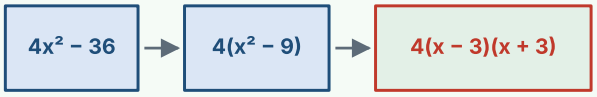


Example 2. **Factor $4x^2 - 36$ completely.**

Solution.

GCF 4 first: $4(x^2 - 9)$, and $x^2 - 9$ is a difference of squares.

= **$4(x - 3)(x + 3)$**. Stopping at $4(x^2 - 9)$ is not *completely* factored.



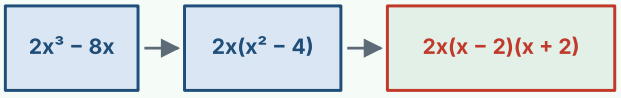
the GCF first, then the pattern underneath it

Example 3. **Factor $2x^3 - 8x$.**

Solution.

GCF $2x$: $2x(x^2 - 4)$.

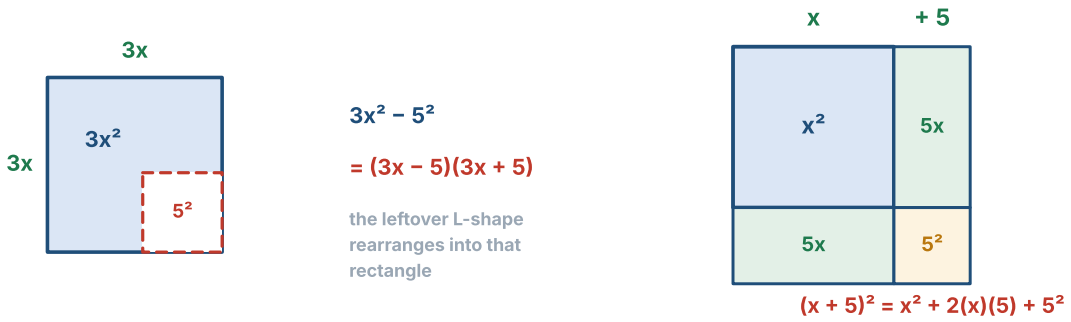
= **$2x(x - 2)(x + 2)$**.



3 Difference of squares and perfect squares

$$a^2 - b^2 = (a - b)(a + b) \mid a^2 \pm 2ab + b^2 = (a \pm b)^2$$

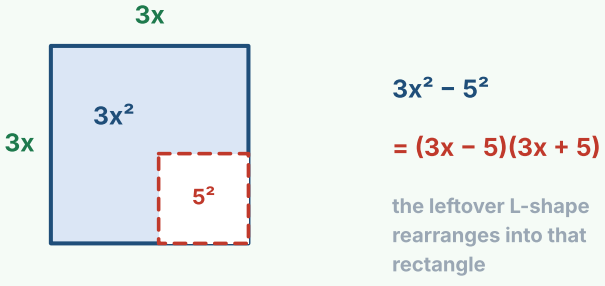
These two patterns appear constantly, often disguised. A sum of squares, $x^2 + 9$, does *not* factor over the real numbers — recognising that saves time on a question designed to waste it.



Left: cut a small square out of a big one and the L-shape left over rearranges into the rectangle (a − b)(a + b). Right: a perfect square has two identical middle pieces, which is where the 2ab comes from.

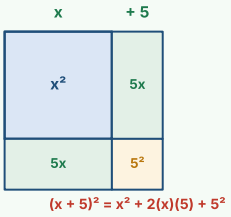
Example 1. **Factor $9x^2 - 25$.**

Solution.
This is $(3x)^2 - 5^2$.
$= \mathbf{(3x - 5)(3x + 5)}$.



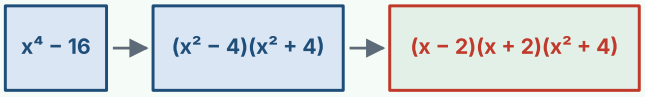
Example 2. **Factor $x^2 + 10x + 25$.**

Solution.
$25 = 5^2$ and $10x = 2(x)(5)$.
$= \mathbf{(x + 5)^2}$.



Example 3. **Factor $x^4 - 16$.**

Solution.
$(x^2)^2 - 4^2 = (x^2 - 4)(x^2 + 4)$, and the first bracket factors again.
$= \mathbf{(x - 2)(x + 2)(x^2 + 4)}$. The sum of squares stops here.



$x^2 + 4$ is a sum of squares — it does not factor

4 Factoring trinomials

$x^2 + bx + c$: find two numbers that multiply to c and add to b

When the leading coefficient is not 1, multiply a by c, split the middle term using two numbers that multiply to ac and add to b, then factor by grouping. Writing out the factor pairs of ac beats guessing.

two numbers that multiply to 12 and add to −7

-1 × -12 = 12 → -1 + -12 = -13

-2 × -6 = 12 → -2 + -6 = -8

-3 × -4 = 12 → -3 + -4 = -7 ✓

= (x − 3)(x − 4)

List the factor pairs and test the sum of each. Only one pair does both jobs, and that pair is the pair of brackets.

Example 1. **Factor $x^2 - 7x + 12$.**

Solution.
Two numbers multiplying to 12 and adding to −7: −3 and −4.
$= \mathbf{(x - 3)(x - 4)}$.

two numbers that multiply to 12 and add to −7

-1 × -12 = 12 → -1 + -12 = -13

-2 × -6 = 12 → -2 + -6 = -8

-3 × -4 = 12 → -3 + -4 = -7 ✓

= (x − 3)(x − 4)

Example 2. **Factor $x^2 + 2x - 15$.**

Solution.

Multiply to −15, add to 2: 5 and −3.

= **(x + 5)(x − 3)**.

two numbers that multiply to −15 and add to 2

-1 × 15 = -15 → -1 + 15 = 14

1 × -15 = -15 → 1 + -15 = -14

3 × -5 = -15 → 3 + -5 = -2

5 × -3 = -15 → 5 + -3 = 2 ✓

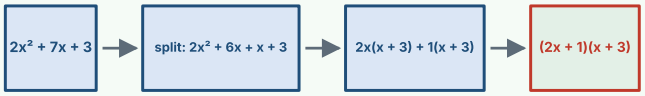
= (x + 5)(x − 3)

Example 3. **Factor $2x^2 + 7x + 3$.**

Solution.

ac = 6; two numbers multiplying to 6 and adding to 7 are 6 and 1.

$2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) =$ **(2x + 1)(x + 3)**.



ac = 6, and 6 + 1 = 7

5 Exponent rules

$$x^m \cdot x^n = x^{m+n} \mid x^m / x^n = x^{m-n} \mid (x^m)^n = x^{mn} \mid x^{-n} = 1/x^n \mid x^0 = 1$$

Add exponents when multiplying, subtract when dividing, multiply when raising a power to a power. A negative exponent is a reciprocal, not a negative number: $2^{-3} = 1/8$, not -8.

five rules, and one worked instance of each

$x^m \cdot x^n = x^{m+n}$	$(3x^2y)(4x^3y^5) = 12x^5y^6$
$x^m / x^n = x^{m-n}$	$12^5/12^3 = 12^2 = 144$
$(x^m)^n = x^{mn}$	$(2x^3)^4 = 16x^{12}$
$x^{-n} = 1/x^n$	$5^{-2} = 1/25 = 0.04$
$x^0 = 1$	$7^0 = 1$

Read each rule with its example beside it. The fourth is the one that costs marks: a negative exponent flips the base, it never makes the value negative.

Example 1. **Simplify $(3x^2y)(4x^3y^5)$.**

Solution.
Multiply coefficients, add exponents.
= **$12x^5y^6$**.

coefficients: 3×4	$= 12$
x: $2 + 3$	$= x^5$
y: $1 + 5$	$= y^6$

Example 2. **Simplify $(2x^3)^4$.**

Solution.
Raise both parts to the fourth: $2^4 = 16$ and x^{12}.
= **$16x^{12}$**.

the 2 is inside the bracket too	$2^4 = 16$
exponents multiply	$(x^3)^4 = x^{12}$

Example 3. **Write 5^{-2} as a fraction.**

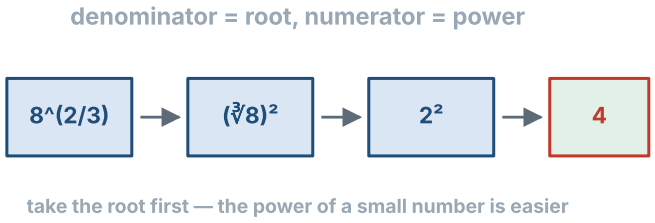
Solution.
1/25. A negative exponent flips the base; it never makes the value negative.

5^{-2}	$= 1/5^2 = 1/25 = 0.04$
not -25	the sign of the base never changes

6 Fractional exponents and radicals

$$x^{1/n} = \sqrt[n]{x} \mid x^{m/n} = \sqrt[n]{(x^m)}$$

The denominator of the exponent is the root; the numerator is the power. This conversion is the whole skill — once an expression is written with exponents, the ordinary exponent rules apply and radicals stop being special.



Denominator 3 means a cube root, numerator 2 means squared. Take the root first and you are squaring 2 instead of cube-rooting 64.

Example 1. **Write √x as a power of x.**

Solution.
$x^{1/2}$.

√x	= x^(1/2)
∛x	= x^(1/3)
ⁿ√x	= x^(1/n)

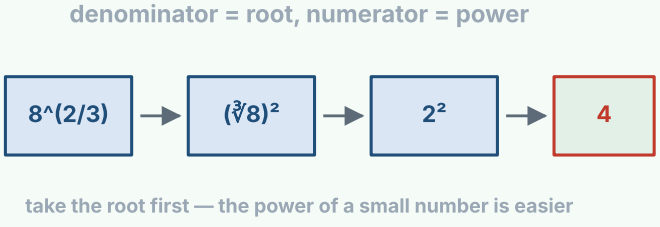
Example 2. **Simplify $x^{1/2} \cdot x^{1/3}$.**

Solution.
Add the exponents: 1/2 + 1/3 = 5/6.
= $x^{5/6}$.

same base → add the exponents	1/2 + 1/3 = 5/6
answer	x^(5/6)

Example 3. **Evaluate $8^{2/3}$.**

Solution.
Cube root first, then square: $\sqrt[3]{8} = 2$, and $2^2 = \mathbf{4}$. Taking the root first keeps the numbers small.



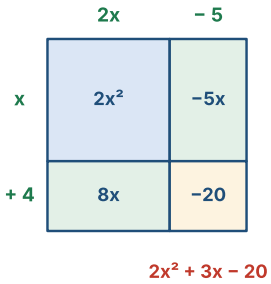
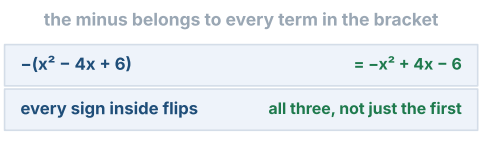
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

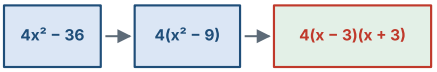
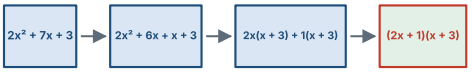
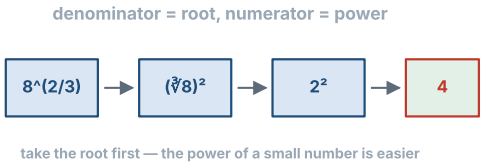
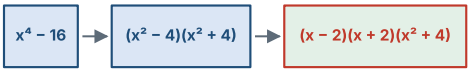
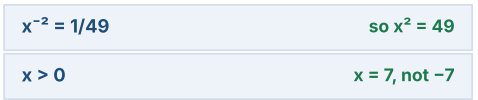
The SAT gives you Desmos on every maths question. For this topic it is **a checker, not a solver**. Desmos will not factor for you, but graphing your answer against the original shows instantly whether they are the same expression.

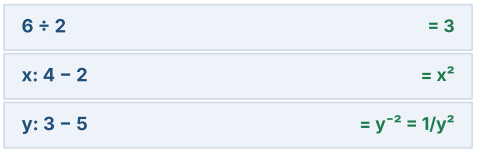
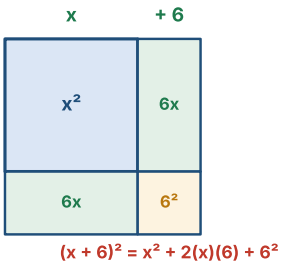
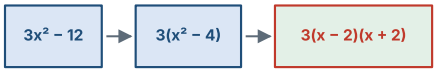
#	Question	Type into Desmos	Then	Answer
D1	Expand $(x + 2)(x + 3)$ and state the coefficient of x.	`(x+2)(x+3)`	Desmos will not expand symbolically; graph both forms to confirm they agree.	5
D2	What is the value of $8^{2/3}$?	`8^(2/3)`	Use the caret for exponents.	4
D3	What is the value of 5^{-2} as a decimal?	`5^(-2)`	A negative exponent gives a reciprocal.	0.04
D4	Simplify $12^5/12^3$.	`12^5/12^3`	Subtracting exponents gives 12^2.	144
D5	Evaluate $(2x^3)^4$ at x = 1.	`(2*1^3)^4`	Checks the coefficient $2^4 = 16$.	16
D6	At x = 5, what is the value of $x^2 - 7x + 12$?	`5^2-7*5+12`	Substitute to test a factorisation.	2
D7	At x = 3, does $(x - 3)(x - 4)$ equal zero?	`(3-3)*(3-4)`	A factor of zero makes the product zero.	0
D8	Evaluate $9x^2 - 25$ at x = 2.	`9*2^2-25`	Compare with $(3x - 5)(3x + 5)$ at x = 2.	11
D9	What is the value of $x^{1/2} \cdot x^{1/3}$ at x = 64?	`64^(1/2)*64^(1/3)`	Equals $64^{5/6}$.	32
D10	Evaluate $2x^3 - 8x$ at x = 2.	`2*2^3-8*2`	x = 2 is a zero, which confirms $(x - 2)$ is a factor.	0

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Read the word **completely** when it appears: more than one option may be equal in value, and only one is fully factored.

#	Level	Question	Answer	Solution
P1	Hard	Write an equivalent expression for $(2x - 5)(x + 4)$. 	$2x^2 + 3x - 20$	$8x - 5x = +3x$. $= 2x^2 + 3x - 20$. A common wrong answer multiplies only the first and last terms; C reverses the sign of the middle term.
P2	Hard	Write an equivalent expression for $(3x^2 + 2x - 1) - (x^2 - 4x + 6)$. 	$2x^2 + 6x - 7$	The minus sign applies to all three terms of the second bracket. $= 2x^2 + 6x - 7$. Two common wrong answers subtract only the first term.

#	Level	Question	Answer	Solution
P3	Hard	Factor completely: $4x^2 - 36$. 	4(x − 3)(x + 3)	Take the GCF 4 first, then the difference of squares. = **4(x − 3)(x + 3)**. Two common wrong answers equal in value but not *completely* factored — read the word 'complete' in the stem.
P4	Hard	What is the complete factorisation of $2x^2 + 7x + 3$? 	(2x + 1)(x + 3)	ac = 6, and 6 + 1 = 7, so split the middle term. 2x(x + 3) + 1(x + 3) = **(2x + 1)(x + 3)**. Expanding that answer gives $2x^2 + 7x + 3$.
P5	Hard	Write an equivalent expression for $(2x^3y^2)^3$. 	$8x^9y^6$	Every factor is cubed: $2^3 = 8$, and the exponents multiply by 3. = **$8x^9y^6$**. A common wrong answer multiplies the coefficient by 3 instead of cubing it.
P6	Hard	What is the value of $8^{2/3}$? 	4	The denominator is the root and the numerator is the power. $(\sqrt[3]{8})^2 = 2^2 = $ **4**.
P7	Very hard	Write an equivalent expression for $x^4 - 16$, factored completely over the real numbers. 	(x − 2)(x + 2)(x² + 4)	$x^4 - 16 = (x^2 - 4)(x^2 + 4)$, and $x^2 - 4$ factors again. = **$(x - 2)(x + 2)(x^2 + 4)$**. A sum of squares does not factor over the reals, so that answer is as far as it goes; A stops one step early.
P8	Very hard	If $x^{-2} = 1/49$ and $x > 0$, what is the value of x? 	7	$x^{-2} = 1/x^2$, so $x^2 = 49$. With $x > 0$, x = **7**.
P9	Very hard	Write an equivalent expression for $(x^{1/2})(x^{1/3})$ for $x > 0$. 	$x^{5/6}$	Multiplying powers of the same base adds the exponents. 1/2 + 1/3 = 5/6, so the answer is **$x^{5/6}$**. A common wrong answer multiplies the exponents.

#	Level	Question	Answer	Solution
P10	Very hard	The expression $(6x^4y^3)/(2x^2y^5)$ is equivalent to which of the following, for x and y nonzero? 	$3x^2/y^2$	Divide the coefficients and subtract the exponents: 6/2 = 3, $x^{4-2} = x^2$, $y^{3-5} = y^{-2}$. = **$3x^2/y^2$**. A common wrong answer leaves the negative exponent in the numerator as a positive one.
P11	Very hard	For what value of k is $x^2 + kx + 36$ a perfect square trinomial, given k > 0? 	12	A perfect square $(x + a)^2$ needs $a^2 = 36$, so a = 6, and the middle term is 2a. k = 2(6) = **12**.
P12	Very hard	The expression $3x^2 − 12$ can be written as $3(x − a)(x + a)$ for a positive constant a. What is the value of a? 	2	$3x^2 − 12 = 3(x^2 − 4) = 3(x − 2)(x + 2)$. So a = **2**. A common wrong answer takes a to be the 4 inside the bracket rather than its square root.

The one habit to build. Take the common factor out first, every single time. Half the ‘hard’ questions in this topic become one-liners once the GCF is gone.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

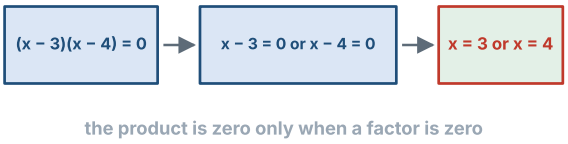
SAT Math 2.2 — Quadratic Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

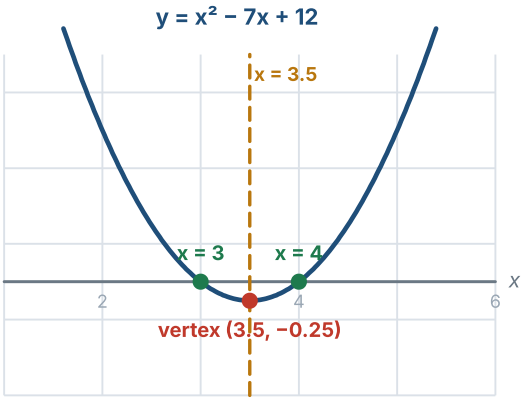
1 Solving by factoring

get = 0 first | factor | set each factor to zero

The zero-product property only works against zero, so move everything to one side before you factor. Dividing both sides by a variable loses a solution — factor it out instead.



the product is zero only when a factor is zero



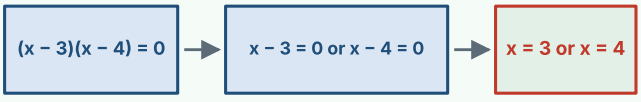
The factored form names the roots directly, and the graph shows them as the two places the curve meets the axis. Both brackets are zero nowhere else, which is why the method works.

Example 1. **Solve $x^2 - 7x + 12 = 0$.**

Solution.

$(x - 3)(x - 4) = 0$.

$x =$ **3 or 4**.



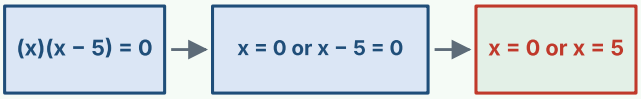
the product is zero only when a factor is zero

Example 2. **Solve $x^2 = 5x$.**

Solution.

$x^2 - 5x = 0$, so $x(x - 5) = 0$.

$x =$ **0 or 5**. Dividing by x at the start would have lost $x = 0$.



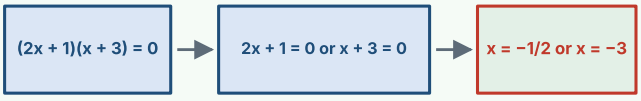
the product is zero only when a factor is zero

Example 3. **Solve $2x^2 + 7x + 3 = 0$.**

Solution.

$(2x + 1)(x + 3) = 0$.

$x =$ **$-1/2$ or -3**.



the product is zero only when a factor is zero

2 The quadratic formula

$x = [-b \pm \sqrt{(b^2 - 4ac)}] \div 2a$ — not on the reference sheet

Learn it: the SAT does not give it to you. Write a, b and c down before substituting, and keep the signs of b and c with their numbers. It works on every quadratic, factorable or not.

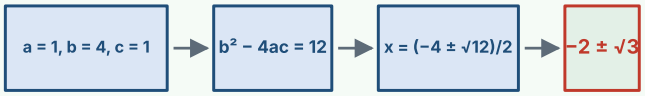
the formula works on every quadratic, factorable or not

$x = [-b \pm \sqrt{(b^2 - 4ac)}] \div 2a$	not on the reference sheet
write a, b and c down first	with their signs
$b^2 - 4ac$ is the discriminant	it decides how many roots

Three lines worth memorising. The sign of b travels with it: for $x^2 - 6x + 5$ the value of b is −6, so −b is +6.

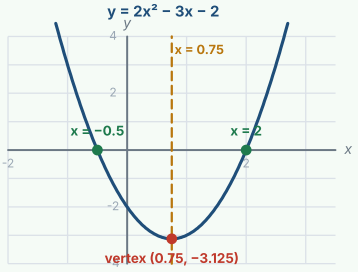
Example 1. **Solve $x^2 + 4x + 1 = 0$.**

Solution.
a = 1, b = 4, c = 1, so the discriminant is 16 − 4 = 12.
$x = (-4 \pm \sqrt{12})/2 =$ **$-2 \pm \sqrt{3}$**.



Example 2. **Solve $2x^2 - 3x - 2 = 0$.**

Solution.
Discriminant = 9 + 16 = 25, and √25 = 5.
x = (3 ± 5)/4, so x = **2 or −1/2**.



Example 3. **Why keep the sign with b?**

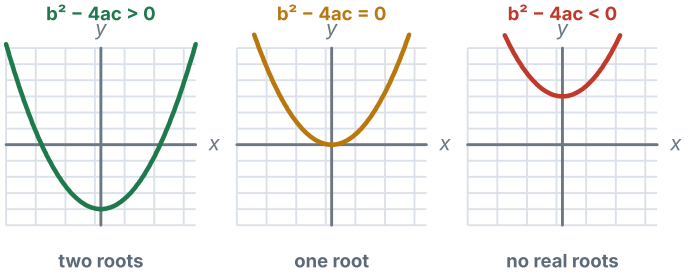
Solution.
For $x^2 - 6x + 5$, b = −6, so −b = +6. Writing b = 6 flips both roots — the single most common slip with the formula.

$x^2 - 6x + 5$	b = −6, so −b = +6
writing b = 6	gives −5 and −1 — both wrong

3 The discriminant

$b^2 - 4ac > 0$ → two real solutions | $= 0$ → one | < 0 → none

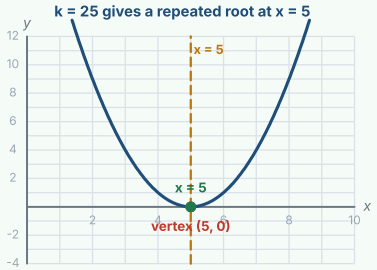
Whenever a question says ‘exactly one solution’, ‘a repeated root’, ‘touches the x-axis once’ or ‘the line is tangent’, it is asking you to set $b^2 - 4ac = 0$. One idea, many disguises.



The discriminant is the quantity under the root, so it counts the crossings: positive gives two, zero gives a touch, negative gives none because the root of a negative is not real.

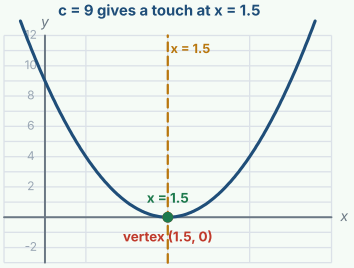
Example 1. **For what value of k does $x^2 - 10x + k = 0$ have exactly one solution?**

Solution.
$100 - 4k = 0$.
$k = \mathbf{25}$.



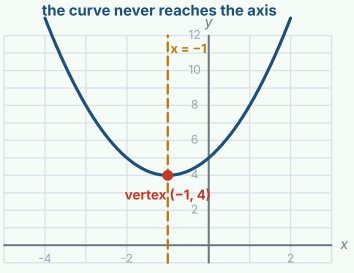
Example 2. **For what value of c does $4x^2 - 12x + c = 0$ have exactly one real solution?**

Solution.
$144 - 16c = 0$.
$c = \mathbf{9}$.



Example 3. **How many real solutions does $x^2 + 2x + 5 = 0$ have?**

Solution.
$4 - 20 = -16$, which is negative.
None.



4 Sum and product of the roots

sum = −b/a | product = c/a

When a question asks for the sum or product of the solutions, you never need the solutions themselves. Watch the sign: for $x^2 - 7x + 12$ the sum is +7, not −7.

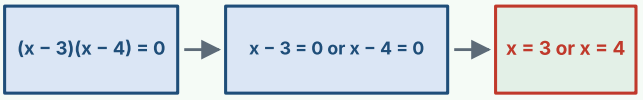
read them straight off the coefficients

sum of the roots	−b/a
product of the roots	c/a
$x^2 - 7x + 12$	sum −(−7)/1 = 7, product 12/1 = 12

The roots of $x^2 - 7x + 12$ are 3 and 4: they add to 7 and multiply to 12, exactly as −b/a and c/a predict — without solving anything.

Example 1. **What is the sum of the solutions of $x^2 - 7x + 12 = 0$?**

Solution.
−b/a = −(−7)/1 = **7**. Checking: 3 + 4 = 7.



the product is zero only when a factor is zero

Example 2. **What is the product of the solutions of $2x^2 + 7x + 3 = 0$?**

Solution.
c/a = 3/2 = **1.5**. Checking: (−1/2)(−3) = 1.5.

$2x^2 + 7x + 3$	a = 2, c = 3
product = c/a	3/2 = 1.5

Example 3. **The solutions of $x^2 + bx + 10 = 0$ sum to 7. What is b?**

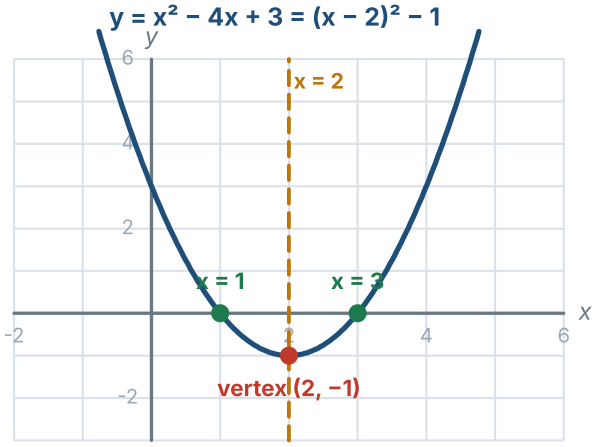
Solution.
−b = 7, so b = **−7**.

sum = −b/a = −b	and the sum is 7
so −b = 7	b = −7

5 Completing the square and vertex form

$x^2 + bx = (x + b/2)^2 - (b/2)^2$ | vertex form $y = a(x - h)^2 + k$ has vertex (h, k)

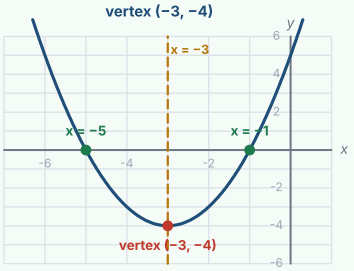
Halve the coefficient of x, square it, add and subtract it. Vertex form hands you the turning point directly, which is why the SAT asks you to produce it.



Both forms draw the same curve. Standard form shows the y-intercept 3; vertex form shows the turning point (2, −1) without any further work.

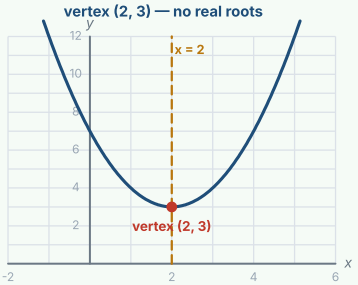
Example 1. **Write $x^2 + 6x + 5$ in vertex form.**

Solution.
Half of 6 is 3, and $3^2 = 9$.
$= (x + 3)^2 - 9 + 5 =$ **$(x + 3)^2 - 4$**, vertex $(-3, -4)$.



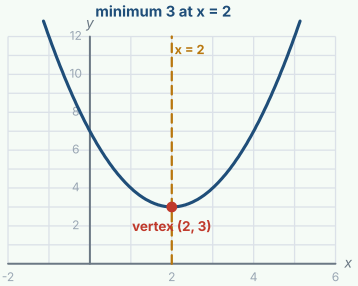
Example 2. **Write $x^2 - 4x + 7$ in vertex form.**

Solution.
Half of −4 is −2, squared is 4.
$=$ **$(x - 2)^2 + 3$**, vertex $(2, 3)$.



Example 3. **What is the minimum value of $y = x^2 - 4x + 7$?**

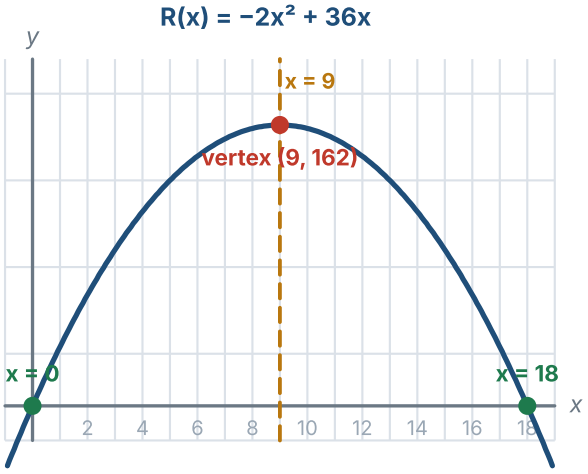
Solution.
From the vertex form, the smallest $(x - 2)^2$ can be is 0.
The minimum is **3**, reached at $x = 2$.



6 Quadratic word problems

maximum or minimum → the vertex | hits the ground / breaks even → a root

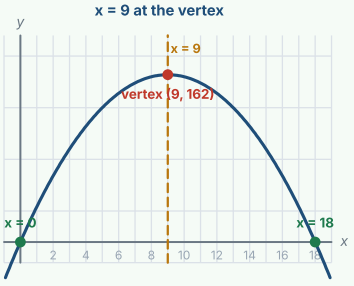
Translate the question into one of two requests. ‘What price gives the greatest revenue?’ wants the vertex’s x. ‘What is the greatest revenue?’ wants its y. Confusing those two costs marks every time.



The vertex (9, 162) answers both questions at once: x = 9 is the price that maximises revenue, y = 162 is the revenue itself. The roots are where revenue falls to zero.

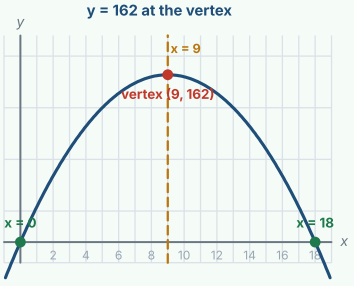
Example 1. **R(x) = −2x² + 36x. What price x gives the maximum revenue?**

Solution.
Vertex at x = −b/2a = −36/(−4) = **$9**.



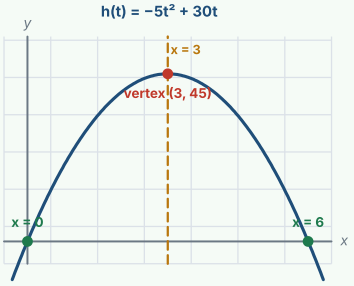
Example 2. **For that same R, what is the maximum revenue?**

Solution.
R(9) = −2(81) + 324 = **$162**. Note this answers a different question from the one above.



Example 3. **A ball’s height is h(t) = −5t² + 30t. When does it land?**

Solution.
−5t(t − 6) = 0, so t = 0 or 6.
It lands at **t = 6 seconds**; t = 0 is the launch.



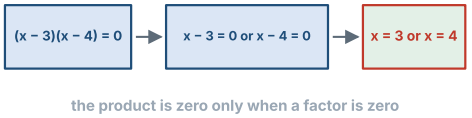
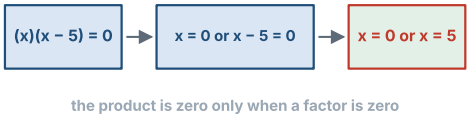
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

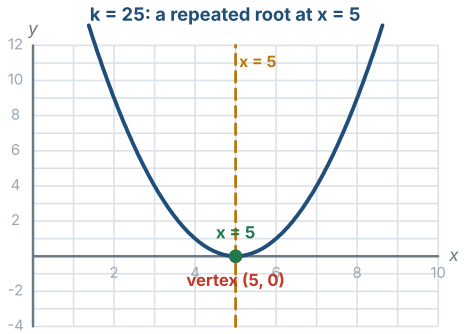
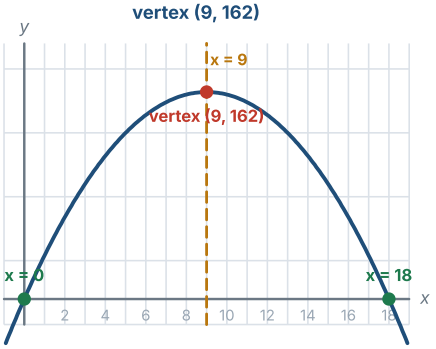
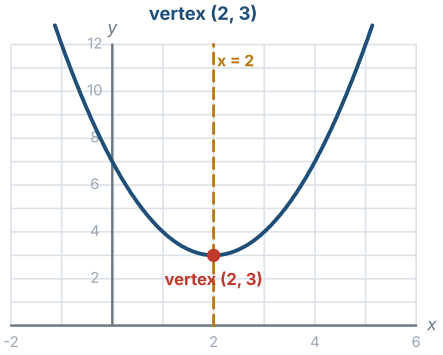
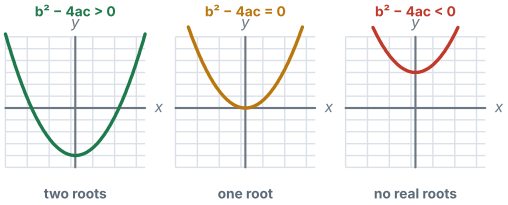
The SAT gives you Desmos on every maths question. Here it is **a root finder and a vertex finder**. Type the equation and read the x-intercepts, or type the function and click the turning point.

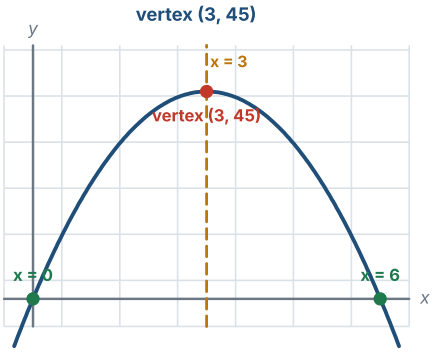
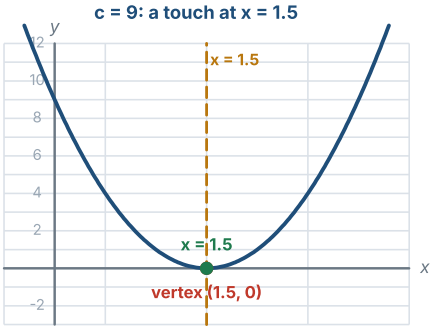
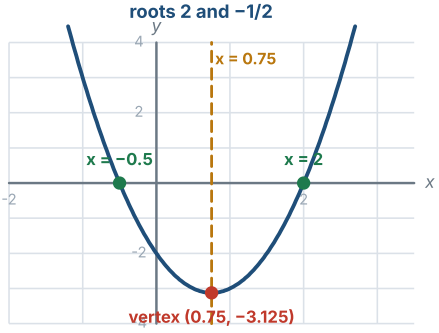
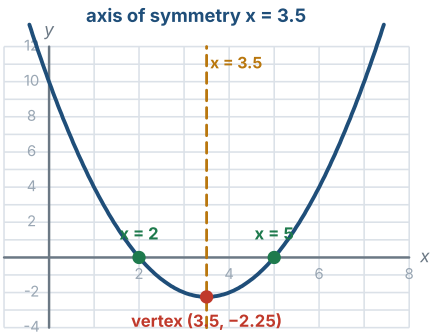
#	Question	Type into Desmos	Then	Answer
D1	What are the solutions of $x^2 - 7x + 12 = 0$?	`x^2-7x+12=0`	Type the equation; Desmos marks the roots on the x-axis.	3 and 4
D2	What is the sum of the solutions of $x^2 - 7x + 12 = 0$?	`3+4`	Or use −b/a directly.	7
D3	For what value of k does $x^2 - 10x + k = 0$ have exactly one solution?	`10^2/4`	Set the discriminant to zero: $k = b^2/4a$.	25
D4	What is the discriminant of $x^2 + 2x + 5 = 0$?	`2^2-4*1*5`	Negative means no real solutions.	-16
D5	Solve $2x^2 - 3x - 2 = 0$.	`2x^2-3x-2=0`	Graph it and read both x-intercepts.	2 and -0.5
D6	What is the minimum value of $y = x^2 - 4x + 7$?	`y=x^2-4x+7`	Graph it and click the turning point.	3
D7	$R(x) = -2x^2 + 36x$. What value of x gives the maximum?	`y=-2x^2+36x`	Click the vertex; Desmos labels it.	9
D8	For that R, what is the maximum value?	`-2*9^2+36*9`	The y of the vertex, not the x.	162
D9	At what positive time does $h(t) = -5t^2 + 30t$ return to zero?	`-5t^2+30t=0`	Read the right-hand root.	6
D10	What is the value of the discriminant of $4x^2 - 12x + 9$?	`12^2-4*4*9`	Zero means one repeated root.	0

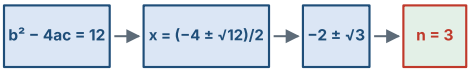
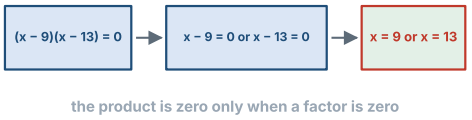
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two questions decide most of these: is it asking **where** or **how much**, and does 'exactly one solution' appear anywhere in the stem?

#	Level	Question	Answer	Solution
P1	Hard	What is the sum of the solutions of the equation $x^2 - 7x + 12 = 0$? 	7	Sum of roots = −b/a = −(−7)/1 = **7**, and indeed 3 + 4 = 7. A common wrong answer applies the formula without the sign of b — the error five of your class made on Test 5.
P2	Hard	If $x^2 = 5x$ and $x \neq 0$, what is the value of x? 	5	$x^2 - 5x = 0$ gives $x(x - 5) = 0$, so x = 0 or 5. The condition $x \neq 0$ leaves **5**. Dividing by x at the start would also reach 5, but it silently discards a solution, which matters when the question does not rule it out.

#	Level	Question	Answer	Solution
P3	Hard	For what value of k does the equation $x^2 - 10x + k = 0$ have exactly one real solution? 	25	Exactly one solution means $b^2 - 4ac = 0$. $100 - 4k = 0$, so k = **25**. A common wrong answer is b/2 and B is b itself, both read off the equation rather than derived.
P4	Hard	A candle shop's daily revenue, in dollars, is $R(x) = -2x^2 + 36x$, where x is the price of one candle. What price gives the maximum daily revenue? 	$9	The vertex is at $x = -b/2a = -36/(-4) =$ **$9**. A common wrong answer is the maximum revenue R(9) = 162, which answers a different question; C is the coefficient read straight off the equation.
P5	Hard	Write an equivalent form of $x^2 - 4x + 7$. 	$(x - 2)^2 + 3$	Half of −4 is −2 and $(-2)^2 = 4$, so $x^2 - 4x = (x - 2)^2 - 4$. $=$ **$(x - 2)^2 + 3$**. A common wrong answer forgets to subtract the 4 back out.
P6	Hard	How many distinct real solutions does $x^2 + 2x + 5 = 0$ have? 	Zero	$b^2 - 4ac = 4 - 20 = -16 < 0$. A negative discriminant means **no real solutions** — the parabola never reaches the x-axis.

#	Level	Question	Answer	Solution
P7	Very hard	A ball is thrown upward and its height, in metres, after t seconds is $h(t) = -5t^2 + 30t$. What is the maximum height reached, in metres? 	45	The vertex is at $t = -30/(-10) = 3$ seconds. $h(3) = -45 + 90 =$ **45 metres**. Answering 3 gives the time, not the height.
P8	Very hard	In the equation $4x^2 - 12x + c = 0$, c is a constant. If the equation has exactly one real solution, what is the value of c? 	9	$144 - 16c = 0$, so $c = 144/16 =$ **9**. A common wrong answer is 12/4, read off the equation rather than derived.
P9	Very hard	The solutions of $2x^2 - 3x - 2 = 0$ are p and q. What is the value of pq? 	−1	Product of roots $= c/a = -2/2 =$ **−1**. Checking: the roots are 2 and −1/2, and $2 \times (-1/2) = -1$. A common wrong answer uses c/a without the sign of c.
P10	Very hard	The graph of $y = x^2 + bx + 10$ has its vertex on the line x = 3.5. What is the value of b? 	−7	The vertex sits at $x = -b/2a = -b/2$. $-b/2 = 3.5$ gives $b =$ **−7**. A common wrong answer drops the minus sign.

#	Level	Question	Answer	Solution
P11	Very hard	The equation $x^2 + 4x + 1 = 0$ has solutions of the form $-2 \pm \sqrt{n}$, where n is a constant. What is the value of n? 	3	Discriminant = 16 − 4 = 12, so $x = (-4 \pm \sqrt{12})/2 = -2 \pm \sqrt{12}/2$. $\sqrt{12}/2 = 2\sqrt{3}/2 = \sqrt{3}$, so n = **3**.
P12	Very hard	A rectangular garden has a perimeter of 44 metres and an area of 117 square metres. What is the length of its longer side, in metres? 	13	Half the perimeter is 22, so the sides are x and 22 − x with $x(22 - x) = 117$. $x^2 - 22x + 117 = 0$ factors as $(x - 9)(x - 13) = 0$, giving sides 9 and **13**. A common wrong answer is the shorter side.

The one habit to build. The quadratic formula is **not** on the reference sheet. Write a, b and c down with their signs before you substitute — losing the sign of b is the most expensive habit in this topic.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

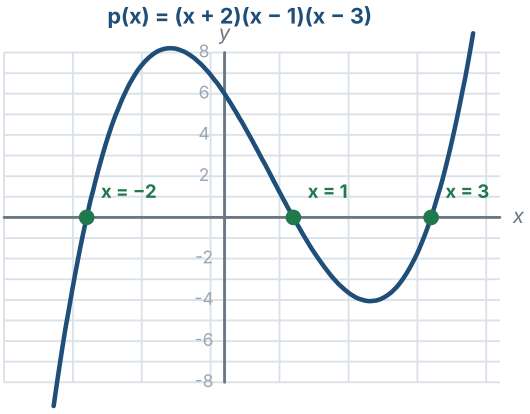
SAT Math 2.3 — Polynomials

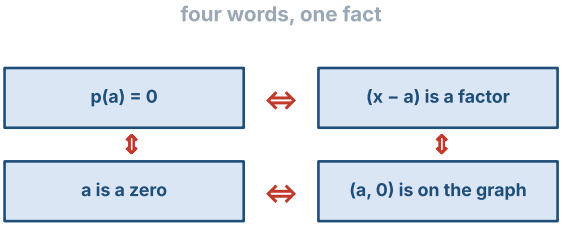
Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Zeros, roots, factors and intercepts

all four words mean the same thing: p(a) = 0 ⇔ (x − a) is a factor ⇔ (a, 0) is on the graph

The SAT swaps these words freely inside one question. Translate whichever you are given into whichever you need. A zero at x = 3 means the factor is (x − 3) — note the sign flip.

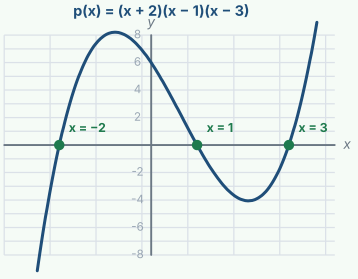




Each bracket switches off at one value of x, and that is where the curve meets the axis. The four statements on the right are the same fact in four vocabularies.

Example 1. **p(x) = (x + 2)(x − 1)(x − 3). What are its zeros?**

Solution.
Set each factor to zero.
x = **−2, 1 and 3**. Each sign flips.



Example 2. **A polynomial has zeros at x = −4 and x = 5. Give two of its factors.**

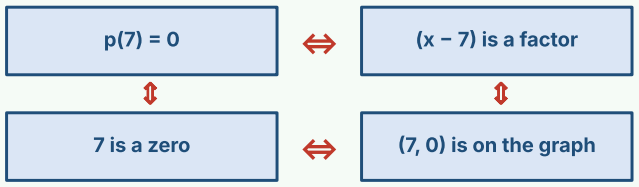
Solution.
(x + 4) and **(x − 5)**.

zero at x = −4	factor (x + 4)
zero at x = 5	factor (x − 5)
the sign always flips	x = a → (x − a)

Example 3. **The graph of p crosses the x-axis at (7, 0). What does that tell you about p?**

Solution.
p(7) = 0, and (x − 7) is a factor of p(x).

four words, one fact



2 The remainder and factor theorems

remainder on dividing by (x − a) is p(a) | remainder 0 ⇔ (x − a) is a factor

You never have to do the long division. To find the remainder, substitute. Watch the sign: dividing by (x + 2) means a = −2, because x + 2 = x − (−2).



and if p(a) = 0, the divisor is a factor

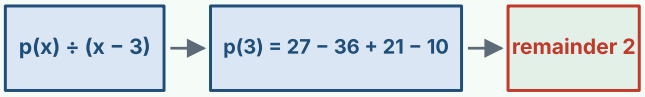
the sign flips on the way in

divide by (x − 3)	substitute x = +3
divide by (x + 2)	substitute x = −2

No division at all: put the number that switches the divisor off into p, and what comes out is the remainder. Zero means the divisor divides exactly.

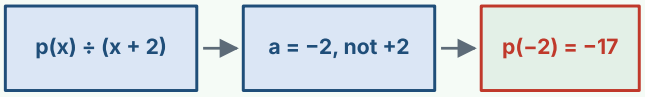
Example 1. **What is the remainder when $p(x) = x^3 - 4x^2 + 7x - 10$ is divided by x − 3?**

Solution.
p(3) = 27 − 36 + 21 − 10 = **2**.



Example 2. **What is the remainder when $p(x) = x^3 + 2x - 5$ is divided by x + 2?**

Solution.
Here a = −2.
p(−2) = −8 − 4 − 5 = **−17**.



Example 3. **Is $(x - 2)$ a factor of $x^3 - 3x^2 + 4$?**

Solution.

$p(2) = 8 - 12 + 4 = 0$.

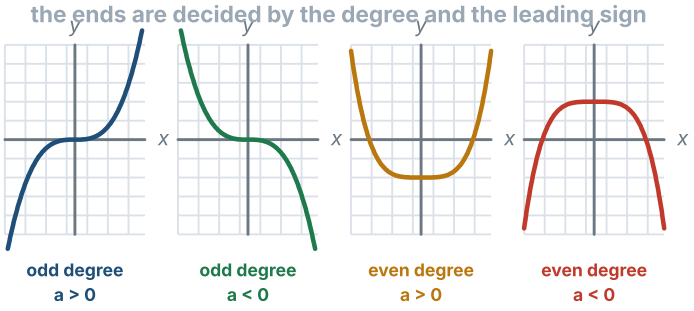
Yes — a remainder of zero means it is a factor.



3 Degree, end behaviour and turning points

degree n has at most n zeros and at most n − 1 turning points

The degree is the highest power. An odd-degree graph runs from bottom-left to top-right (or the reverse if the leading coefficient is negative); an even-degree graph has both ends pointing the same way.

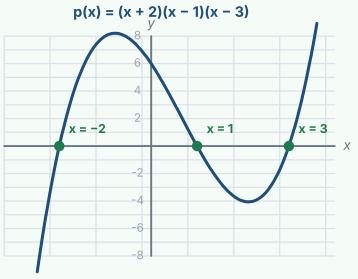


Four pictures cover every case. Odd degree sends the ends opposite ways; even degree sends them the same way; the leading sign says which way up.

Example 1. **How many turning points can a cubic have at most?**

Solution.

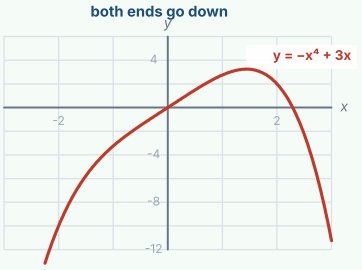
Degree 3, so at most **2** turning points.



Example 2. **Describe the end behaviour of $p(x) = -x^4 + 3x$.**

Solution.

Even degree with a negative leading coefficient, so **both ends go down**.



Example 3. **A polynomial has degree 5. What is the greatest number of x-intercepts it can have?**

Solution.

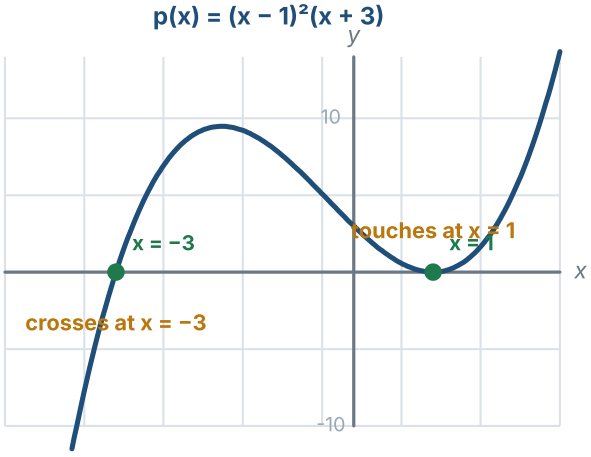
5. It could have fewer, but never more than its degree.

degree n	at most n zeros
degree n	at most n − 1 turning points
degree 5	5 zeros, 4 turning points at most

4 Multiplicity: crossing or touching

odd multiplicity → the graph crosses | even multiplicity → it touches and turns back

A repeated factor changes the shape at that zero. $(x - 2)^2$ makes the curve bounce off the axis at $x = 2$; $(x - 2)^3$ flattens but still passes through.

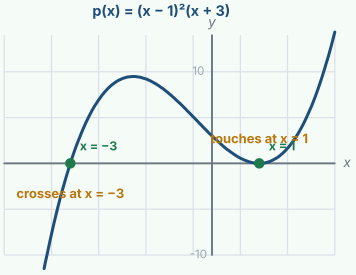


The squared factor at $x = 1$ turns the curve back without letting it through; the single factor at $x = -3$ lets it cross. Count the power on the bracket and you know which happens.

Example 1. **$p(x) = (x - 1)^2(x + 3)$. Describe the graph at each zero.**

Solution.

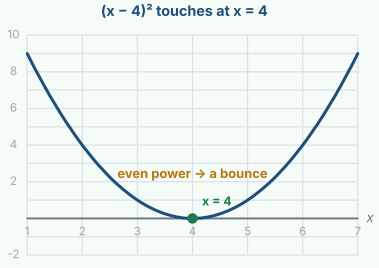
At $x = 1$ the multiplicity is 2, so it **touches and turns**. At $x = -3$ it is 1, so it **crosses**.



Example 2. **A graph touches the x-axis at $x = 4$ without crossing. What does that say about the factor?**

Solution.

$(x - 4)$ appears an **even** number of times — at least squared.



Example 3. **What is the degree of $p(x) = (x - 1)^2(x + 3)$?**

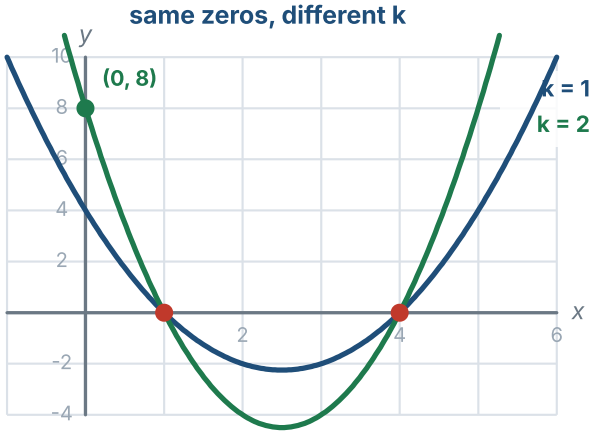
Solution.
Add the multiplicities: 2 + 1 = **3**.

$(x - 1)^2$	contributes 2
$(x + 3)$	contributes 1
degree	2 + 1 = 3

5 Building a polynomial from its zeros

zeros a, b, c → $p(x) = k(x - a)(x - b)(x - c)$

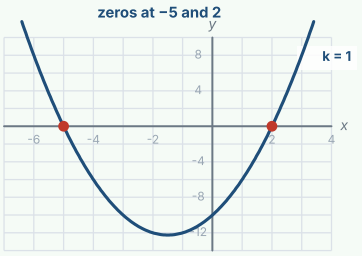
The zeros fix the shape but not the vertical scale, so a constant k remains. One extra point — usually the y-intercept — pins k down.



Both curves have zeros at 1 and 4; only the green one passes through (0, 8). That extra point is what fixes k = 2.

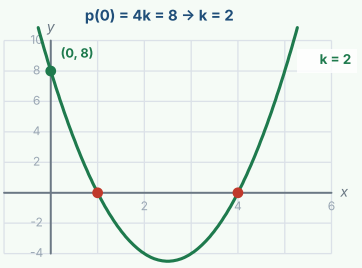
Example 1. **Write a quadratic with zeros 2 and −5.**

Solution.
$k(x - 2)(x + 5)$ for any nonzero k; with k = 1, $x^2 + 3x - 10$.



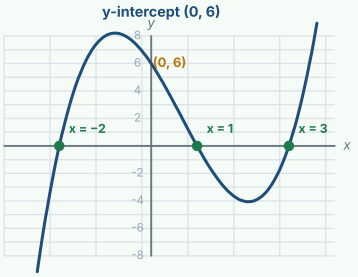
Example 2. **A quadratic has zeros 1 and 4 and passes through (0, 8). Find it.**

Solution.
$p(x) = k(x - 1)(x - 4)$, and $p(0) = 4k = 8$, so k = 2.
$p(x) = 2(x - 1)(x - 4)$.



Example 3. **What is the y-intercept of $p(x) = (x + 2)(x - 1)(x - 3)$?**

Solution.
$p(0) = (2)(-1)(-3) = \mathbf{6}$.



6 Dividing polynomials

$p(x) = (\text{divisor})(\text{quotient}) + \text{remainder}$

If a question gives you a quotient and a remainder, reassemble the original with this identity rather than dividing again. When the remainder is zero the divisor and quotient are factors.

reassemble instead of dividing again

$p(x) =$ [$x - 1$] [$x + 4$] + [3]

$= x^2 + 3x - 1$

Multiply the divisor by the quotient and add the remainder: the original polynomial reappears, with no long division anywhere.

Example 1. **$x^2 + 5x + 6$ divided by $(x + 2)$ gives what quotient?**

Solution.
$x^2 + 5x + 6 = (x + 2)(x + 3)$, so the quotient is $\mathbf{x + 3}$ with remainder 0.

reassemble instead of dividing again

$p(x) =$ [$x + 2$] [$x + 3$] + [0]

$= x^2 + 5x + 6$

Example 2. **$p(x)$ divided by $(x - 1)$ gives quotient $x + 4$ and remainder 3. Find $p(x)$.**

Solution.
$p(x) = (x - 1)(x + 4) + 3 = x^2 + 3x - 4 + 3$.
$= \mathbf{x^2 + 3x - 1}$.

reassemble instead of dividing again

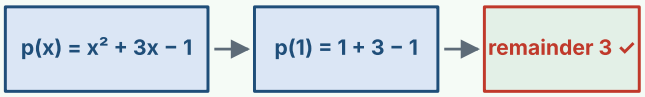
$p(x) =$ [$x - 1$] [$x + 4$] + [3]

$= x^2 + 3x - 1$

Example 3. **Check that result using the remainder theorem.**

Solution.

p(1) = 1 + 3 − 1 = **3**, which is the stated remainder. The identity holds.



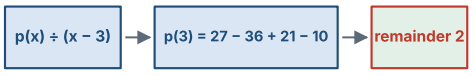
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

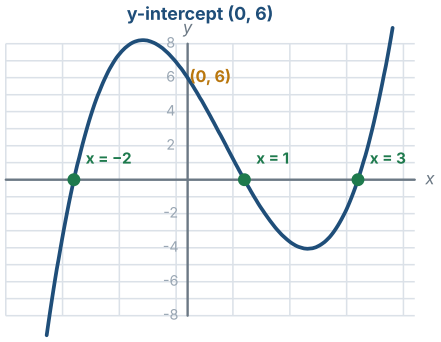
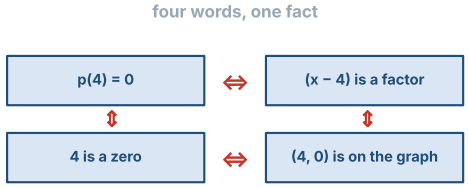
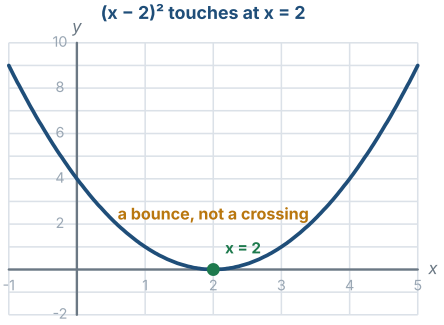
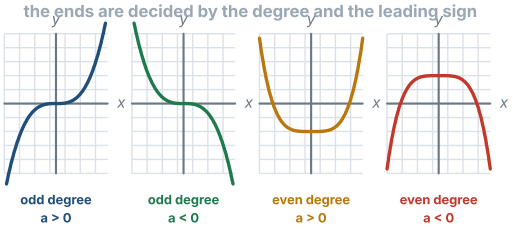
The SAT gives you Desmos on every maths question. For polynomials it is **a zero finder**. Graph p and read where it meets the axis; substitute a value to test a factor without dividing.

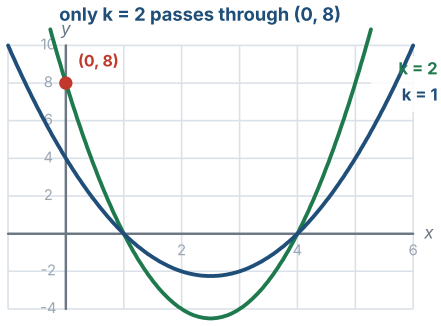
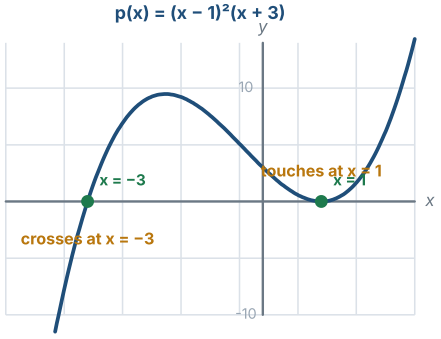
#	Question	Type into Desmos	Then	Answer
D1	What is the value of $x^3 - 4x^2 + 7x - 10$ at $x = 3$?	`3^3-4*3^2+7*3-10`	The remainder theorem in one line.	2
D2	What is the value of $x^3 + 2x - 5$ at $x = -2$?	`(-2)^3+2*(-2)-5`	Dividing by $x + 2$ means substituting -2.	-17
D3	Is $x = 2$ a zero of $x^3 - 3x^2 + 4$?	`2^3-3*2^2+4`	A value of 0 confirms the factor.	0 (yes)
D4	What is the y-intercept of $y = (x + 2)(x - 1)(x - 3)$?	`(0+2)*(0-1)*(0-3)`	Substitute $x = 0$.	6
D5	Graph $y = (x + 2)(x - 1)(x - 3)$. How many x-intercepts does it have?	`y=(x+2)(x-1)(x-3)`	Count where the curve meets the axis.	3
D6	For $p(x) = k(x - 1)(x - 4)$ with $p(0) = 8$, what is k?	`8/((0-1)*(0-4))`	Solve $4k = 8$.	2
D7	Expand $(x - 1)(x + 4) + 3$ at $x = 5$ to check a division.	`(5-1)*(5+4)+3`	Compare with $p(5)$ for $p(x) = x^2 + 3x - 1$.	39
D8	What is $x^2 + 3x - 1$ at $x = 5$?	`5^2+3*5-1`	Matches the line above, confirming the identity.	39
D9	Graph $y = (x - 1)^2(x + 3)$. At which zero does the curve only touch the axis?	`y=(x-1)^2(x+3)`	The squared factor makes it bounce.	x = 1
D10	What is the remainder when $x^4 - 1$ is divided by $x - 1$?	`1^4-1`	Zero means $(x - 1)$ is a factor.	0

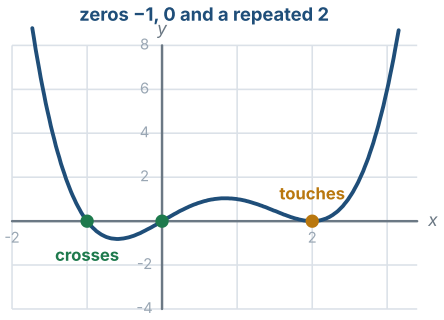
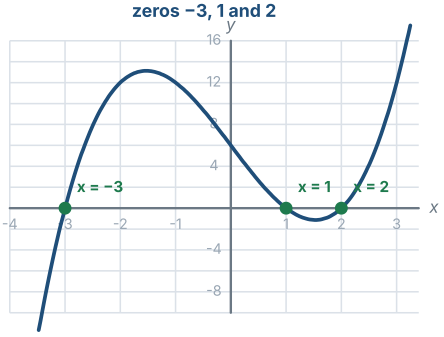
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Nearly every question here is the same idea in different words: **p(a) = 0** means $(x - a)$ is a factor means the graph crosses at $(a, 0)$.

#	Level	Question	Answer	Solution
P1	Hard	A shop's weekly profit index is modelled by $p(x) = x^3 - 4x^2 + 7x - 10$. What is the remainder when $p(x)$ is divided by $x - 3$? 	2	By the remainder theorem the answer is $p(3)$. $27 - 36 + 21 - 10 =$ **2**. No division is needed.

#	Level	Question	Answer	Solution
P2	Hard	The polynomial p is defined by p(x) = (x + 2)(x − 1)(x − 3). What is the y-intercept of the graph of y = p(x)? 	6	The y-intercept is p(0) = (2)(−1)(−3) = **6**. A common wrong answer loses one of the two negative signs.
P3	Hard	If (x − 4) is a factor of the polynomial p, which of the following must be true? 	p(4) = 0	A factor (x − a) corresponds to a zero at x = a, so **p(4) = 0**. A common wrong answer flips the sign, which is the single most common error in this topic.
P4	Hard	The graph of a polynomial p touches the x-axis at x = 2 without crossing it. Which could be a factor of p(x)? 	(x − 2)²	Touching without crossing means **even multiplicity**, so (x − 2)². Two common wrong answers use odd multiplicities, and both would cross.
P5	Hard	What is the greatest possible number of turning points on the graph of a polynomial of degree 5? 	4	A polynomial of degree n has at most n − 1 turning points. 5 − 1 = **4**.

#	Level	Question	Answer	Solution
P6	Hard	A quadratic function has zeros at x = 1 and x = 4 and passes through the point (0, 8). What is the value of the leading coefficient? 	2	p(x) = k(x − 1)(x − 4), so p(0) = k(−1)(−4) = 4k. 4k = 8 gives k = **2**.
P7	Very hard	When the polynomial p is divided by x − 1, the quotient is x + 4 and the remainder is 3. What is p(x)? reassemble instead of dividing again p(x) = (x − 1)(x + 4) + 3 = x² + 3x − 1	x² + 3x − 1	p(x) = (x − 1)(x + 4) + 3 = x² + 3x − 4 + 3. = **x² + 3x − 1**. A common wrong answer forgets to add the remainder.
P8	Very hard	The polynomial p(x) = x³ + kx² − 5x + 6 has a factor of (x − 2). What is the value of k? (x − 2) is a factor → p(2) = 0 → 8 + 4k − 10 + 6 = 0 → k = −1	−1	A factor (x − 2) means p(2) = 0. 8 + 4k − 10 + 6 = 0, so 4k + 4 = 0 and k = **−1**. Check: 8 − 4 − 10 + 6 = 0.
P9	Very hard	The graph of y = p(x) crosses the x-axis at x = −3, touches it at x = 1, and p has degree 3. Which could be p(x)? 	(x + 3)(x − 1)²	Crossing at −3 needs (x + 3) to odd power; touching at 1 needs (x − 1) to even power; the degrees must total 3. = **(x + 3)(x − 1)²**. A common wrong answer swaps which zero is repeated; C has both signs wrong.
P10	Very hard	For p(x) = x³ + 2x − 5, what is the remainder when p(x) is divided by x + 2? divide by (x + 2) → a = −2 p(−2) → −8 − 4 − 5 = −17 p(+2) → 8 + 4 − 5 = 7 — the trap	−17	x + 2 = x − (−2), so substitute −2. −8 − 4 − 5 = **−17**. A common wrong answer substitutes +2 instead, which is the sign trap this question is built on.

#	Level	Question	Answer	Solution
P11	Very hard	A polynomial p of degree 4 has zeros at x = −1, 0, 2 and 2. State what is true here. 	**The graph crosses at x = −1 and x = 0 and touches at x = 2.**	x = 2 appears twice, so it has **even multiplicity** and the curve turns back there; the other two zeros are simple and the curve passes through. A common wrong answer confuses the degree with the number of distinct intercepts.
P12	Very hard	The polynomial p(x) = x^3 − 7x + 6 has x = 1 as a zero. What is the sum of its other two zeros? 	**−1**	Dividing by (x − 1) gives x^2 + x − 6, whose roots sum to −b/a = **−1** (they are 2 and −3). Alternatively all three zeros sum to 0 (no x^2 term), so the other two sum to 0 − 1 = −1.

The one habit to build. Watch the sign when you convert. A factor of (x + 2) means a zero at x = −2, and dividing by (x + 2) means substituting −2. Your class lost more marks to that flip than to any calculation.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

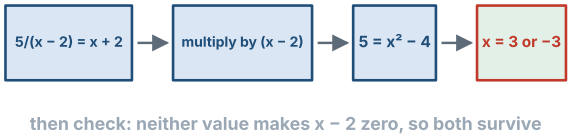
SAT Math 2.4 — Rational & Radical Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving rational equations

multiply both sides by the denominator | then check nothing was made zero

Clearing the fraction turns the problem into something familiar. The one extra step is checking: any value that makes an original denominator zero must be thrown out, however correctly it was derived.



then check: neither value makes x − 2 zero, so both survive



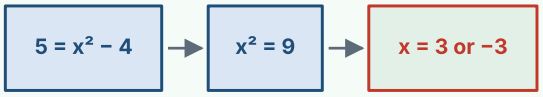
Clearing the denominator is a legal move, but it quietly forgets that x = 2 was never allowed. The number line keeps that exclusion in view while you solve.

Example 1. **Solve 5/(x − 2) = x + 2.**

Solution.

Multiply by (x − 2): 5 = (x + 2)(x − 2) = x² − 4.

x² = 9, so x = **3 or −3**. Neither makes x − 2 zero.



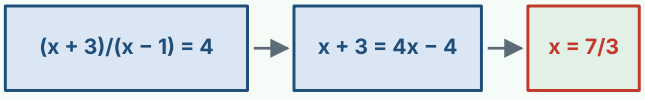
both check out

Example 2. **Solve (x + 3)/(x − 1) = 4.**

Solution.

x + 3 = 4(x − 1) = 4x − 4, so 7 = 3x.

x = **7/3**.



Example 3. **Solve 1/x + 1/(2x) = 3.**

Solution.

Multiply by 2x: 2 + 1 = 6x.

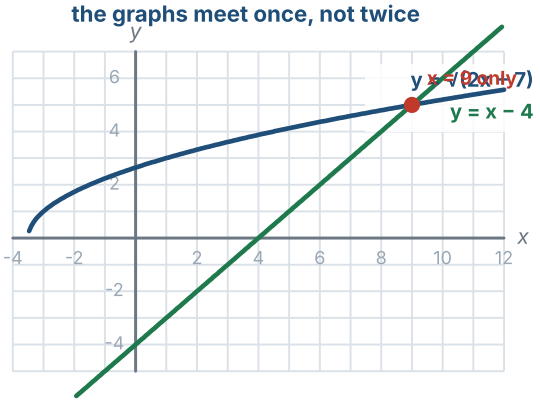
x = **1/2**. Check x ≠ 0, which it is not.



2 Extraneous solutions

squaring or clearing a denominator can invent solutions — always substitute back

This is the highest-value habit in the topic. Both squaring and multiplying out can produce values that satisfy the new equation but not the original. The check is part of the method, not an optional tidy-up.



substitute each candidate back into the original

x = 1	√9 = 3 against 1 − 4 = −3	× reject
x = 9	√25 = 5 against 9 − 4 = 5	✓ keep

Squaring turned one equation into another with an extra root. The graphs cross only at x = 9; x = 1 solves the squared equation and nothing else.

Example 1. **Solve √(2x + 7) = x − 4.**

Solution.

Squaring: 2x + 7 = x^2 − 8x + 16, so x^2 − 10x + 9 = 0 and x = 1 or 9.

Check x = 1: √9 = 3 but 1 − 4 = −3, so it fails. The solution set is **{9}**.

substitute each candidate back into the original

x = 1	√9 = 3 against 1 − 4 = −3	× reject
x = 9	√25 = 5 against 9 − 4 = 5	✓ keep

Example 2. **Why did x = 1 appear at all?**

Solution.

Squaring turns 3 = −3 into 9 = 9. The step is not reversible, so it can create solutions that were never there.

3 = −3	is false
square both sides	9 = 9 is true
so squaring	can invent a solution

Example 3. **Solve √(x + 5) = x − 1.**

Solution.

$x + 5 = x^2 - 2x + 1$ gives $x^2 - 3x - 4 = 0$, so x = 4 or −1.

x = −1 gives √4 = 2 but −2, so it fails. Answer **x = 4**.

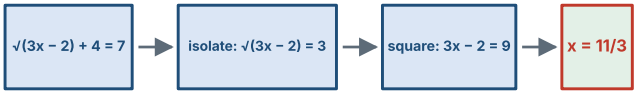
substitute each candidate back into the original

x = −1	√4 = 2 against −2	× reject
x = 4	√9 = 3 against 3	✓ keep

3 Radical equations

isolate the radical first, then square once

Get the root alone on one side before squaring, or you will square a sum and create cross terms. If two radicals appear, isolate one, square, then isolate and square again.



squaring before isolating leaves a radical behind

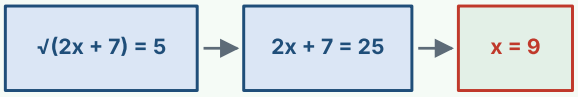
Four steps in order. Squaring the left side as it stands would give 3x − 2 + 8√(3x − 2) + 16 — still a radical, and worse than where you started.

Example 1. **Solve √(2x + 7) = 5.**

Solution.

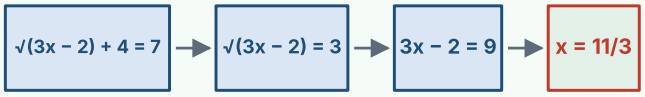
Square both sides: 2x + 7 = 25.

x = **9**. Check: √25 = 5.



Example 2. **Solve √(3x − 2) + 4 = 7.**

Solution.
Isolate first: √(3x − 2) = 3, so 3x − 2 = 9.
x = **11/3**.



Example 3. **Why not square √(3x − 2) + 4 = 7 immediately?**

Solution.
The left side would become 3x − 2 + 8√(3x − 2) + 16, which still has a radical in it — **worse than where you started**.

$(\sqrt{a} + 4)^2$	$= a + 8\sqrt{a} + 16$
the radical survives	so isolate it first

4 Domain restrictions

a denominator cannot be zero | an even root cannot take a negative

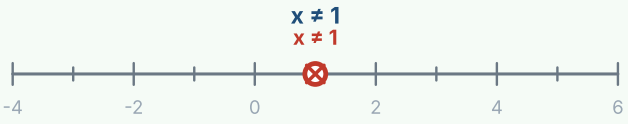
Questions often ask which values are excluded rather than asking you to solve anything. Set the denominator to zero and solve; for a square root, set the inside greater than or equal to zero.



A hollow crossed circle marks a value x may not take; a filled dot with a ray marks where a square root starts being defined. At x = 3 the root is √0 = 0, which is allowed.

Example 1. **For what value of x is (x + 3)/(x − 1) undefined?**

Solution.
x − 1 = 0.
x = 1.



Example 2. **What is the domain of $\sqrt{(2x - 6)}$?**

Solution.

$2x - 6 \geq 0$, so $2x \geq 6$.

$x \geq 3$.



Example 3. **For what values of x is $1/(x^2 - 9)$ undefined?**

Solution.

$x^2 - 9 = 0$ gives $x = \pm 3$.

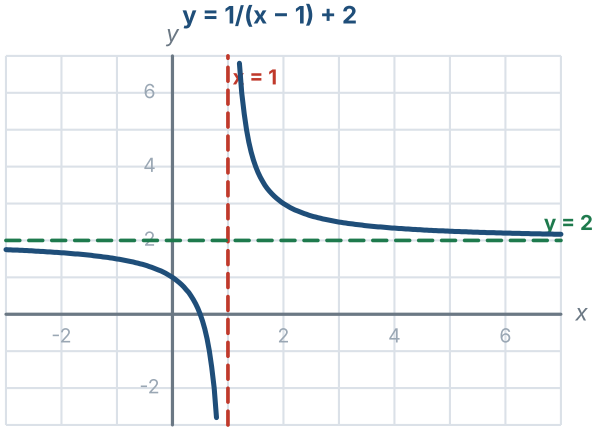
Undefined at **$x = 3$ and $x = -3$**.



5 Asymptotes of a rational function

vertical asymptote where the denominator is zero | horizontal asymptote from comparing degrees

A vertical asymptote is a value x cannot take. For the horizontal one: if the top's degree is smaller the asymptote is y = 0; if the degrees match it is the ratio of leading coefficients.

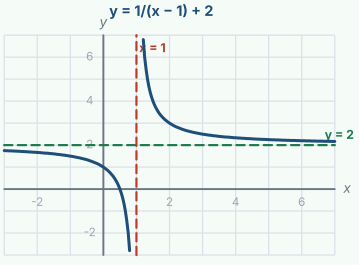


The curve runs away to infinity at x = 1, where the denominator vanishes, and flattens towards y = 2, the constant that was added on. Neither line is ever reached.

Example 1. **Find the vertical asymptote of $y = 1/(x - 1) + 2$.**

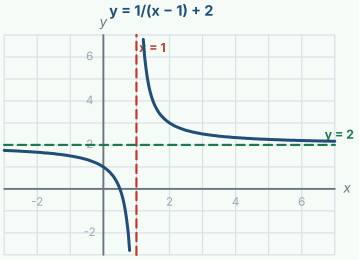
Solution.

The denominator vanishes at **$x = 1$**.



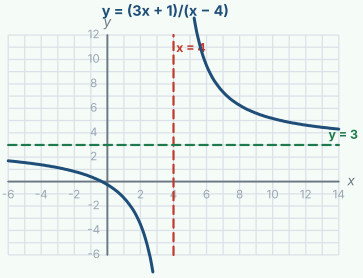
Example 2. **Find the horizontal asymptote of that same function.**

Solution.
As x grows, 1/(x − 1) tends to 0, so y tends to **2**.



Example 3. **Find the horizontal asymptote of y = (3x + 1)/(x − 4).**

Solution.
Degrees match, so take the ratio of leading coefficients.
y = 3.



6 Simplifying rational expressions

factor top and bottom, then cancel common factors — never cancel across a + sign

You may cancel a factor, never a term. $(x^2 + 3x)/x$ cancels to x + 3 because x is a factor of both terms on top; (x + 3)/x cancels to nothing at all.

cancel a factor, never a term

$$\frac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}} = x - 3$$

valid for $x \neq -3$

The bracket (x + 3) multiplies everything above and below it, so it cancels. A loose x added to something else does not multiply the whole line and cannot.

Example 1. **Simplify $(x^2 - 9)/(x + 3)$.**

Solution.
Factor the top: (x − 3)(x + 3)/(x + 3).
= **x − 3**, for x ≠ −3.

cancel a factor, never a term

$$\frac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}} = x - 3$$

$x \neq -3$

Example 2. **Simplify $(x^2 + 5x + 6)/(x^2 - 4)$.**

Solution.

$= (x + 2)(x + 3) / [(x - 2)(x + 2)].$

$=$ **$(x + 3)/(x - 2)$**, for $x \neq -2$.

cancel a factor, never a term

$$\frac{\cancel{(x+2)}\,(x+3)}{\cancel{(x+2)}} = (x+3)/(x-2), \quad x \neq -2$$

Example 3. **Can you cancel the x in (x + 4)/x?**

Solution.

No. On top x is a term, not a factor. The expression is already simplest.

$(x^2 + 3x)/x$	$= x + 3$ ✓ x is a factor
$(x + 4)/x$	cannot cancel — x is a term

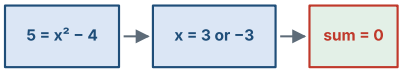
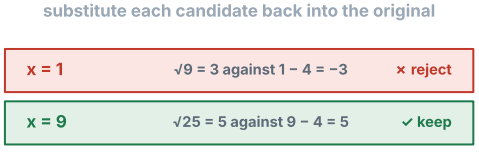
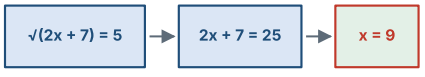
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

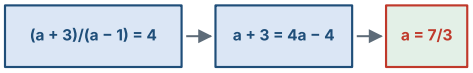
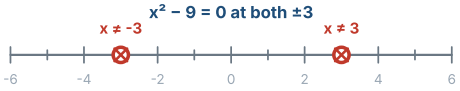
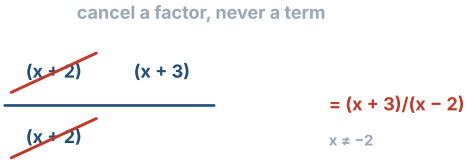
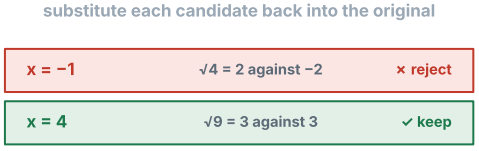
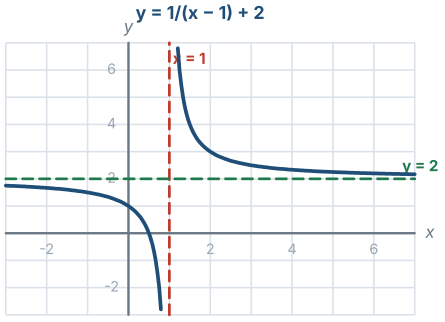
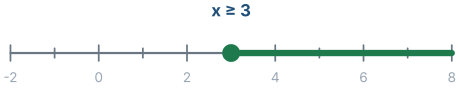
The SAT gives you Desmos on every maths question. Here it is **an extraneous-root filter**. Graph both sides: Desmos shows only the real intersections, so a root it does not show is one your algebra invented.

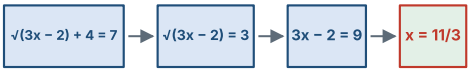
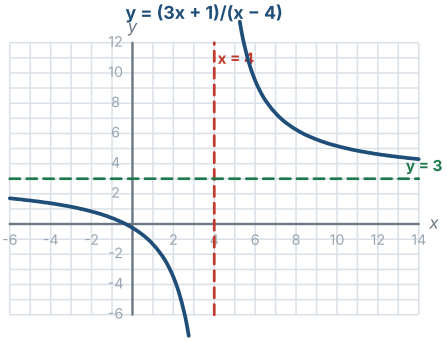
#	Question	Type into Desmos	Then	Answer
D1	Solve 5/(x − 2) = x + 2. What is the sum of the solutions?	`5/(x-2)=x+2`	Graph both sides; the roots are 3 and −3.	0
D2	Solve √(2x + 7) = x − 4.	`sqrt(2x+7)=x-4`	Desmos shows the genuine intersection, not the extraneous root.	9
D3	Solve √(2x + 7) = 5.	`sqrt(2x+7)=5`	Square both sides mentally to check.	9
D4	Solve (x + 3)/(x − 1) = 4.	`(x+3)/(x-1)=4`	Desmos gives the exact fraction.	7/3
D5	For what value of x is (x + 3)/(x − 1) undefined?	`x-1=0`	Graph it and look for the vertical asymptote.	1
D6	What is the horizontal asymptote of y = (3x + 1)/(x − 4)?	`y=(3x+1)/(x-4)`	Zoom out until the curve flattens.	3
D7	Solve 1/x + 1/(2x) = 3.	`1/x+1/(2x)=3`	Clear the denominators first if you prefer.	0.5
D8	Solve √(3x − 2) + 4 = 7.	`sqrt(3x-2)+4=7`	Isolate the radical first.	11/3
D9	Evaluate (x² − 9)/(x + 3) at x = 10.	`(10^2-9)/(10+3)`	Compare with x − 3 at x = 10.	7
D10	Solve √(x + 5) = x − 1.	`sqrt(x+5)=x-1`	Only one intersection appears; −1 is extraneous.	4

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Every radical equation on this sheet needs a check, and every rational one needs you to ask which values the denominator forbids.

#	Level	Question	Answer	Solution
P1	Hard	What is the sum of all solutions to the equation 5/(x − 2) = x + 2? 	0	Multiplying by (x − 2) gives 5 = x² − 4, so x² = 9 and x = ±3. 3 + (−3) = **0**. A common wrong answer is the positive root alone — the word *sum* means there is more than one.
P2	Hard	What is the solution set of the equation √(2x + 7) = x − 4? 	{9}	Squaring gives x² − 10x + 9 = 0, so x = 1 or 9. But x = 1 gives √9 = 3 against 1 − 4 = −3, which is false. Only **x = 9** survives. A common wrong answer is what you get by never checking — six of your class chose it on Test 5.
P3	Hard	If √(2x + 7) = 5, what is the value of x? 	9	Square both sides: 2x + 7 = 25, so 2x = 18 and x = **9**.

#	Level	Question	Answer	Solution
P4	Hard	The function f is defined by f(x) = (x + 3)/(x − 1). If f(a) = 4, what is the value of a? 	7/3	a + 3 = 4(a − 1) = 4a − 4, so 7 = 3a. a = **7/3**.
P5	Hard	For what value of x is the expression (x + 5)/(x² − 9) undefined, given x > 0? 	3	The expression is undefined where the denominator is zero: x² = 9 gives x = ±3. With x > 0, x = **3**.
P6	Hard	Write an equivalent expression for (x² + 5x + 6)/(x² − 4) for x ≠ ±2. 	(x + 3)/(x − 2)	Factor both: (x + 2)(x + 3) over (x − 2)(x + 2), and cancel (x + 2). = **(x + 3)/(x − 2)**.
P7	Very hard	What is the solution to √(x + 5) = x − 1? 	4	Squaring gives x² − 3x − 4 = 0, so x = 4 or −1. Testing x = −1: √4 = 2 but −1 − 1 = −2, so it fails. Only **x = 4** works. A common wrong answer keeps the extraneous root.
P8	Very hard	The graph of y = 1/(x − 1) + 2 has a vertical asymptote and a horizontal asymptote. What are they? 	x = 1 and y = 2	The denominator vanishes at x = 1; as x grows, 1/(x − 1) → 0, leaving y → 2. **x = 1 and y = 2.** A common wrong answer gives the asymptotes of y = 1/(x − 1) before the shift.
P9	Very hard	State the complete domain of f(x) = √(2x − 6). 	x ≥ 3	An even root needs a non-negative inside: 2x − 6 ≥ 0. **x ≥ 3.** The equals sign belongs: at x = 3 the value is √0 = 0, which is defined.
P10	Very hard	If 1/x + 1/(2x) = 3 and x ≠ 0, what is the value of x? 	1/2	Multiply through by 2x: 2 + 1 = 6x, so 6x = 3. x = **1/2**.

#	Level	Question	Answer	Solution
P11	Very hard	A student solves √(3x − 2) + 4 = 7 by squaring both sides immediately, obtaining 3x − 2 + 16 = 49. What is wrong, and what is the correct value of x? 	**The radical must be isolated first; x = 11/3.**	Squaring a sum is not squaring each piece: $(\sqrt{a} + 4)^2 = a + 8\sqrt{a} + 16$, so the radical survives. Isolating first gives √(3x − 2) = 3, then 3x − 2 = 9 and x = **11/3**.
P12	Very hard	What is the horizontal asymptote of the graph of y = (3x + 1)/(x − 4)? 	**y = 3**	The numerator and denominator have the same degree, so the asymptote is the ratio of leading coefficients. 3/1 = **y = 3**. A common wrong answer is the rule for when the top's degree is smaller.

The one habit to build. Substitute your answer back into the **original** equation. Squaring and clearing denominators are not reversible steps, and six of your class lost M2 Q25 on Test 5 by skipping the check.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

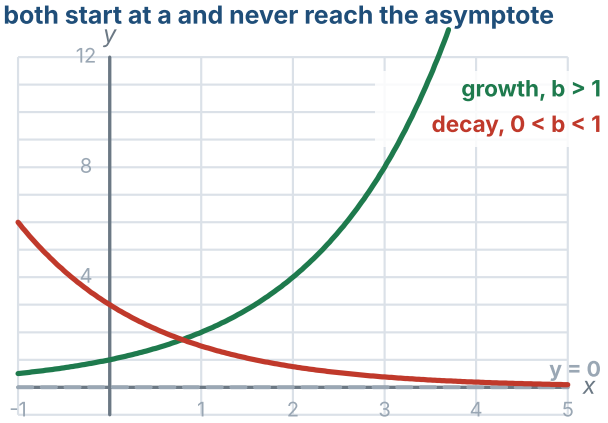
SAT Math 2.5 — Exponential Functions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 The exponential model

$y = a \cdot b^x$ | a is the starting value, b is the growth factor per step

Read a at x = 0 and b as the multiplier for one step. Exponential means repeated multiplication, which is what separates it from a linear model's repeated addition.



Both curves cut the y-axis at their starting value a. Above b = 1 the curve climbs ever faster; below it the curve falls towards y = 0 without ever arriving.

Example 1. **A shop opens with 150 members and the number doubles each year. Write the model.**

Solution.
a = 150 and b = 2.
$M(t) = 150(2)^t$.



Example 2. **How many members after 3 years?**

Solution.
$150(2)^3 = 150 \times 8 =$ **1200**. Listing 150 → 300 → 600 → 1200 stops you counting one doubling short.

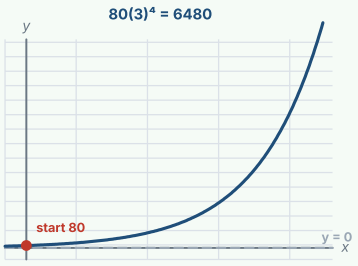
x = 0	x = 1	x = 2	x = 3
150	300	600	1200

differences	+150	+300	+600
ratios	×2	×2	×2

equal ratios → exponential

Example 3. **A culture starts at 80 and triples every hour. How many after 4 hours?**

Solution.
$80(3)^4 = 80 \times 81 = \mathbf{6480}$.



2 Growth, decay and the percentage

rise of r% → b = 1 + r | fall of r% → b = 1 − r

Convert between the base and the percentage in both directions. A base of 0.88 is a fall of 12%, not a fall of 88% and not 'keeps 12%'. A base of 1.07 is a rise of 7%.

the percentage is the distance from 1, never the base

rise of 15% →	b = 1 + 0.15 = 1.15
rise of 7% →	b = 1 + 0.07 = 1.07
fall of 8% →	b = 1 − 0.08 = 0.92
fall of 12% →	b = 1 − 0.12 = 0.88

The percentage is the distance from 1, never the base itself. 0.88 keeps 88% and therefore loses 12%.

Example 1. **A population falls 8% a year. What is the growth factor?**

Solution.
$1 - 0.08 = \mathbf{0.92}$.

the percentage is the distance from 1, never the base

fall of 8% → b = 1 − 0.08 = 0.92

Example 2. **V(t) = 4200(0.88)ᵗ. What does 0.88 mean?**

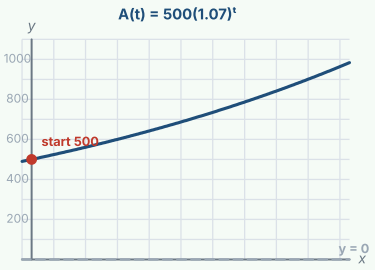
Solution.
The value **falls by 12% each year**, because 1 − 0.88 = 0.12. It keeps 88%, not 12%.

the percentage is the distance from 1, never the base

fall of 12% → b = 1 − 0.12 = 0.88

Example 3. **A deposit grows 7% a year. Write the model for $500.**

Solution.
$A(t) = 500(1.07)^t$.



3 Changing the time unit

$b^t = (b^{1/n})^{nt}$ — take the n-th root of the base, multiply the exponent by n

This is the subtopic your class scored lowest on. Changing from years to months never changes the base alone and never the exponent alone — it does both, in opposite directions. The tell in the question is a word like *monthly* or *quarterly*.

one year = 12 months

$V = a\cdot(0.88)^t$ → $a\cdot((0.88)^{(1/12)})^{12t}$

per year | per month

base → its 12-th root; exponent → ×12, both at once

$(0.88)^{(1/12)} \approx 0.9894$

Twelve monthly steps must come to one yearly step, so the monthly base is the twelfth root and the exponent is multiplied by twelve. Do one without the other and the model no longer matches.

Example 1. **$V(t) = 84{,}000(0.88)^t$ with t in years. Rewrite it with a monthly factor.**

Solution.
One year is 12 months, so t years is 12t months.
$84{,}000((0.88)^{1/12})^{12t}$, and $(0.88)^{1/12} \approx 0.9894$.

one year = 12 months

$V = a\cdot(0.88)^t$ → $a\cdot((0.88)^{(1/12)})^{12t}$

per year | per month

base → its 12-th root; exponent → ×12, both at once

$(0.88)^{(1/12)} \approx 0.9894$

Example 2. **Why is $84{,}000(0.88)^{12t}$ wrong?**

Solution.
That multiplies by 0.88 twelve times a *year* — a fall of about 78% annually instead of 12%.

$(0.88)^{12}$	≈ 0.216 — a 78% fall
$(0.88^{1/12})^{12}$	$= 0.88$ ✓ the same 12% fall

Example 3. **P = 200(1.6)^t with t in years. Write it with a quarterly factor.**

Solution.

Four quarters in a year.

200((1.6)$^{1/4}$)^{4t}.

one year = 4 quarters

$V = a\cdot(1.6)^t$	→	$a\cdot((1.6)^{(1/4)})^{4t}$
per year		per quarter

base → its 4-th root; exponent → ×4, both at once

(1.6)^(1/4) ≈ 1.1247

4 Exponential against linear

constant difference → linear | constant ratio → exponential

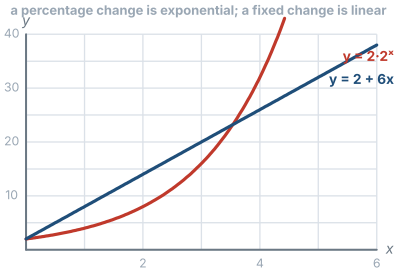
Test a table both ways. Subtract consecutive values: equal differences mean linear. Divide them: equal ratios mean exponential. Exponential growth eventually overtakes any linear growth, however slow it starts.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

	x = 0	x = 1	x = 2	x = 3
	5	10	20	40
differences		+5	+10	+20
ratios		×2	×2	×2

equal ratios → exponential



Same first value, two different engines. Adding a fixed amount draws a straight line; multiplying by a fixed amount bends upward and always wins in the end.

Example 1. **y = 5, 8, 11, 14. Which model?**

Solution.

Differences 3, 3, 3.

Linear, y = 3x + 2.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

Example 2. **y = 5, 10, 20, 40. Which model?**

Solution.

Ratios 2, 2, 2.

Exponential, $y = 5(2)^x$ counting from x = 0.

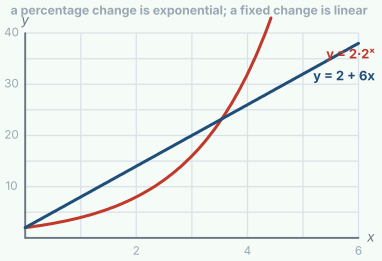
	x = 0	x = 1	x = 2	x = 3
	5	10	20	40
differences		+5	+10	+20
ratios		×2	×2	×2

equal ratios → exponential

Example 3. **A saver adds $50 a month; an investment grows 3% a month. Which is exponential?**

Solution.

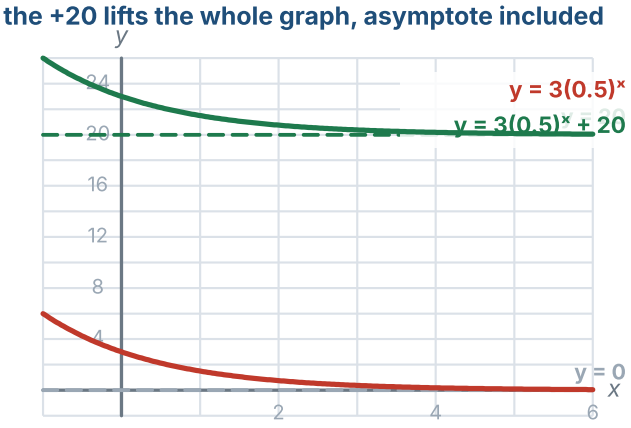
The **investment** — a percentage of a changing amount. Adding a fixed $50 is linear.



5 The horizontal asymptote

$y = a \cdot b^x$ approaches $y = 0$ | adding c shifts the asymptote to $y = c$

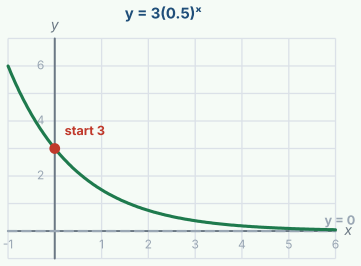
A decaying exponential gets arbitrarily close to its asymptote but never reaches it, which is why a model like 'cools towards room temperature' is written with a constant added on.



Both curves have the same shape; the green one sits 20 higher, so it levels off at 20 instead of 0. Whatever is added on is the value the model approaches.

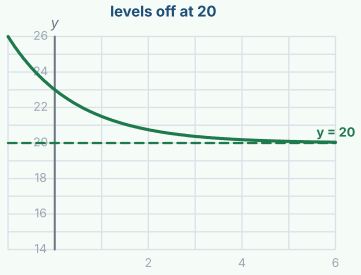
Example 1. What is the horizontal asymptote of $y = 3(0.5)^x$?

Solution.
$y = 0$. The curve falls towards the x-axis without touching it.



Example 2. What is it for $y = 3(0.5)^x + 20$?

Solution.
$y = 20$ — the whole graph is lifted 20.



Example 3. A drink cools according to $T(t) = 65(0.9)^t + 22$. What does 22 represent?

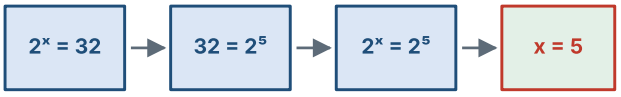
Solution.
The **room temperature**, 22 degrees — the value the drink approaches but never quite reaches.



6 Solving exponential equations

same base → equate the exponents | otherwise graph it in Desmos

Rewrite both sides as powers of the same number whenever you can. When the bases cannot be matched, the SAT expects you to read the intersection off a graph rather than use logarithms.

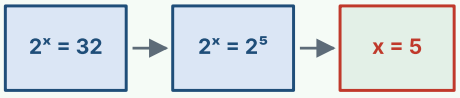


matching bases turns it into an ordinary equation

Once both sides are powers of 2, the exponents must agree. That is the whole method — no logarithms required.

Example 1. **Solve $2^x = 32$.**

Solution.
$32 = 2^5$.
$x = \mathbf{5}$.



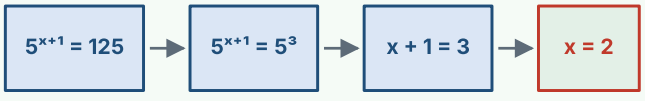
Example 2. **Solve $3^{2x} = 81$.**

Solution.
$81 = 3^4$, so $2x = 4$.
$x = \mathbf{2}$.



Example 3. **Solve $5^{x+1} = 125$.**

Solution.
$125 = 5^3$, so $x + 1 = 3$.
$x = \mathbf{2}$.



Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

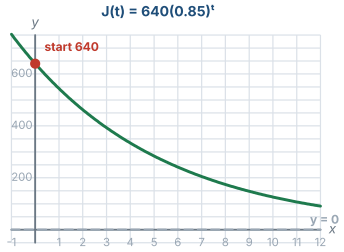
The SAT gives you Desmos on every maths question. Here it is **a growth calculator**. Type the whole model and evaluate it, or graph both sides of an exponential equation and read the crossing.

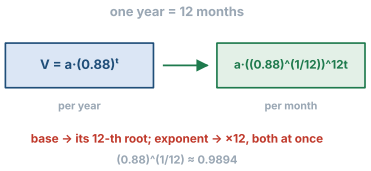
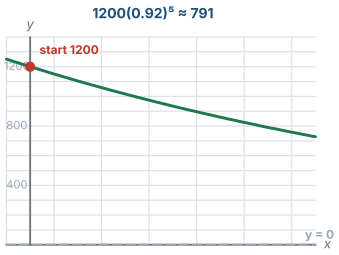
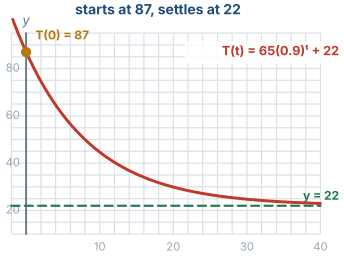
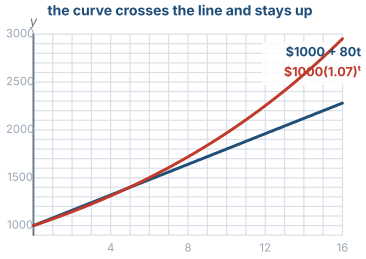
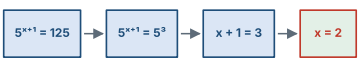
#	Question	Type into Desmos	Then	Answer
D1	A shop opens with 150 members and the number doubles each year. How many after 3 years?	`150*2^3`	Count the doublings carefully.	1200
D2	A town of 1200 falls 8% a year. What is the population after 5 years, to the nearest whole number?	`1200*0.92^5`	A fall of 8% means a factor of 0.92.	791
D3	What is $(0.88)^{1/12}$, to four decimal places?	`0.88^(1/12)`	The monthly factor behind an annual 0.88.	0.9894
D4	$500 grows at 7% a year. What is it worth after 3 years, to the nearest cent?	`500*1.07^3`	Repeated multiplication, not 500 × 1.21.	612.52
D5	Solve $2^x = 32$.	`2^x=32`	Graph both sides and read the intersection.	5
D6	A culture starts at 80 and triples hourly. How many after 4 hours?	`80*3^4`	Powers grow fast; let Desmos do it.	6480
D7	What is the growth factor for a rise of 15% per period?	`1+0.15`	A rise adds to 1.	1.15
D8	250 subscribers grow 20% monthly. How many after 2 months?	`250*1.2^2`	1.2 squared is 1.44.	360
D9	Solve $3^{2x} = 81$.	`3^(2x)=81`	Both sides are powers of 3.	2
D10	What does $65(0.9)^t + 22$ approach as t grows large?	`65*0.9^100+22`	Try a large t to see the asymptote.	22

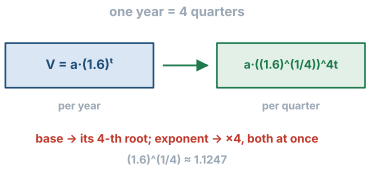
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Read the base carefully every time: it tells you growth or decay, and the percentage is its distance from 1 — not the base itself.

#	Level	Question	Answer	Solution
P1	Hard	When a tea shop opened it had 150 loyalty members, and the number has doubled every year since. How many members does the shop have 3 years after opening? 	1200	$150 \times 2^3 = 150 \times 8 =$ **1200**. A common wrong answer counts only two doublings — four of your class chose it on Test 5.

#	Level	Question	Answer	Solution
P2	Hard	The function $V(t) = 4200(0.88)^t$ gives the value, in talens, of an inventory t years after purchase. What is the best interpretation of 0.88? the percentage is the distance from 1, never the base fall of 12% → b = 1 − 0.12 = 0.88	**The value decreases by 12% each year.**	A base below 1 is decay, and 1 − 0.88 = 0.12. The value **falls 12% a year** and keeps 88%. A common wrong answer reverses which figure is kept.
P3	Hard	A spice stall opens with 640 jars and the number of unsold jars decreases by 15% each week. Which function gives the number of jars after t weeks? 	**$J(t) = 640(0.85)^t$**	A fall of 15% leaves 85%, so the base is 1 − 0.15 = 0.85. **$J(t) = 640(0.85)^t$.** A common wrong answer uses the rate as the base; C is linear.
P4	Hard	A subscription service has 250 subscribers and grows by 20% each month. How many subscribers will it have after 2 months? x = 0: 250 · x = 1: 300 · x = 2: 360 differences +50 +60 ratios ×1.2 ×1.2 equal ratios → exponential	**360**	$250(1.2)^2 = 250 \times 1.44 =$ **360**. Adding 20% of 250 twice would give 350, which is the common error.
P5	Hard	If $3^{2x} = 81$, what is the value of x? $3^{2x} = 81$ → $3^{2x} = 3^4$ → $2x = 4$ → $x = 2$	**2**	$81 = 3^4$, so $2x = 4$ and $x =$ **2**. A common wrong answer stops at $2x = 4$ without halving.
P6	Hard	A table shows y = 5, 10, 20, 40 for x = 0, 1, 2, 3. Which function models the data? x = 0: 5 · x = 1: 10 · x = 2: 20 · x = 3: 40 differences +5 +10 +20 ratios ×2 ×2 ×2 equal ratios → exponential	**$y = 5(2)^x$**	The ratios are all 2, so the model is exponential with a = 5 and b = 2. **$y = 5(2)^x$.** A common wrong answer swaps the starting value and the base; at x = 0 it gives 2, not 5.

#	Level	Question	Answer	Solution
P7	Very hard	The value V(t), in dollars, of a tour bus t years after purchase is $V(t) = 84{,}000(0.88)^t$. Which expression is equivalent and uses the approximate monthly decay factor? 	$84{,}000(0.88^{1/12})^{12t}$	One year is 12 months, so take the twelfth root of the base and multiply the exponent by 12. **$84{,}000(0.88^{1/12})^{12t}$**, where $0.88^{1/12} \approx 0.9894$. Two common wrong answers decay twelve times too fast; C treats 0.12 as a factor rather than a rate.
P8	Very hard	A population of 1,200 decreases by 8% each year. To the nearest whole number, what is the population after 5 years? 	791	$1200(0.92)^5 = 790.89\ldots \approx$ **791**. Subtracting 8% of 1,200 five times would give 720 — that treats the decrease as linear.
P9	Very hard	A hot drink cools according to $T(t) = 65(0.9)^t + 22$, where T is in degrees Celsius and t in minutes. What does the 22 represent? 	The room temperature the drink approaches.	As t grows, $65(0.9)^t \to 0$, so T → **22**, the horizontal asymptote. The drink's initial temperature is T(0) = 87, which is that answer's trap.
P10	Very hard	Two investments start at \$1,000. The first adds \$80 each year; the second grows by 7% each year. State what is true here. 	The first leads at first, but the second eventually overtakes it.	After one year: \$1,080 against \$1,070. But 7% of a growing balance eventually exceeds a fixed \$80. **Exponential growth always overtakes linear growth in the end**, whatever the head start.
P11	Very hard	If $5^{x+1} = 125$, what is the value of x? 	2	$125 = 5^3$, so x + 1 = 3. x = **2**.

#	Level	Question	Answer	Solution
P12	Very hard	A quantity P is modelled by $P = 200(1.6)^t$, where t is in years. Which expression gives P using a quarterly growth factor? 	$200(1.6^{1/4})^{4t}$	A year holds 4 quarters, so take the fourth root of the base and multiply the exponent by 4. **$200(1.6^{1/4})^{4t}$.** A common wrong answer is algebraically equal to the original but uses an *annual* factor raised to a quarter power — it is not the quarterly factor the question asks for.

The one habit to build. Changing the time unit changes **both** the base and the exponent, in opposite directions. Your class scored 23% on that one subtopic, the second worst on the whole paper, so drill it until the shape of the answer is automatic.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

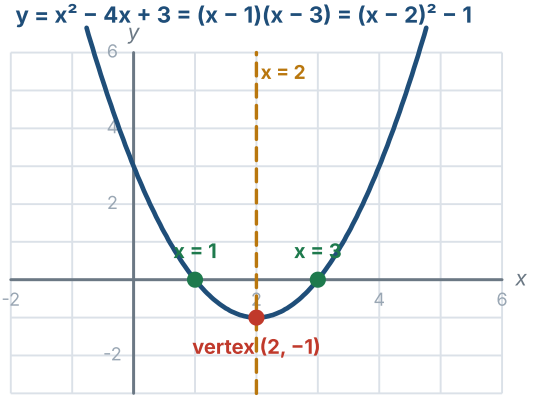
SAT Math 2.6 — Quadratic & Polynomial Graphs

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Three forms, three facts

standard ax^2+bx+c → y-intercept c | factored → x-intercepts | vertex $a(x-h)^2+k$ → vertex (h, k)

Each form hands you one fact for free. The question tells you which fact it wants, and that tells you which form to produce. Converting between them is the work; knowing which to aim for is the skill.



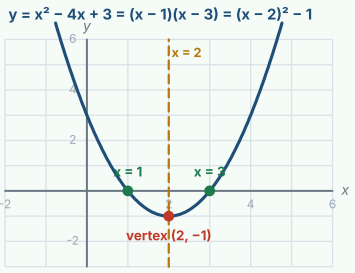
one curve, three ways of writing it

standard: $x^2 - 4x + 3$	y-intercept (0, 3)
factored: $(x - 1)(x - 3)$	x-intercepts 1 and 3
vertex: $(x - 2)^2 - 1$	vertex (2, −1)

All three expressions describe the identical parabola. Which one you write depends only on which number the question is asking for.

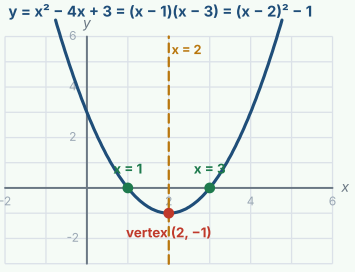
Example 1. **$y = x^2 - 4x + 3$. What is the y-intercept?**

Solution.
Standard form gives it immediately: c = **3**, so (0, 3).



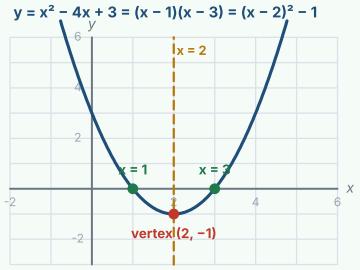
Example 2. **Write the same function in factored form and give the x-intercepts.**

Solution.
$(x - 1)(x - 3)$, so the x-intercepts are **(1, 0) and (3, 0)**.



Example 3. **Write it in vertex form and give the vertex.**

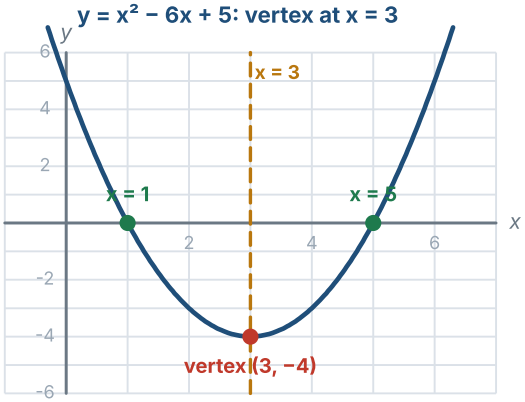
Solution.
$(x - 2)^2 - 1$, so the vertex is **(2, −1)**.



2 Finding the vertex

x = −b/2a | or the midpoint of the two x-intercepts

Both routes give the same x. Substitute it back to get the y. When a question gives you the factored form, the midpoint of the roots is faster than the formula.



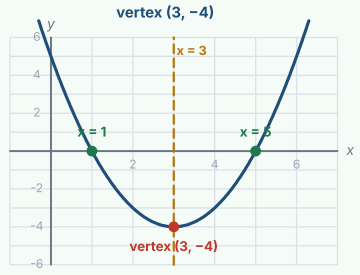
two routes, one answer

from standard form	x = −b/2a = 6/2 = 3
from the roots 1 and 5	midpoint (1 + 5)/2 = 3
then substitute	y = 9 − 18 + 5 = −4

The parabola is symmetric about its vertex, so the turning point sits exactly halfway between the two crossings — which is why both routes agree.

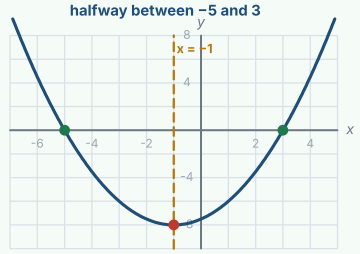
Example 1. **Find the vertex of $y = x^2 - 6x + 5$.**

Solution.
x = 6/2 = 3, and y = 9 − 18 + 5 = −4.
Vertex **(3, −4)**.



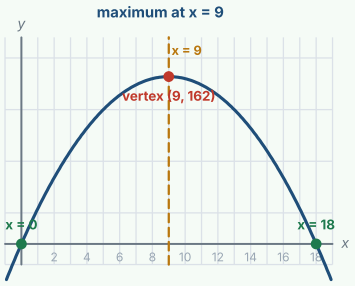
Example 2. **Find the vertex of $y = a(x - 3)(x + 5)$.**

Solution.
Midpoint of 3 and −5 is (3 − 5)/2 = **x = −1**.



Example 3. **$R(x) = -2x^2 + 36x$. Find the x that maximises R.**

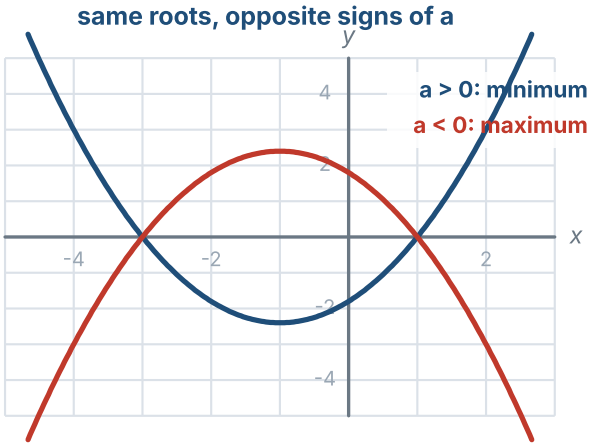
Solution.
$x = -36/(2 \times -2) = \mathbf{9}$.



3 Which way does it open, and what does that mean?

a > 0 → opens up, vertex is a minimum | a < 0 → opens down, vertex is a maximum

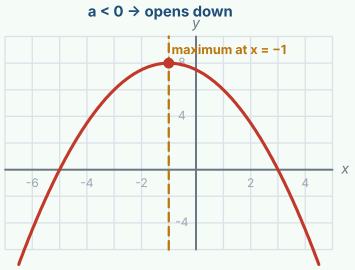
The sign of a settles maximum against minimum on its own. A negative a does not make everything else negative — the y-intercept, for instance, can still be positive.



Flipping the sign of a turns the curve over: the lowest point becomes the highest. The roots stay exactly where they were.

Example 1. **$y = a(x - 3)(x + 5)$ with $a < 0$. Does the vertex give a maximum or a minimum?**

Solution.
$a < 0$ means the parabola opens downward, so the vertex is a **maximum**, at $x = -1$.



Example 2. **For that same function, is the y-intercept positive or negative?**

Solution.

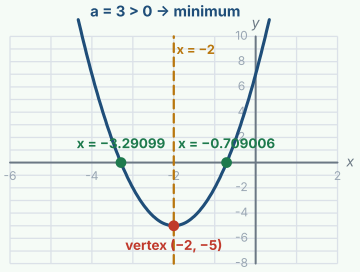
$y(0) = a(-3)(5) = -15a$. With $a < 0$ this is **positive**. A negative a does not make the intercept negative — seven of your class assumed it did.

$y(0) = a(-3)(5)$	$= -15a$
$a < 0$	so $-15a > 0$
the intercept	is positive

Example 3. **$y = 3x^2 + 12x + 7$ has what kind of turning point?**

Solution.

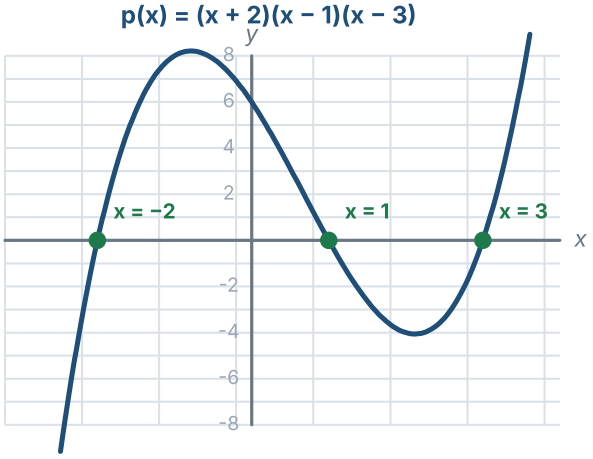
$a = 3 > 0$, so a **minimum**.



4 Reading intercepts off a graph

x-intercepts are where y = 0 | the y-intercept is where x = 0

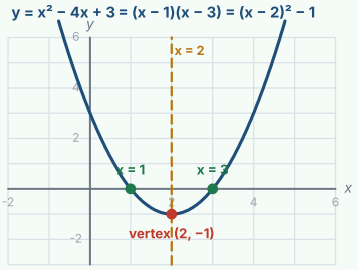
Substituting zero is the whole method. For the y-intercept of a product, multiply the constants: the y-intercept of (x + 2)(x − 1)(x − 3) is 2 × −1 × −3 = 6.



Where the curve meets the x-axis, y is zero; where it meets the y-axis, x is zero. Putting x = 0 into the three brackets gives the height 6.

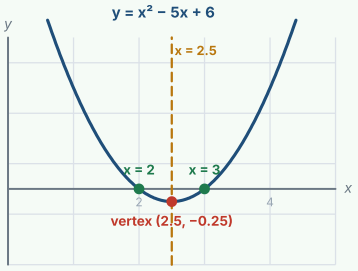
Example 1. **Find the y-intercept of y = (x − 1)(x − 3).**

Solution.
y(0) = (−1)(−3) = **3**.



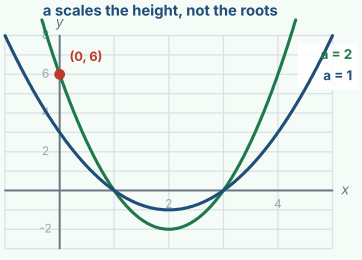
Example 2. **Find the x-intercepts of y = x² − 5x + 6.**

Solution.
(x − 2)(x − 3) = 0.
(2, 0) and (3, 0).



Example 3. **A parabola has x-intercepts 1 and 3 and passes through (0, 6). Find a in y = a(x−1)(x−3).**

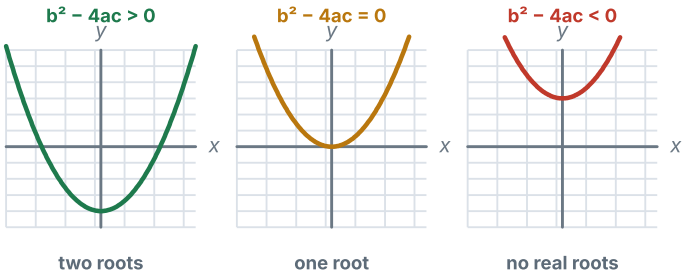
Solution.
6 = a(−1)(−3) = 3a, so a = **2**.



5 How many times does it cross?

the number of x-intercepts is decided by the discriminant $b^2 - 4ac$

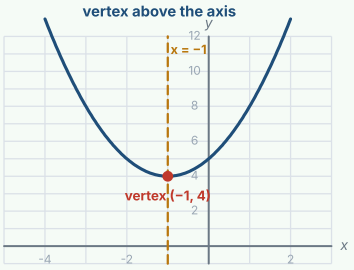
Two intercepts, one (a touch), or none. A question that says ‘touches the x-axis at exactly one point’ or ‘is tangent to the x-axis’ is asking you to set the discriminant to zero.



The same three cases as in 2.2, now read off the picture: cutting twice, grazing once, or missing the axis altogether.

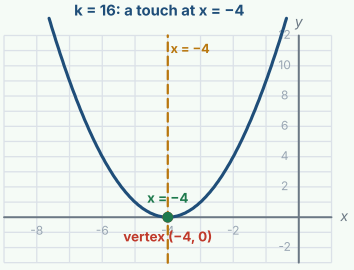
Example 1. **How many x-intercepts does $y = x^2 + 2x + 5$ have?**

Solution.
$4 - 20 = -16 < 0$.
None — the parabola sits entirely above the axis.



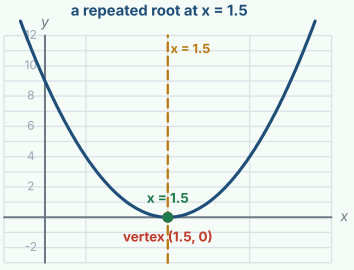
Example 2. **For what k does $y = x^2 + 8x + k$ touch the x-axis exactly once?**

Solution.
$64 - 4k = 0$.
$k = \mathbf{16}$.



Example 3. **How many x-intercepts does $y = 4x^2 - 12x + 9$ have?**

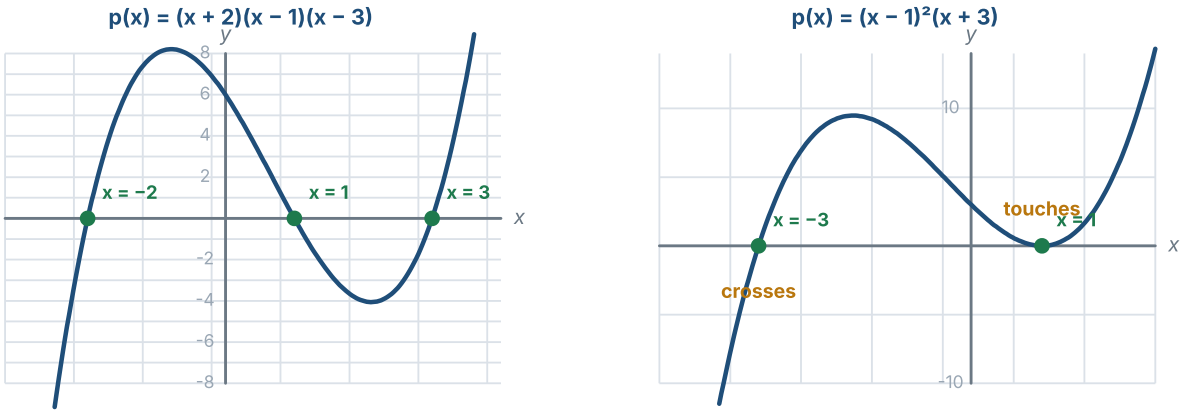
Solution.
$144 - 144 = 0$.
One, at the vertex $x = 1.5$.



6 Cubics and higher

the graph crosses at odd multiplicity and touches at even | degree n has at most n − 1 turning points

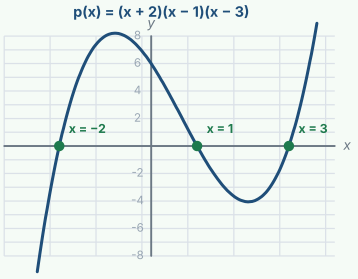
The factored form of a polynomial is a map of its graph. Each factor gives a zero, and the power on that factor says whether the curve passes through or bounces off.



Three single factors on the left: three crossings and two turning points. A squared factor on the right: the curve meets the axis at x = 1 and turns straight back.

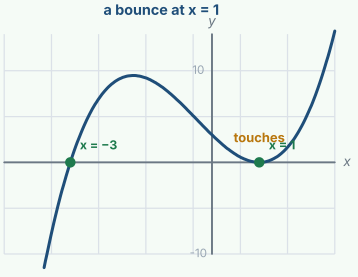
Example 1. **Describe the graph of y = (x + 2)(x − 1)(x − 3).**

Solution.
Three simple zeros, so it **crosses at −2, 1 and 3**, with at most 2 turning points.



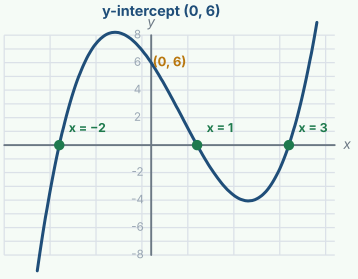
Example 2. **Describe the graph of y = (x − 1)²(x + 3) at x = 1.**

Solution.
Multiplicity 2, so the curve **touches the axis and turns back**.



Example 3. **What is the y-intercept of y = (x + 2)(x − 1)(x − 3)?**

Solution.
(2)(−1)(−3) = **6**.



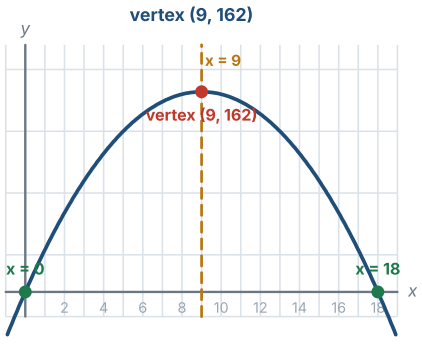
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

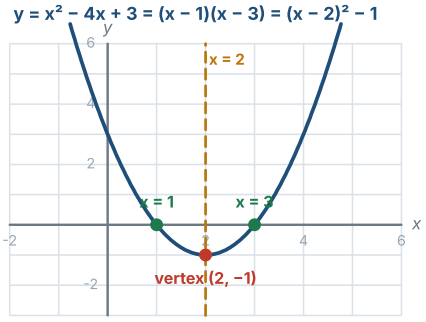
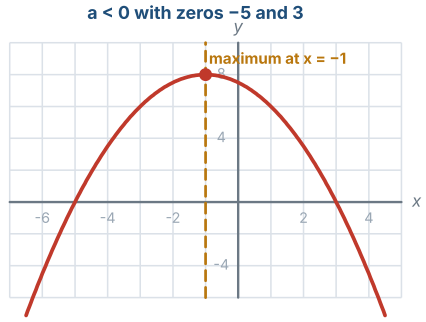
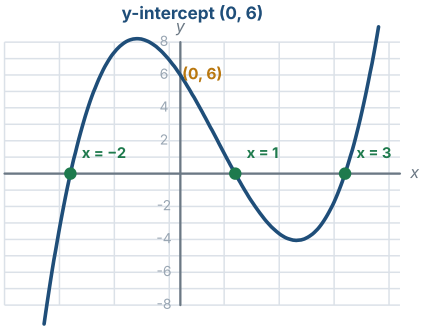
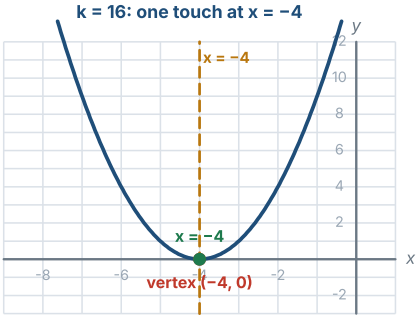
The SAT gives you Desmos on every maths question. This topic is **where Desmos earns its keep**. Graph the function, click the vertex and the intercepts, and the question usually answers itself.

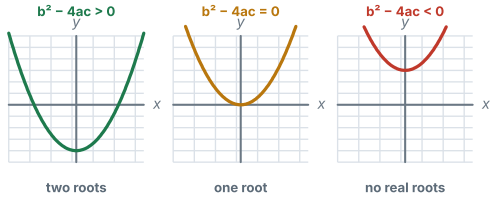
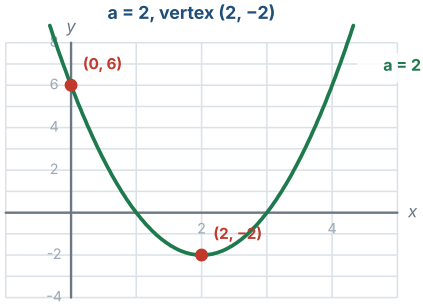
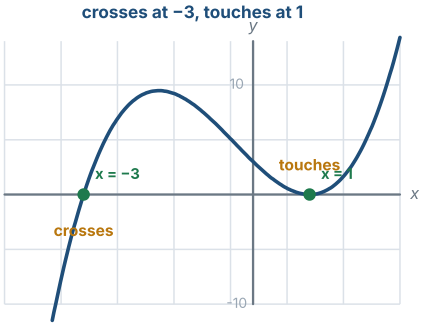
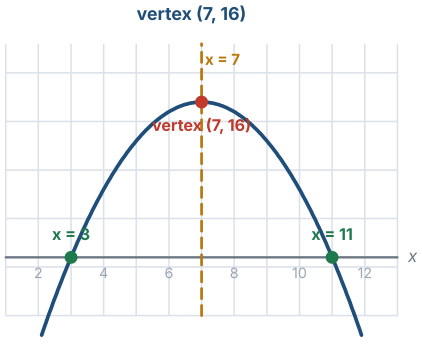
#	Question	Type into Desmos	Then	Answer
D1	Graph $y = x^2 - 4x + 3$. What is the vertex?	`y=x^2-4x+3`	Click the turning point; Desmos labels it.	(2, -1)
D2	For that graph, what are the x-intercepts?	`y=x^2-4x+3`	Click where the curve meets the axis.	1 and 3
D3	What is the minimum value of $y = x^2 - 6x + 5$?	`y=x^2-6x+5`	Read the y of the vertex.	-4
D4	$R(x) = -2x^2 + 36x$. What x gives the maximum?	`y=-2x^2+36x`	The x of the vertex, not the y.	9
D5	How many x-intercepts does $y = x^2 + 2x + 5$ have?	`y=x^2+2x+5`	The curve never reaches the axis.	0
D6	For what k does $x^2 + 8x + k = 0$ have exactly one solution?	`8^2/4`	Set the discriminant to zero.	16
D7	What is the y-intercept of $y = (x + 2)(x - 1)(x - 3)$?	`(0+2)*(0-1)*(0-3)`	Substitute x = 0.	6
D8	Graph $y = (x - 1)^2(x + 3)$. How many turning points does it have?	`y=(x-1)^2(x+3)`	Degree 3 allows at most 2.	2
D9	A parabola has x-intercepts 1 and 3 and passes through (0, 6). What is a in $y = a(x-1)(x-3)$?	`6/((0-1)*(0-3))`	Solve 3a = 6.	2
D10	What is the value of $-2x^2 + 36x$ at x = 9?	`-2*9^2+36*9`	The maximum revenue itself.	162

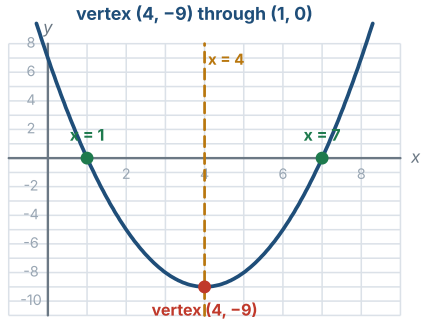
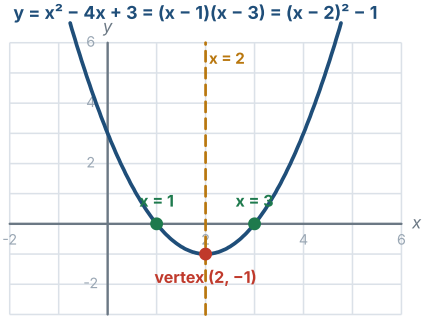
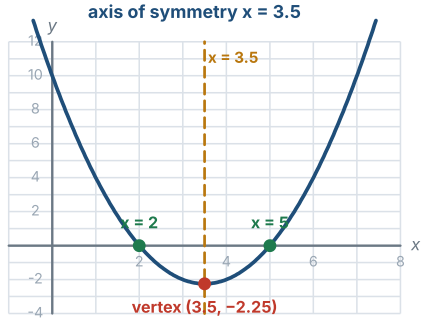
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide first whether the question wants **where** something happens (an x) or **how much** it is (a y). Most of the lost marks in this topic are the right number answering the wrong question.

#	Level	Question	Answer	Solution
P1	Hard	A candle shop's daily revenue, in dollars, is modelled by $R(x) = -2x^2 + 36x$, where x is the price of one candle in dollars. What price gives the maximum daily revenue? 	$9	The vertex sits at $x = -b/2a = -36/(-4) =$ **$9**. A common wrong answer is the maximum revenue R(9) = 162 — the answer to a different question. Five of your class chose C, reading 36 straight off the equation.

#	Level	Question	Answer	Solution
P2	Hard	The graph of $y = x^2 - 4x + 3$ has its vertex at which point? 	(2, −1)	$x = -b/2a = 4/2 = 2$, and $y = 4 - 8 + 3 = -1$. Vertex **(2, −1)**. A common wrong answer drops the sign on −b.
P3	Hard	A candle stall's weekly profit is modelled by $y = a(x - 3)(x + 5)$, where $a < 0$. Which must be true about the graph? 	The graph has a maximum at x = −1.	The zeros are 3 and −5, so the vertex is at their midpoint $x = -1$, and $a < 0$ makes it a **maximum**. The trap seven of your class took: the y-intercept is $a(-3)(5) = -15a$, which is *positive* when $a < 0$.
P4	Hard	What is the y-intercept of the graph of $y = (x + 2)(x - 1)(x - 3)$? 	6	Substitute $x = 0$: $(2)(-1)(-3) =$ **6**.
P5	Hard	For what value of k does the graph of $y = x^2 + 8x + k$ touch the x-axis at exactly one point? 	16	Touching once means the discriminant is zero: $64 - 4k = 0$. $k =$ **16**.

#	Level	Question	Answer	Solution
P6	Hard	How many x-intercepts does the graph of $y = x^2 + 2x + 5$ have? 	Zero	$b^2 - 4ac = 4 - 20 = -16 < 0$. The parabola opens upward with vertex above the axis, so it has **no** x-intercepts.
P7	Very hard	A parabola has x-intercepts at (1, 0) and (3, 0) and passes through (0, 6). What is the y-value of its vertex? 	−2	$y = a(x - 1)(x - 3)$, and $6 = a(-1)(-3) = 3a$ gives $a = 2$. The vertex sits at $x = 2$. $y = 2(1)(-1) = \mathbf{-2}$. A common wrong answer is the vertex y when $a = 1$.
P8	Very hard	The graph of a polynomial p crosses the x-axis at $x = -3$ and touches it at $x = 1$. Which could be p(x)? 	$(x + 3)(x - 1)^2$	Crossing needs odd multiplicity, touching needs even. $= \mathbf{(x + 3)(x - 1)^2}$. A common wrong answer swaps which zero is repeated; D has both signs wrong.
P9	Very hard	The function f is defined by $f(x) = -x^2 + 14x - 33$. What is the maximum value of f? 	16	The vertex is at $x = -14/(-2) = 7$. $f(7) = -49 + 98 - 33 = \mathbf{16}$. Answering 7 gives where the maximum occurs, not the maximum itself.

#	Level	Question	Answer	Solution
P10	Very hard	The graph of $y = a(x - h)^2 + k$ has its vertex at $(4, -9)$ and passes through $(1, 0)$. What is the value of a? 	1	$y = a(x - 4)^2 - 9$, and at $(1, 0)$: $0 = a(9) - 9$, so $9a = 9$. $a =$ **1**.
P11	Very hard	Two functions are given by $f(x) = x^2 - 4x + 3$ and $g(x) = (x - 2)^2 - 1$. State what is true here. 	f and g are the same function.	Expanding g: $x^2 - 4x + 4 - 1 = x^2 - 4x + 3 = f(x)$. They are **the same function written in two forms** — standard and vertex.
P12	Very hard	The graph of $y = x^2 + bx + 10$ has its axis of symmetry at $x = 3.5$. What is the value of b? 	−7	The axis of symmetry is $x = -b/2a = -b/2$. $-b/2 = 3.5$ gives $b =$ **−7**. A common wrong answer drops the minus sign.

The one habit to build. The x of the vertex is where the maximum occurs; the y is the maximum itself. On Test 5 five students answered 36 and one answered 162 when the question wanted 9 — all three numbers were in the problem, and only one answered what was asked.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

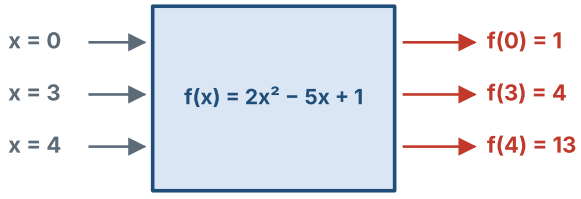
SAT Math 2.7 — Function Notation & Transformations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Function notation

f(3) means put 3 in | f(x) = 3 means the output is 3 — solve for x

The brackets say substitute; they are not multiplication. f(a + b) is not f(a) + f(b), and the difference between f(3) = ? and f(x) = 3 is the difference between evaluating and solving.



put the number in, read the number out

two different jobs, one notation

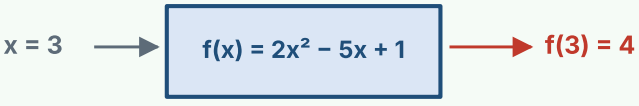
f(3) = ?	put 3 in, read the output
f(x) = 3	the output is 3, solve for x

The machine takes an input and returns an output. ‘Find f(3)’ runs it forwards; ‘solve f(x) = 3’ runs it backwards and can have two answers.

Example 1. **f(x) = 2x² − 5x + 1. Find f(3).**

Solution.

2(9) − 15 + 1 = 18 − 15 + 1 = **4**.



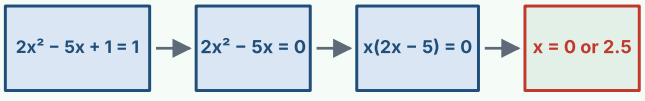
put the number in, read the number out

Example 2. **For that f, solve f(x) = 1.**

Solution.

2x² − 5x + 1 = 1 gives 2x² − 5x = 0, so x(2x − 5) = 0.

x = **0 or 2.5**.

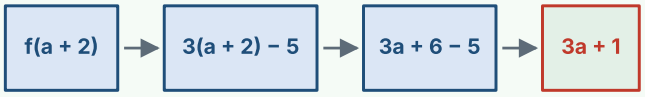


Example 3. **f(x) = 3x − 5. Find f(a + 2).**

Solution.

Substitute the whole expression: 3(a + 2) − 5.

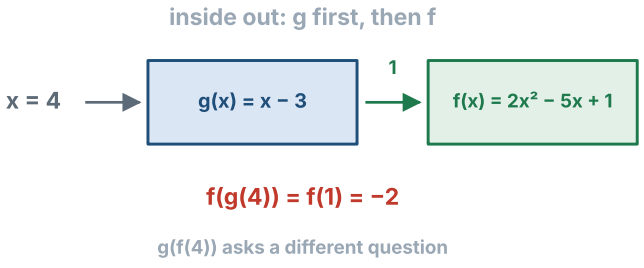
= **3a + 1**.



2 Composition

f(g(x)): work the inner function first, then feed that number outward

Composition is done from the inside out, and the order matters: f(g(2)) is usually not g(f(2)). Write the inner value down before touching the outer function.



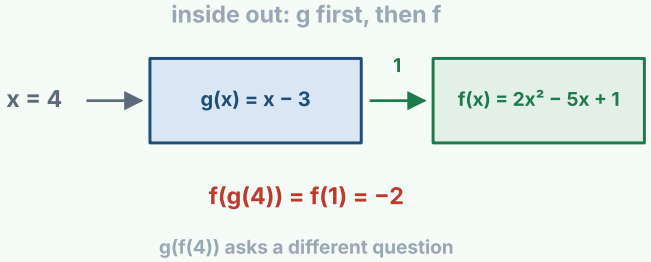
g(4) = 1 comes out of the first box and goes straight into the second, where f(1) = −2. Reversing the boxes gives a different number altogether.

Example 1. **f(x) = 2x² − 5x + 1 and g(x) = x − 3. Find f(g(4)).**

Solution.

g(4) = 1 first.

f(1) = 2 − 5 + 1 = **−2**.

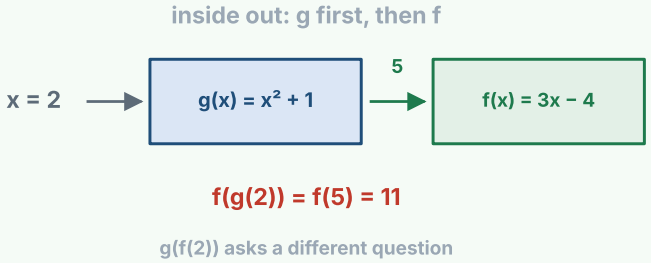


Example 2. **f(x) = 3x − 4 and g(x) = x² + 1. Find f(g(2)).**

Solution.

g(2) = 5.

f(5) = 15 − 4 = **11**.

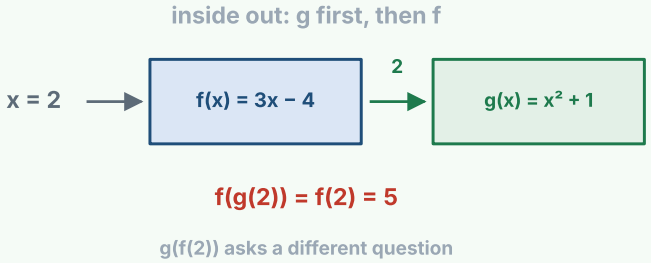


Example 3. **For those same functions, find g(f(2)).**

Solution.

f(2) = 2.

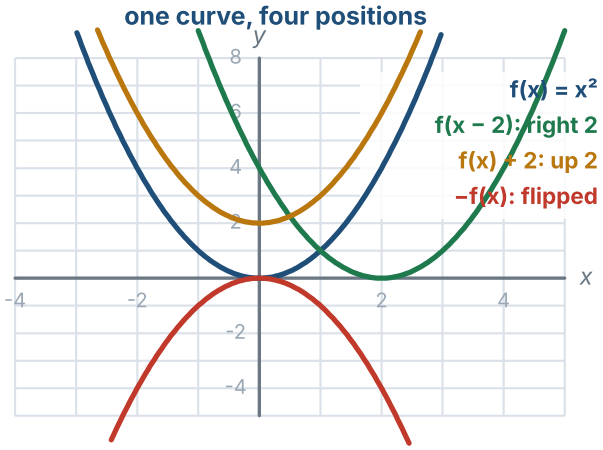
g(2) = 5. Different from f(g(2)) = 11 — **order matters**.



3 Vertical shifts and reflections

f(x) + k moves up k | f(x) − k moves down | −f(x) flips over the x-axis

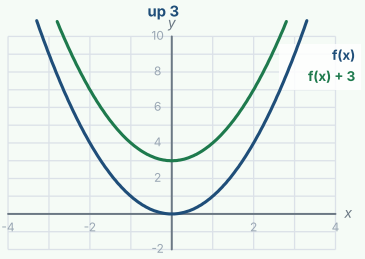
Changes made *outside* the function act on the output, so they behave exactly as you expect: add to go up, negate to flip vertically.



Everything outside the bracket does what it says: +2 lifts the curve, the minus sign turns it upside down. Only the green curve, whose change is inside the bracket, moves sideways.

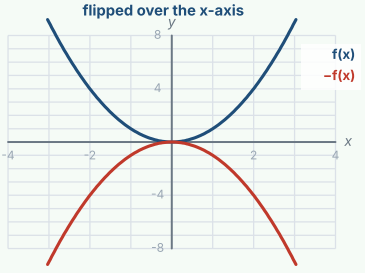
Example 1. **The graph of y = f(x) is shifted up 3. Write the new function.**

Solution.
y = f(x) + 3.



Example 2. **f(x) = x². Describe y = −f(x).**

Solution.
A reflection over the **x-axis**: the upward parabola becomes a downward one.



Example 3. **The point (2, 7) is on y = f(x). Where is it on y = f(x) − 4?**

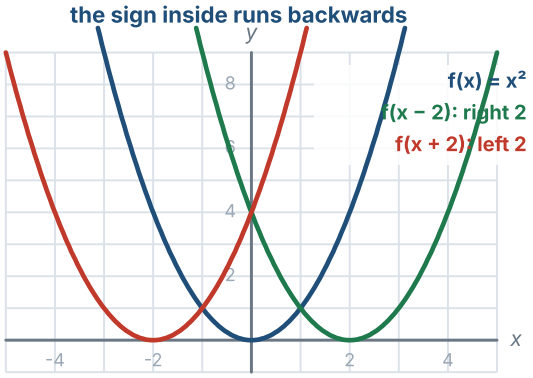
Solution.
Only the output changes.
(2, 3).

(2, 7) on f(x)	the input stays 2
f(x) − 4	the output drops by 4
new point	(2, 3)

4 Horizontal shifts — the opposite way

f(x − h) moves right h | f(x + h) moves left h

Changes made *inside* the brackets act on the input, and they run backwards. f(x − 2) shifts right, not left, because x must be 2 larger to produce the same output. This reversal is the single most-missed idea in transformations.



why inside runs backwards

f(x − 2)	x must be 2 bigger → right 2
f(x + 5)	x must be 5 smaller → left 5

To get the same output from f(x − 2) you must feed in an x that is 2 larger — so every point slides right.
The minus sign moves the graph the opposite way to how it reads.

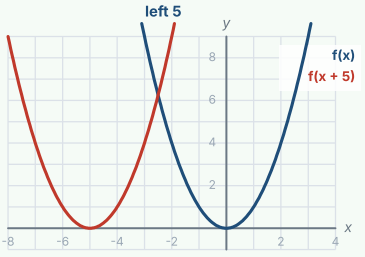
Example 1. **f(x) = x². Describe the graph of y = f(x − 2).**

Solution.
A shift **2 to the right**, giving y = (x − 2)² with vertex (2, 0).



Example 2. **Describe y = f(x + 5).**

Solution.
A shift **5 to the left** — the sign inside is the opposite of the direction.



Example 3. **The point (3, 8) is on y = f(x). Where is the matching point on y = f(x − 4)?**

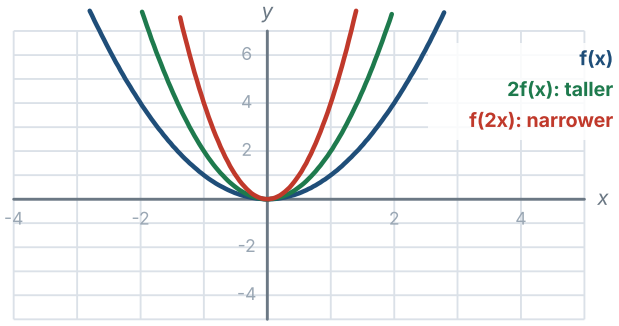
Solution.
The input must be 4 larger to give the same output.
(7, 8).

(3, 8) on f(x)	output 8 at input 3
f(x − 4)	needs x − 4 = 3
so x = 7	new point (7, 8)

5 Stretches

a·f(x) stretches vertically by a | f(bx) squeezes horizontally by 1/b

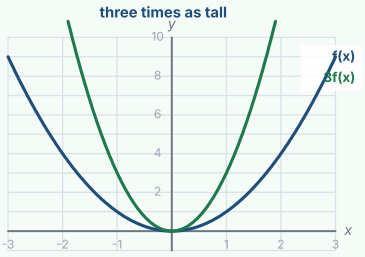
Outside multiplies the outputs, so heights change. Inside multiplies the inputs, so widths change — and again it runs backwards: f(2x) makes the graph half as wide, not twice.



Tripling outside makes every height three times as big. Doubling inside pulls the curve in to half its width, because x only has to reach half as far.

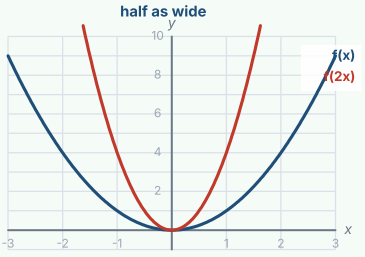
Example 1. **f(x) = x². Describe y = 3f(x).**

Solution.
Every y is tripled: a **vertical stretch by 3**, giving y = 3x².



Example 2. **Describe y = f(2x).**

Solution.
A **horizontal squeeze by a factor of 2** — the graph is half as wide.



Example 3. **The point (4, 6) is on y = f(x). Where is it on y = 2f(x)?**

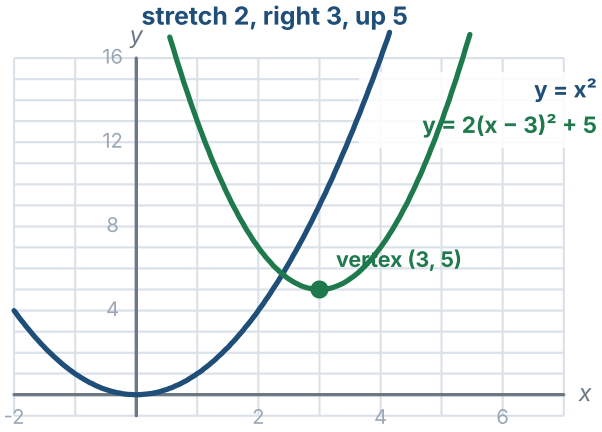
Solution.
The output doubles, the input does not.
(4, 12).

(4, 6) on f(x)	input 4, output 6
2f(x)	doubles the output only
new point	(4, 12)

6 Reading transformations from an equation

$y = a \cdot f(x - h) + k$ → stretch by a, right h, up k

Take the pieces in order and read the sign inside backwards. From vertex form $y = a(x - h)^2 + k$ you can state the whole transformation of $y = x^2$ in one sentence.

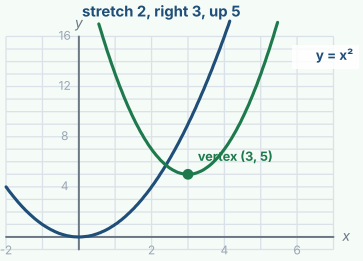


Read a, then h, then k: stretched by 2, moved 3 right (sign reversed) and 5 up. The vertex lands exactly at (h, k).

Example 1. **Describe $y = 2(x - 3)^2 + 5$ as a transformation of $y = x^2$.**

Solution.

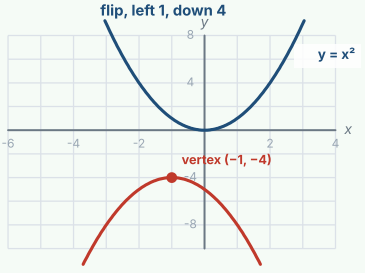
Stretched vertically by 2, moved 3 right and 5 up; vertex (3, 5).



Example 2. **Describe $y = -(x + 1)^2 - 4$.**

Solution.

Flipped over the x-axis, moved 1 left and 4 down; vertex (−1, −4).



Example 3. **$y = f(x)$ has a maximum at (2, 9). Where is the maximum of $y = f(x - 1) + 3$?**

Solution.

Right 1 and up 3.

(3, 12).

maximum (2, 9)	**on y = f(x)**
f(x − 1)	**moves it right 1 → x = 3**
+ 3	**moves it up 3 → y = 12**

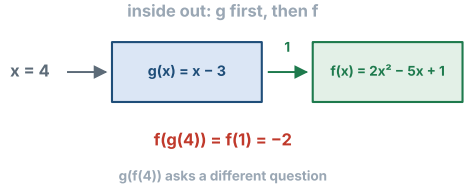
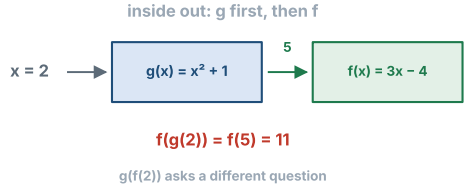
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

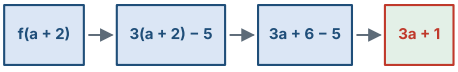
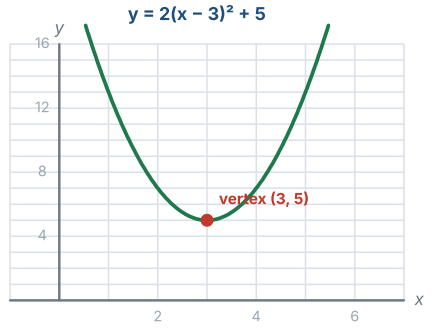
The SAT gives you Desmos on every maths question. Here it is **a transformation viewer**. Graph f and the transformed version together and watch which way the curve actually moves — it settles the inside-sign question for good.

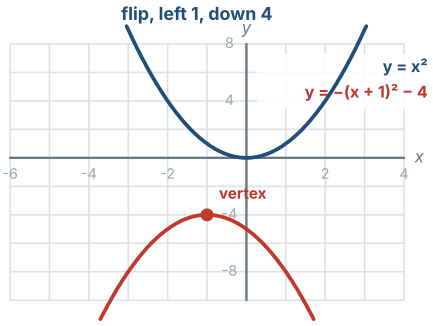
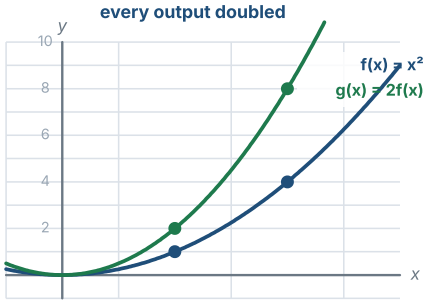
#	Question	Type into Desmos	Then	Answer
D1	$f(x) = 2x^2 - 5x + 1$. What is $f(3)$?	`2*3^2-5*3+1`	Substitute and evaluate in one line.	4
D2	$f(x) = 2x^2 - 5x + 1$ and $g(x) = x - 3$. What is $f(g(4))$?	`2*(4-3)^2-5*(4-3)+1`	Inner function first.	-2
D3	$f(x) = 3x - 4$ and $g(x) = x^2 + 1$. What is $f(g(2))$?	`3*(2^2+1)-4`	$g(2) = 5$, then $f(5)$.	11
D4	For those same functions, what is $g(f(2))$?	`(3*2-4)^2+1`	Different from D3 — order matters.	5
D5	Graph $y = x^2$ and $y = (x - 2)^2$. Which way did it move?	`y=(x-2)^2`	The sign inside runs backwards.	right 2
D6	What is the vertex of $y = 2(x - 3)^2 + 5$?	`y=2(x-3)^2+5`	Read h and k straight off.	(3, 5)
D7	$f(x) = 3x - 5$. What is $f(a + 2)$ when $a = 4$?	`3*(4+2)-5`	Substitute the whole expression.	13
D8	Solve $2x^2 - 5x + 1 = 1$.	`2x^2-5x+1=1`	Two solutions; the question asks where the output is 1.	0 and 2.5
D9	The point (2, 7) is on $y = f(x)$. What is the y-value on $y = f(x) - 4$?	`7-4`	Only the output shifts.	3
D10	What is the vertex of $y = -(x + 1)^2 - 4$?	`y=-(x+1)^2-4`	Inside sign flips; outside does not.	(-1, -4)

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Sort each change into **outside** (acts on the output, behaves as expected) or **inside** (acts on the input, runs backwards) before you answer.

#	Level	Question	Answer	Solution
P1	Hard	The functions f and g are defined by $f(x) = 2x^2 - 5x + 1$ and $g(x) = x - 3$. What is the value of $f(g(4))$? 	−2	$g(4) = 1$ first, then $f(1) = 2 - 5 + 1 = \mathbf{-2}$. A common wrong answer is $f(4)$ — the inner function g was never applied.
P2	Hard	For $f(x) = 3x - 4$ and $g(x) = x^2 + 1$, what is the value of $f(g(2))$? 	11	$g(2) = 5$, then $f(5) = 15 - 4 = \mathbf{11}$. A common wrong answer works the functions in the wrong order: $g(f(2)) = 5$ is a different value again.

#	Level	Question	Answer	Solution
P3	Hard	The graph of y = f(x) is transformed to y = f(x − 2). Which describes the transformation? 	Shift 2 units right	A change inside the brackets acts on the input and runs *backwards*: x must be 2 larger to give the same output. The graph moves **2 right**. A common wrong answer is the instinctive one and the most common error in this topic.
P4	Hard	The point (2, 7) lies on the graph of y = f(x). Which point must lie on the graph of y = f(x) − 4? f(x) − 4 → outside → output drops 4 (2, 7) → becomes (2, 3) f(x − 4) → would give (6, 7) instead	(2, 3)	Subtracting outside the function changes only the output. **(2, 3).** A common wrong answer moves the input instead, which is what f(x − 4) would do.
P5	Hard	If f(x) = 3x − 5, which expression is equal to f(a + 2)? 	3a + 1	Substitute the whole expression: 3(a + 2) − 5 = 3a + 6 − 5. = **3a + 1**. A common wrong answer forgets to subtract the 5.
P6	Hard	What is the vertex of the graph of y = 2(x − 3)² + 5? 	(3, 5)	Vertex form y = a(x − h)² + k has vertex (h, k). **(3, 5)** — note h is +3 even though the bracket reads −3.
P7	Very hard	The graph of y = f(x) has a maximum at the point (2, 9). What are the coordinates of the maximum of y = f(x − 1) + 3? maximum (2, 9) → on y = f(x) f(x − 1) → right 1 → x = 3 + 3 → up 3 → y = 12	(3, 12)	Inside, −1 moves the graph right 1; outside, +3 moves it up 3. **(3, 12).** A common wrong answer moves left instead of right — the inside sign reverses.

#	Level	Question	Answer	Solution
P8	Very hard	Which describes the graph of $y = -(x + 1)^2 - 4$ as a transformation of $y = x^2$? 	Reflected over the x-axis, 1 left, 4 down	The minus outside reflects over the **x-axis**; (x + 1) moves **left** 1; −4 moves down 4. Vertex (−1, −4). A common wrong answer gets the direction of the shift backwards.
P9	Very hard	The point (4, 6) lies on the graph of y = f(x). Which point must lie on the graph of y = 2f(x)? 2f(x) — outside → output doubles (4, 6) — becomes (4, 12) f(2x) — would halve the input instead	(4, 12)	Multiplying outside the function doubles the **output** only. **(4, 12).** A common wrong answer doubles the input, which is what f(x/2) would do.
P10	Very hard	If $f(x) = 2x^2 - 5x + 1$, what is the greatest value of x for which f(x) = 1? f(x) = 1 → 2x² − 5x = 0 → x(2x − 5) = 0 → x = 0 or 2.5	2.5	$2x^2 - 5x + 1 = 1$ gives $2x^2 - 5x = 0$, so x(2x − 5) = 0. x = 0 or 2.5, and the greater is **2.5**. Note this question asks you to *solve*, not to evaluate.
P11	Very hard	The graph of y = f(x) passes through (0, 0), (1, 1) and (2, 4). The graph of y = g(x) passes through (0, 0), (1, 2) and (2, 8). Which relationship is consistent with these points? 	g(x) = 2f(x)	Each output of g is exactly twice the matching output of f: 0, 2, 8 against 0, 1, 4. **g(x) = 2f(x)**, a vertical stretch. A common wrong answer would double the inputs, giving g(1) = f(2) = 4, not 2.
P12	Very hard	The function h is defined by h(x) = f(x) + k, where k is a constant. If f(3) = 7 and h(3) = 2, what is the value of k? h(3) = f(3) + k — 2 = 7 + k k = −5 — the graph moved down 5	−5	h(3) = f(3) + k, so 2 = 7 + k. k = **−5**.

The one habit to build. Inside the brackets, everything is the opposite of what it looks like: f(x − 2) goes right, f(x + 5) goes left, f(2x) gets narrower. Your class scored 13% on transformations, the lowest figure on the whole syllabus, and this single rule is most of it.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

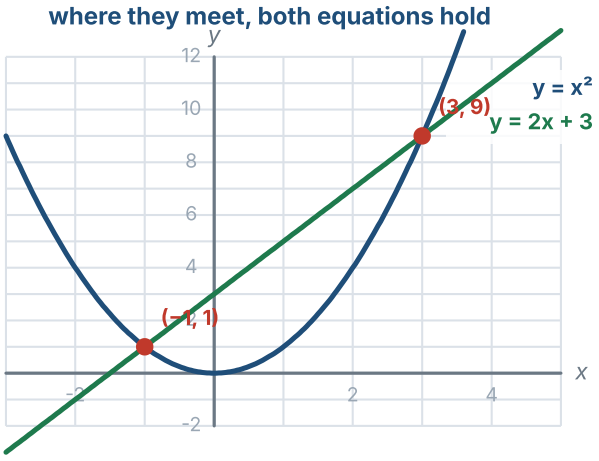
SAT Math 2.8 — Nonlinear Systems

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving by substitution

set the two expressions for y equal | solve the resulting equation | find each y

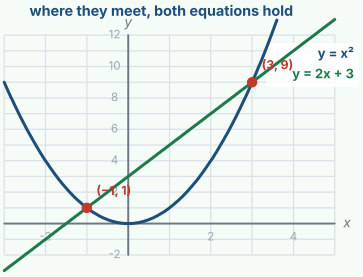
A nonlinear system is one curve and one line, or two curves. Where they meet, both equations hold, so equating them turns the pair into a single equation you already know how to solve. Do not stop at x — the solutions are points.



The two crossings are the two solutions. Equating $x^2 = 2x + 3$ finds their x-values; substituting back into either equation gives the matching y.

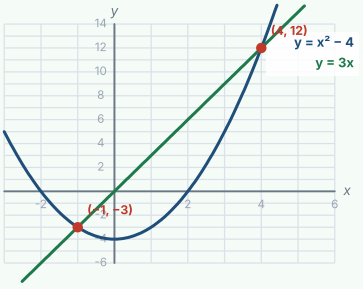
Example 1. **Solve $y = x^2$ and $y = 2x + 3$.**

Solution.
$x^2 = 2x + 3$, so $x^2 - 2x - 3 = 0$ and $(x - 3)(x + 1) = 0$.
$x = 3$ or -1, giving **(3, 9) and (−1, 1)**.



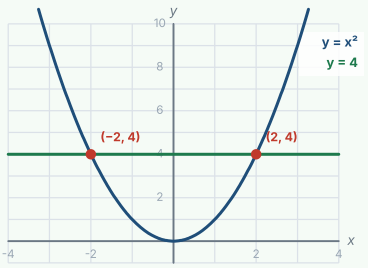
Example 2. **Solve $y = x^2 - 4$ and $y = 3x$.**

Solution.
$x^2 - 3x - 4 = 0$, so $(x - 4)(x + 1) = 0$.
(4, 12) and (−1, −3).



Example 3. **Solve $y = x^2$ and $y = 4$.**

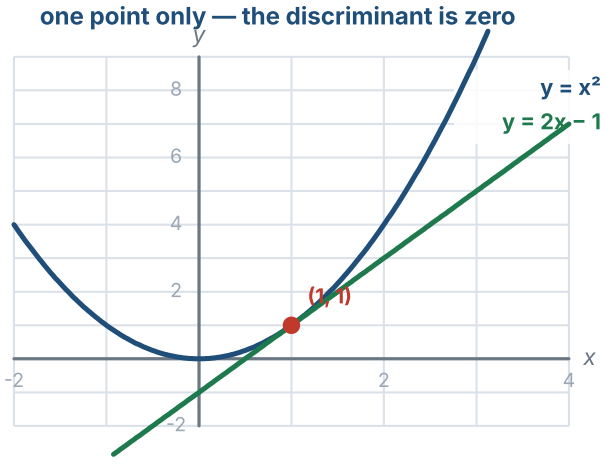
Solution.
$x^2 = 4$, so $x = \pm 2$.
(2, 4) and (−2, 4).



2 How many intersection points?

after substituting, the discriminant of the resulting quadratic counts the meetings

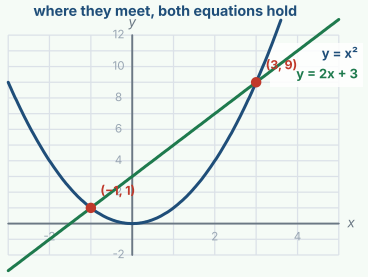
Two real solutions means the line cuts the curve twice; one means it is tangent; none means they miss entirely. This is the same discriminant test as always, wearing a different hat.



Substituting $y = 2x - 1$ into $y = x^2$ gives $x^2 - 2x + 1 = 0$ with discriminant 0 — and the picture shows the line just grazing the curve.

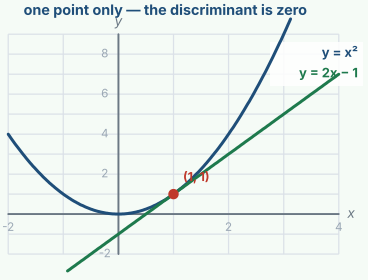
Example 1. **How many times does $y = 2x + 3$ meet $y = x^2$?**

Solution.
$x^2 - 2x - 3 = 0$ has discriminant $4 + 12 = 16 > 0$.
Twice.



Example 2. **How many times does $y = 2x - 1$ meet $y = x^2$?**

Solution.
$x^2 - 2x + 1 = 0$ has discriminant $4 - 4 = 0$.
Once — the line is tangent.

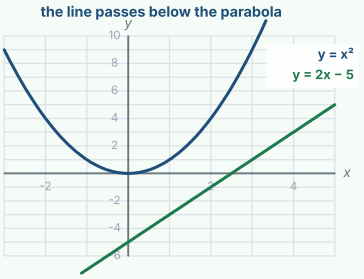


Example 3. **How many times does $y = 2x - 5$ meet $y = x^2$?**

Solution.

$x^2 - 2x + 5 = 0$ has discriminant $4 - 20 = -16 < 0$.

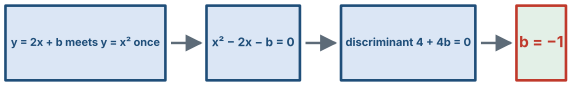
Never.



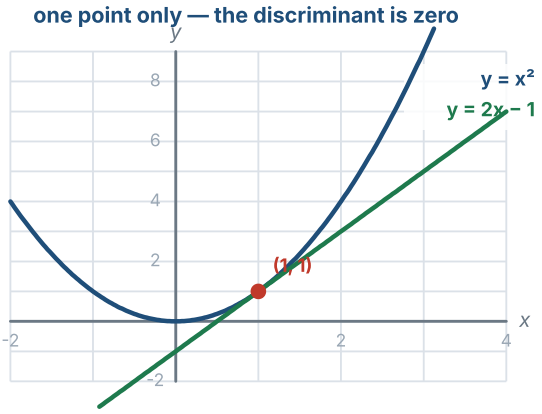
3 Tangency questions

‘meets at exactly one point’ ⇔ discriminant = 0

Whenever a question fixes a constant so that a line touches a curve once, substitute, collect into $ax^2 + bx + c = 0$, and set $b^2 - 4ac = 0$. Watch the sign at the last step — that is where the marks go.



the final sign is where five of your class lost the mark



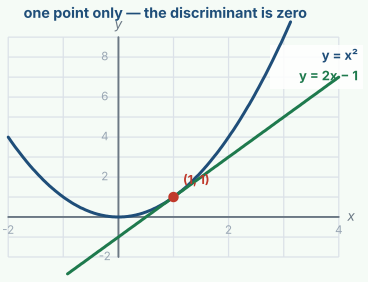
Set up, substitute, set the discriminant to zero, then solve for the constant — and check the sign of the answer against the picture.

Example 1. **For what b does $y = 2x + b$ meet $y = x^2$ exactly once?**

Solution.

$x^2 - 2x - b = 0$ with discriminant $4 + 4b = 0$.

$b = \mathbf{-1}$. Five of your class reached +1 here.

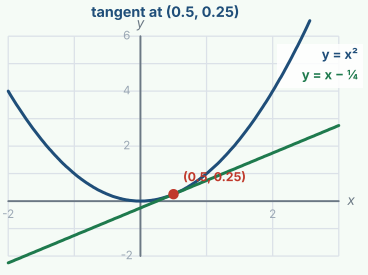


Example 2. **For what c does $y = x + c$ meet $y = x^2$ exactly once?**

Solution.

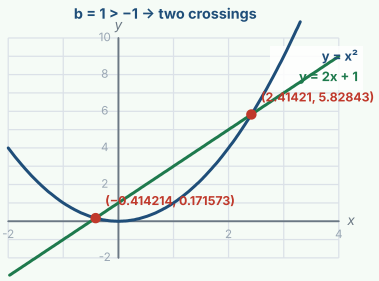
$x^2 - x - c = 0$, so $1 + 4c = 0$.

$c = \mathbf{-1/4}$.



Example 3. **Why does a positive b give two intersections in the first example?**

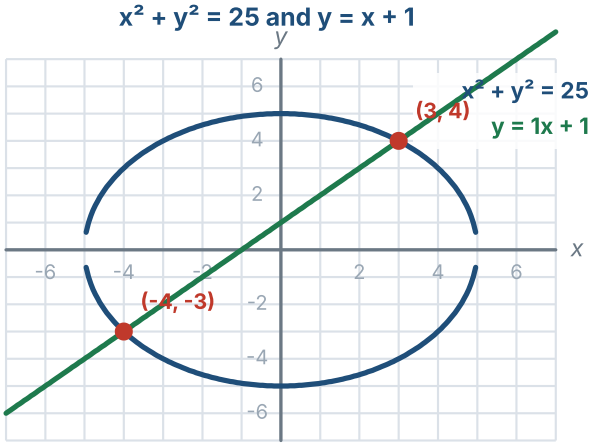
Solution.
$4 + 4b > 0$ whenever $b > -1$, so the discriminant is positive and the line cuts the parabola **twice**.



4 Circle and line systems

substitute the line into the circle equation, then solve the quadratic

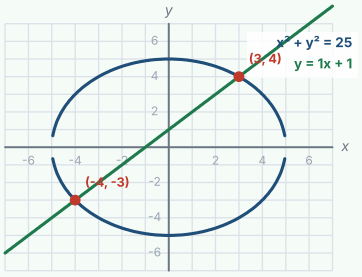
The method does not change when one equation is a circle. Replace y in $x^2 + y^2 = r^2$ using the line, expand carefully, and solve.



The line cuts the circle at two lattice points. Substituting $y = x + 1$ turns the system into $2x^2 + 2x - 24 = 0$, whose roots are those two x-values.

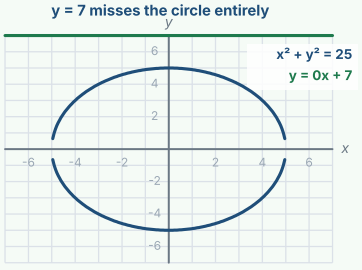
Example 1. **Solve $x^2 + y^2 = 25$ and $y = x + 1$.**

Solution.
$x^2 + (x + 1)^2 = 25$ gives $2x^2 + 2x - 24 = 0$, so $x^2 + x - 12 = 0$.
$x = 3$ or -4, giving **(3, 4) and (−4, −3)**.



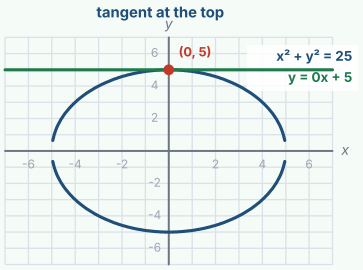
Example 2. **How many points does $y = 7$ share with $x^2 + y^2 = 25$?**

Solution.
$x^2 = 25 - 49 = -24$, which is impossible.
None — the line lies outside the circle.



Example 3. **How many points does y = 5 share with that circle?**

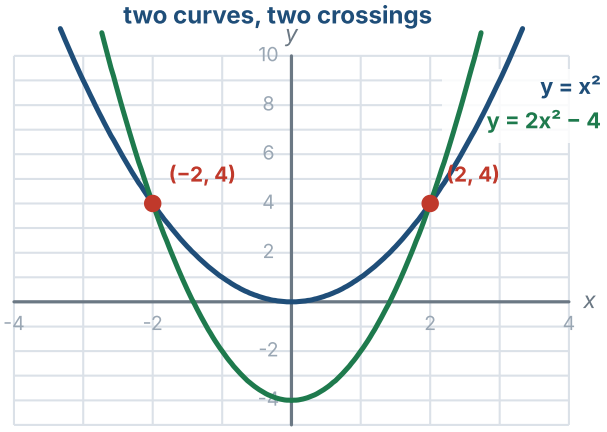
Solution.
$x^2 = 0$, so $x = 0$ only.
One: the line is tangent at (0, 5).



5 Systems with two curves

equate the two expressions | you may get a higher-degree equation

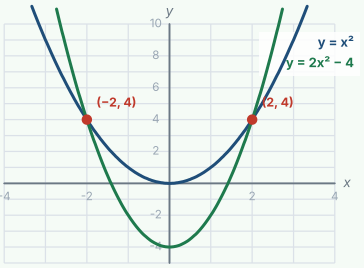
Two parabolas, or a parabola and a reciprocal, follow the same rule. The resulting equation may be cubic or quartic, so factor fully and keep every root.



Both curves are even, so the crossings come in a symmetric pair. Equating them gives $x^2 = 4$ and both roots count.

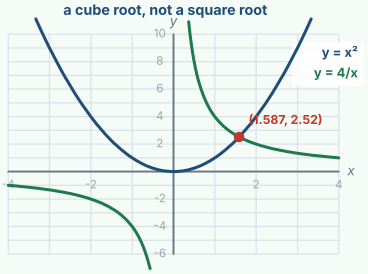
Example 1. **Solve $y = x^2$ and $y = 2x^2 - 4$.**

Solution.
$x^2 = 2x^2 - 4$ gives $x^2 = 4$.
(2, 4) and (−2, 4).



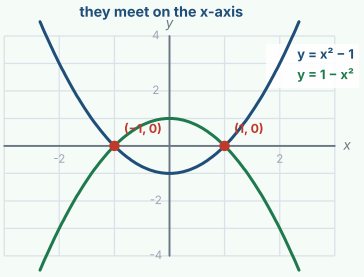
Example 2. **Solve $y = x^2$ and $y = 4/x$ for $x \neq 0$.**

Solution.
$x^2 = 4/x$, so $x^3 = 4$ and $x = \sqrt[3]{4} \approx 1.587$.
One real intersection, near **(1.587, 2.520)**.



Example 3. **Solve $y = x^2 - 1$ and $y = 1 - x^2$.**

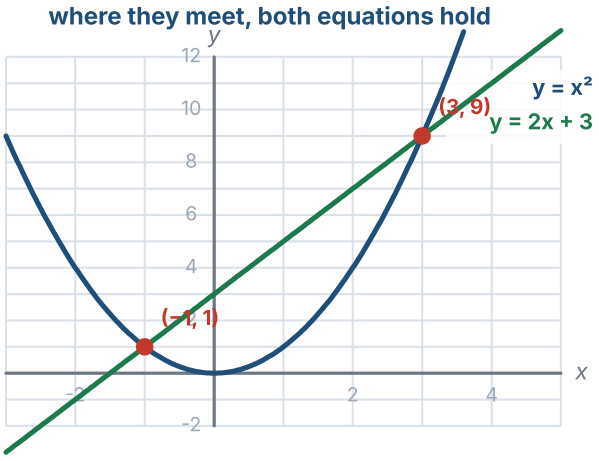
Solution.
$x^2 - 1 = 1 - x^2$ gives $2x^2 = 2$.
(1, 0) and (−1, 0).



6 Reading a system off a graph

a solution is a point the two graphs share | count intersections, not curves

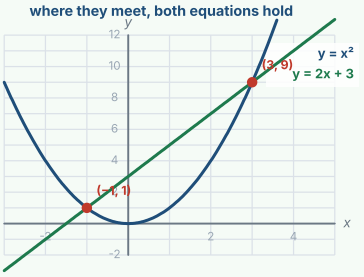
Desmos makes this almost free: enter both equations and click each crossing. The SAT often asks only for the number of solutions, or for one coordinate of one solution.



Two crossings means two solutions, and each one is a pair (x, y). Reading off only the x-values answers half the question.

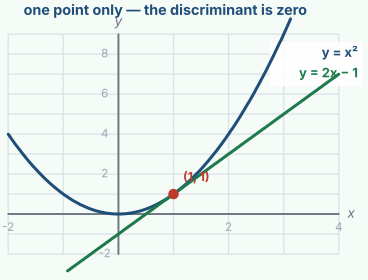
Example 1. **Two graphs cross at (3, 9) and (−1, 1). How many solutions does the system have?**

Solution.
Two — one for each shared point.



Example 2. **A line touches a parabola at one point. How many solutions does that system have?**

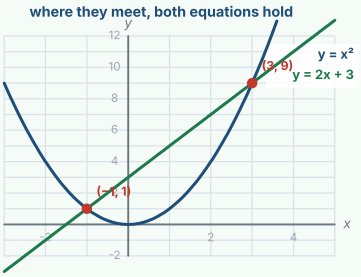
Solution.
One. A tangency is a single repeated solution.



Example 3. **A system's solutions are (3, 9) and (−1, 1). What is the sum of the x-coordinates?**

Solution.

3 + (−1) = **2**. This also equals −b/a for the quadratic you solved.



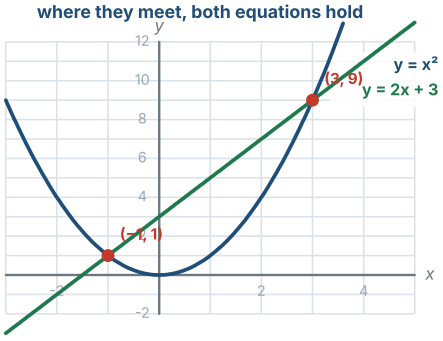
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

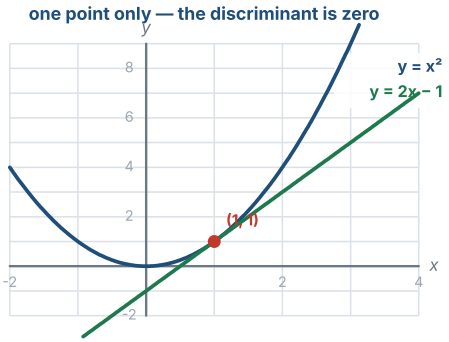
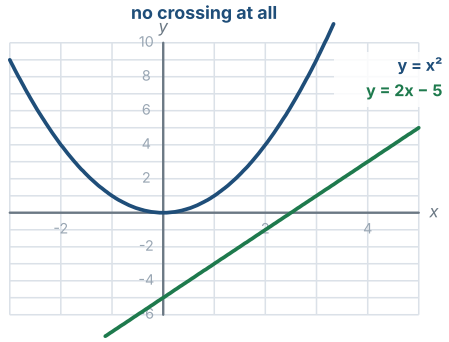
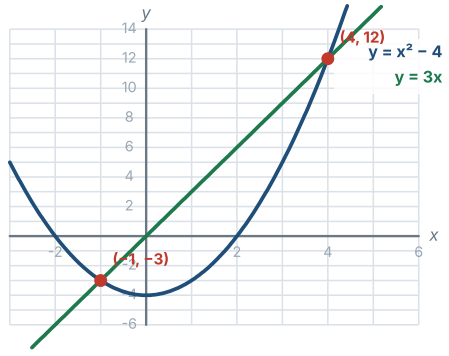
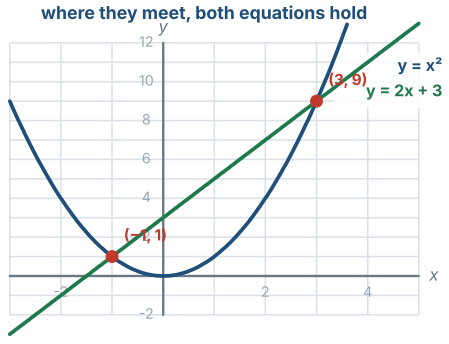
The SAT gives you Desmos on every maths question. Here it is **an intersection finder**. Enter both equations and click every crossing; Desmos gives you each point exactly.

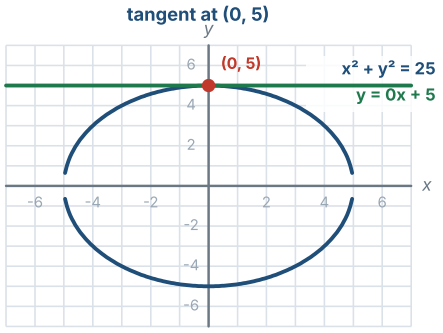
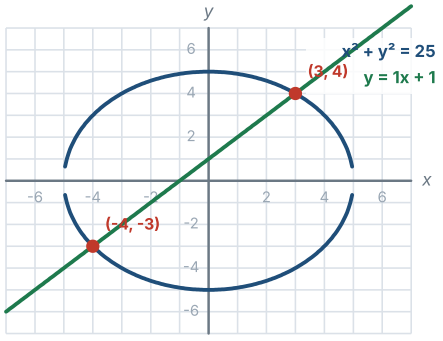
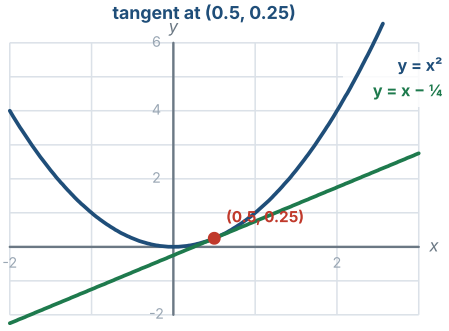
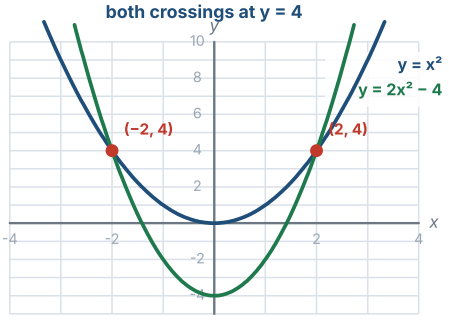
#	Question	Type into Desmos	Then	Answer
D1	Solve the system $y = x^2$ and $y = 2x + 3$. What is the larger x-value?	`y=x^2 and y=2x+3`	Graph both and click the intersections.	3
D2	For that system, what is the sum of the two x-values?	`3+(-1)`	Equals $-b/a$ for $x^2 - 2x - 3$.	2
D3	For what value of b does $y = 2x + b$ meet $y = x^2$ exactly once?	`-(2^2)/4`	Set $4 + 4b = 0$.	-1
D4	How many times does $y = 2x - 5$ meet $y = x^2$?	`y=x^2 and y=2x-5`	The line passes below the parabola.	0
D5	Solve $x^2 + y^2 = 25$ and $y = x + 1$. What is the larger x-value?	`x^2+(x+1)^2=25`	Substitute, then solve the quadratic.	3
D6	Solve $y = x^2$ and $y = 2x^2 - 4$. What are the x-values?	`x^2=2x^2-4`	Both curves are even, so the answers are symmetric.	2 and -2
D7	Solve $y = x^2$ and $y = 4/x$. What is x, to three decimal places?	`x^3=4`	A cube root, not a square root.	1.587
D8	How many points does the line $y = 5$ share with $x^2 + y^2 = 25$?	`25-5^2`	Zero under the root means tangency.	1
D9	Solve $y = x^2 - 4$ and $y = 3x$. What is the larger x-value?	`x^2-4=3x`	Factor $x^2 - 3x - 4$.	4
D10	For what value of c does $y = x + c$ meet $y = x^2$ exactly once?	`-1/4`	Set $1 + 4c = 0$.	-0.25

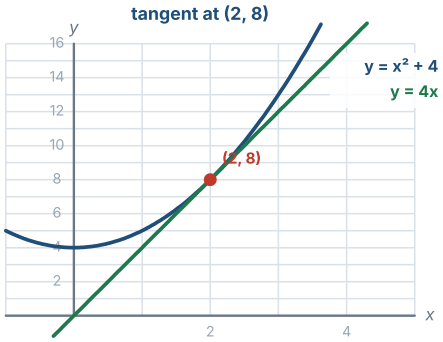
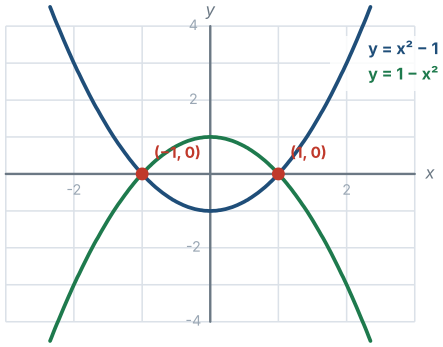
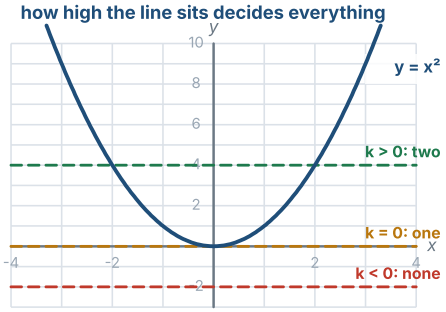
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. When a question fixes a constant so two graphs meet **exactly once**, it is a discriminant question wearing a disguise.

#	Level	Question	Answer	Solution
P1	Hard	In the xy-plane, the graphs of $y = x^2$ and $y = 2x + 3$ intersect at two points. What is the sum of the x-coordinates of those points? 	2	$x^2 = 2x + 3$ gives $x^2 - 2x - 3 = 0$, whose roots sum to $-b/a$ = **2**. The roots are 3 and −1. A common wrong answer is the larger root alone.

#	Level	Question	Answer	Solution
P2	Hard	In the xy-plane, the line $y = 2x + b$, where b is a constant, intersects the parabola $y = x^2$ at exactly one point. What is the value of b? 	−1	$x^2 - 2x - b = 0$ must have discriminant zero: $4 + 4b = 0$. b = **−1**. Five of your class answered +1 — the method was right and the sign was lost at the last step.
P3	Hard	How many solutions does the system $y = x^2$ and $y = 2x - 5$ have? 	Zero	$x^2 - 2x + 5 = 0$ has discriminant $4 - 20 = -16 < 0$. **No real solutions** — the line passes entirely below the parabola.
P4	Hard	The system $y = x^2 - 4$ and $y = 3x$ has two solutions. What is the greater of the two x-values? 	4	$x^2 - 3x - 4 = 0$ factors as $(x - 4)(x + 1) = 0$. x = 4 or −1, so the greater is **4**.
P5	Hard	Which ordered pair is a solution to the system $y = x^2$ and $y = 2x + 3$? 	(−1, 1)	Substituting x = −1: y = 1 from the parabola and y = −2 + 3 = 1 from the line, so both hold. **(−1, 1).** A common wrong answer uses the right y but the wrong sign of x.

#	Level	Question	Answer	Solution
P6	Hard	How many points of intersection do the line $y = 5$ and the circle $x^2 + y^2 = 25$ have? 	One	Substituting gives $x^2 + 25 = 25$, so $x^2 = 0$ and $x = 0$ only. **One** point, (0, 5): the line is tangent at the top of the circle.
P7	Very hard	The system $x^2 + y^2 = 25$ and $y = x + 1$ has two solutions. What is the greater of the two x-values? 	3	$x^2 + (x + 1)^2 = 25$ gives $2x^2 + 2x − 24 = 0$, so $x^2 + x − 12 = 0$ and $(x + 4)(x − 3) = 0$. $x = 3$ or $−4$, so the greater is **3**. A common wrong answer is the circle's radius, not a solution.
P8	Very hard	For what value of c does the line $y = x + c$ intersect the parabola $y = x^2$ at exactly one point? 	−1/4	$x^2 − x − c = 0$ must have discriminant zero: $1 + 4c = 0$. $c = \mathbf{-1/4}$. A common wrong answer drops the minus sign.
P9	Very hard	The graphs of $y = x^2$ and $y = 2x^2 − 4$ intersect at two points. What is the y-coordinate of both points? 	4	$x^2 = 2x^2 − 4$ gives $x^2 = 4$, so $x = ±2$. In either case $y = x^2 = \mathbf{4}$. Both points share the same height.

#	Level	Question	Answer	Solution
P10	Very hard	In the xy-plane, the line y = mx intersects the parabola y = x² + 4 at exactly one point, where m is a positive constant. What is the value of m? 	4	x² − mx + 4 = 0 needs m² − 16 = 0, so m = ±4. With m positive, m = **4**. A common wrong answer is m², stopped one step early.
P11	Very hard	The graphs of y = x² − 1 and y = 1 − x² intersect at how many points, and where? 	Two points, at (1, 0) and (−1, 0)	x² − 1 = 1 − x² gives 2x² = 2, so x = ±1, and then y = 1 − 1 = 0. **(1, 0) and (−1, 0)** — the two parabolas meet on the x-axis.
P12	Very hard	A system consists of y = x² and y = k, where k is a constant. For which values of k does the system have exactly two real solutions? 	k > 0	x² = k has two real solutions exactly when k is positive, one when k = 0, and none when k is negative. **k > 0.** The horizontal line must sit above the vertex.

The one habit to build. Set the discriminant to zero, then check the sign of your answer against the graph. On Test 5 ten students reached the right equation for b and five of them wrote +1 instead of −1.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

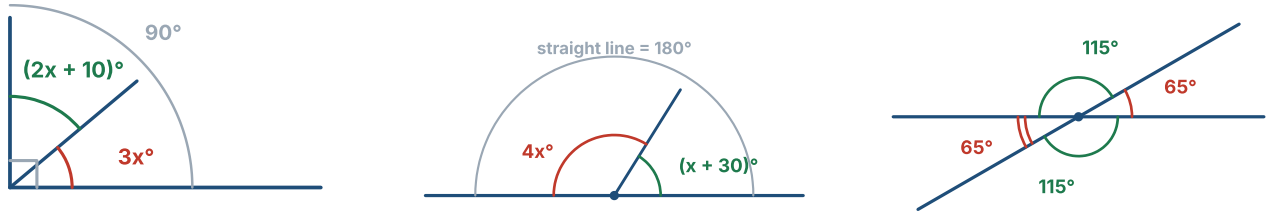
SAT Math 4.1 — Lines, Angles & Triangles

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Angles on a line and at a point

straight line = 180° | full turn = 360° | vertical angles are equal

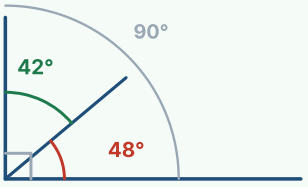
Complementary angles add to 90°; supplementary to 180°. Vertical (opposite) angles made by two crossing lines are always equal. Most angle-chasing questions are one of these three facts applied twice.



Three pictures, three facts: a right angle split in two parts that add to 90°; a straight line split in two that add to 180°; and two crossing lines, where opposite angles are equal and neighbours add to 180°.

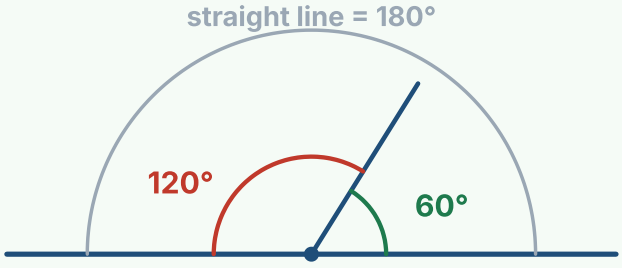
Example 1. **Two angles are complementary. One is 3x° and the other (2x + 10)°. Find x.**

Solution.
3x + 2x + 10 = 90, so 5x = 80.
x = **16**.



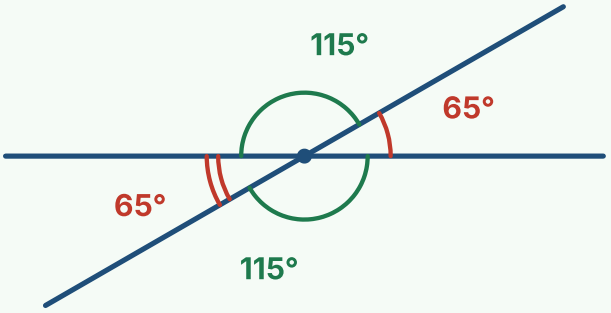
Example 2. **Two angles on a straight line are (4x)° and (x + 30)°. Find x.**

Solution.
4x + x + 30 = 180, so 5x = 150.
x = **30**.



Example 3. **Two lines cross. One angle is 65°. Find the other three.**

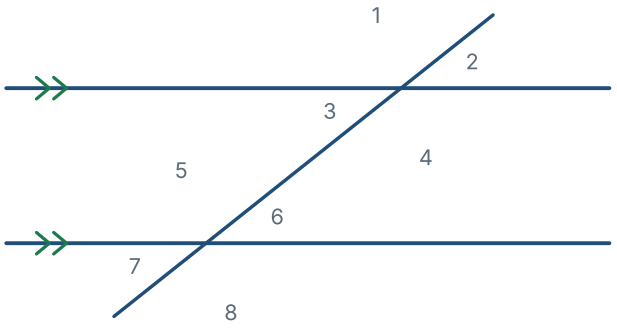
Solution.
The vertical angle is **65°**; the two adjacent ones are each 180 − 65 = **115°**.



2 Parallel lines and a transversal

corresponding and alternate angles are equal | co-interior angles add to 180°

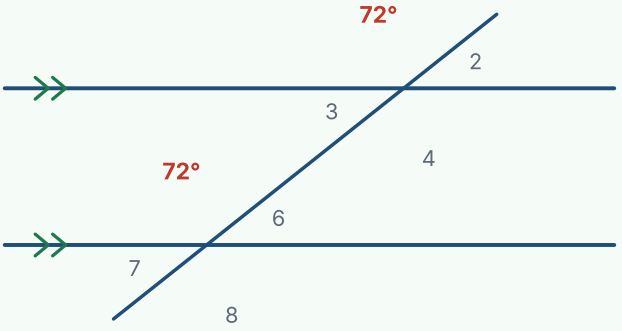
When a line crosses two parallels, every angle is either equal to a given one or supplementary to it. Find the equal pairs first; everything else follows from 180°.



The arrow marks say the two lines are parallel. Angles 1 and 5 are corresponding (equal), 4 and 5 are alternate interior (equal), and 3 and 5 are co-interior (they add to 180°).

Example 1. **Two parallel lines are cut by a transversal and one angle is 72°. What is its corresponding angle?**

Solution.
Corresponding angles are equal.
72°.

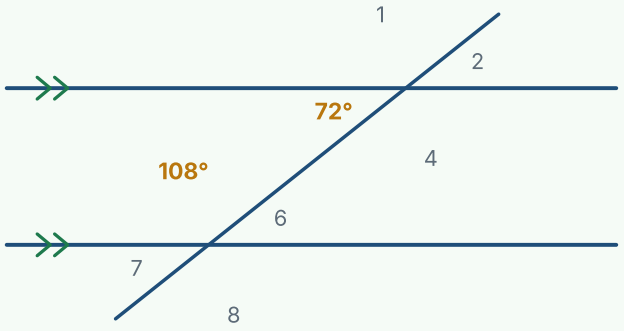


Example 2. **What is the co-interior angle to that 72°?**

Solution.

Co-interior angles add to 180°.

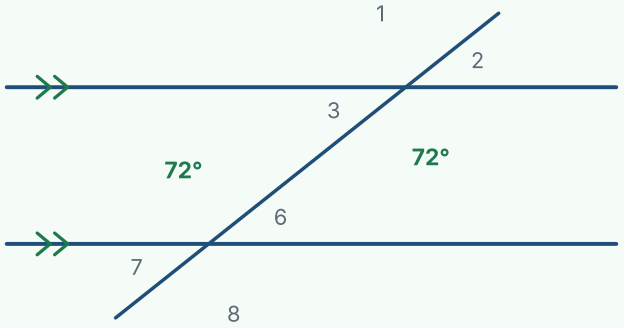
180 − 72 = **108°**.



Example 3. **In the figure above, angles 4 and 5 are what kind of pair?**

Solution.

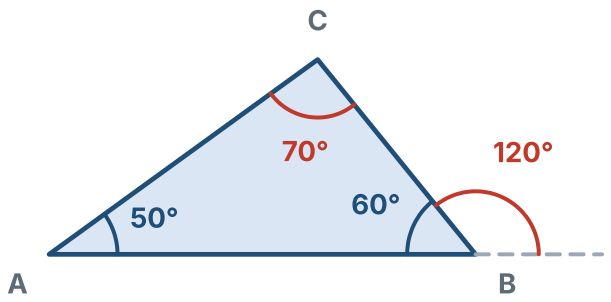
They sit between the parallels on opposite sides of the transversal: **alternate interior**, so they are equal.



3 Triangle angle facts

angles sum to 180° | an exterior angle equals the two remote interior angles

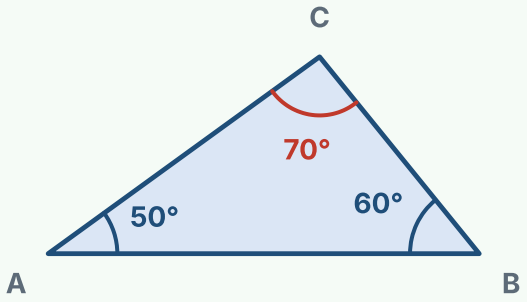
The exterior-angle shortcut saves a step: instead of finding the third angle and subtracting from 180, add the two far angles directly.



The three angles inside total 180°. The exterior angle at B is 120°, which is both 180 − 60 and 50 + 70 — the two angles far away from it.

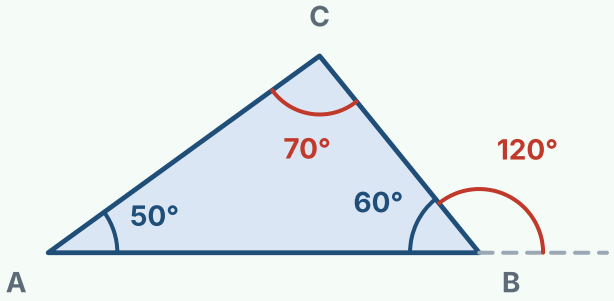
Example 1. **A triangle has angles 50° and 60°. Find the third.**

Solution.
180 − 50 − 60 = **70°**.



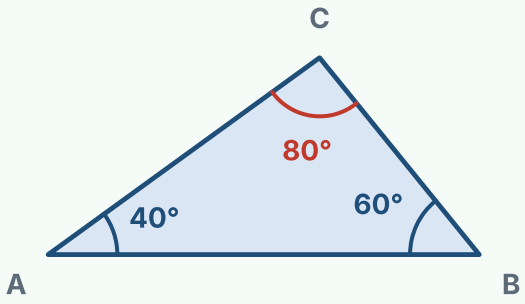
Example 2. **For that triangle, what is the exterior angle at the 60° vertex?**

Solution.
It equals the two remote interior angles: 50 + 70 = **120°**. Checking: 180 − 60 = 120.



Example 3. **A triangle's angles are in the ratio 2 : 3 : 4. Find the largest.**

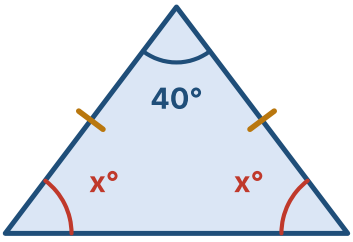
Solution.
The parts total 9, and 180/9 = 20° per part.
The largest is 4 × 20 = **80°**.

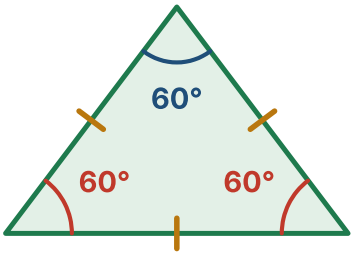


4 Isosceles and equilateral triangles

equal sides face equal angles | equilateral → every angle 60°

The tick marks on a diagram are doing real work: two marked sides mean two equal base angles. This turns most isosceles questions into a single equation.

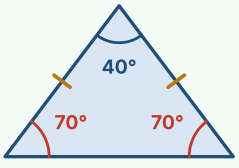




Two equal sides (the tick marks) force two equal base angles, so 2x + 40 = 180. When all three sides are marked, every angle is 60°.

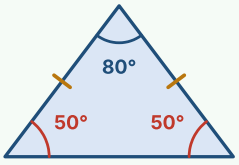
Example 1. **An isosceles triangle has apex angle 40°. Find each base angle.**

Solution.
The two base angles are equal: (180 − 40)/2.
70° each.



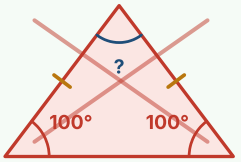
Example 2. **An isosceles triangle has a base angle of 50°. Find the apex angle.**

Solution.
Both base angles are 50°.
180 − 100 = **80°**.



Example 3. **Why can an isosceles triangle not have two angles of 100°?**

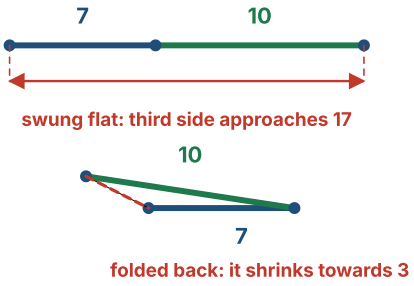
Solution.
They would total 200°, already above 180°. The equal angles in an isosceles triangle must each be **below 90°**.



5 The triangle inequality

any two sides must add to more than the third

Given two sides a and b, the third side x satisfies $|a - b| < x < a + b$. Questions ask for the possible values, or for the largest or smallest whole number that works.





Hinge the two known sides open and shut. Straightened out they reach 7 + 10 = 17; folded back they close to 10 − 7 = 3. The third side lives strictly between, which is the red stretch of the number line.

Example 1. **Two sides are 7 and 10. What values can the third side take?**

Solution.
$10 - 7 < x < 10 + 7$.
$3 < x < 17$.



Example 2. **For those sides, what is the largest possible whole-number third side?**

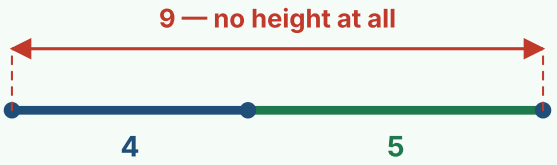
Solution.
x must be below 17.
16.



largest whole number: 16

Example 3. **Can a triangle have sides 4, 5 and 9?**

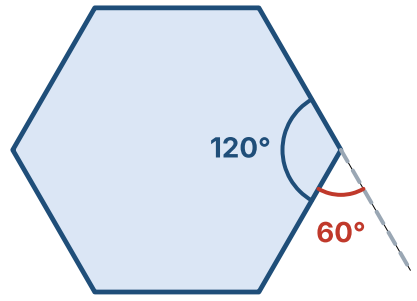
Solution.
4 + 5 = 9, which is not *more* than 9.
No — it collapses flat.



6 Polygons and quadrilaterals

interior angles of an n-gon sum to (n − 2) × 180° | exterior angles always sum to 360°

The exterior-angle fact is the quicker route for regular polygons: each exterior angle is 360/n, and the interior angle is its supplement. In a parallelogram opposite angles are equal and adjacent ones add to 180°.



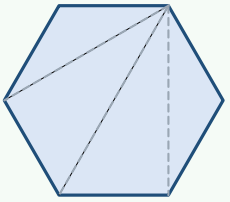
interior + exterior = 180°



Extend one side of the hexagon and the exterior angle appears: 60°, with 120° inside. Walk right round and the exterior angles total 360°. In the parallelogram, 65° sits opposite 65° and beside 115°.

Example 1. **Find the sum of the interior angles of a hexagon.**

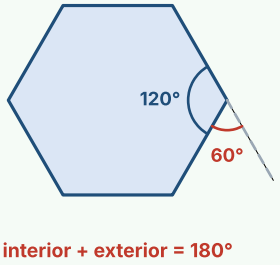
Solution.
(6 − 2) × 180 = **720°**.



4 triangles × 180° = 720°

Example 2. **Find each interior angle of a regular hexagon.**

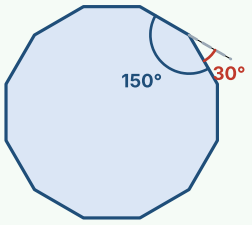
Solution.
Each exterior angle is 360/6 = 60°.
Interior = 180 − 60 = **120°**.



interior + exterior = 180°

Example 3. **A regular polygon has interior angles of 150°. How many sides?**

Solution.
Each exterior angle is 30°, and they total 360°.
360/30 = **12 sides**.



interior + exterior = 180°

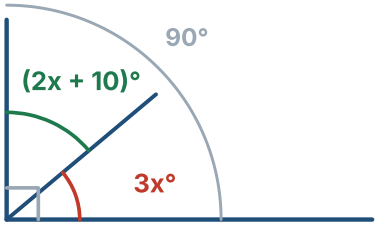
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

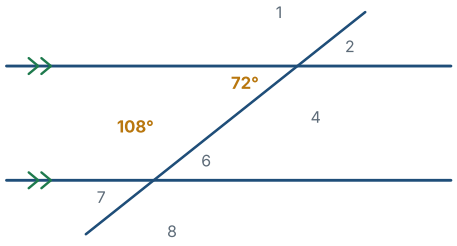
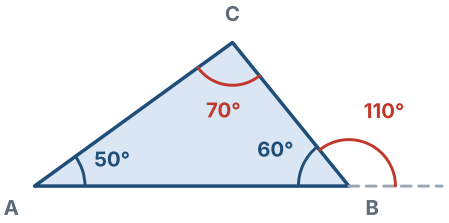
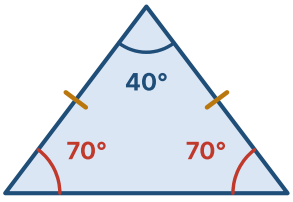
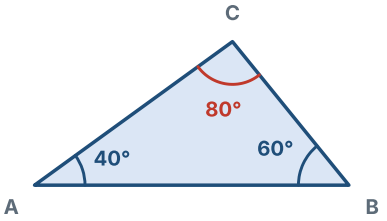
The SAT gives you Desmos on every maths question. For angle work it is **an equation solver**. Write the angle relationship as an equation and let Desmos finish it — the thinking is choosing which relationship applies.

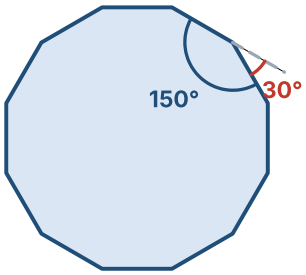
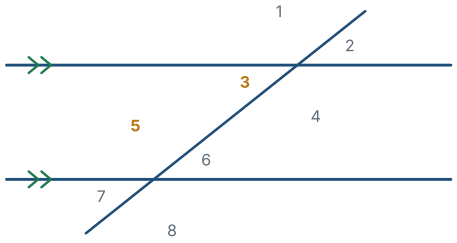
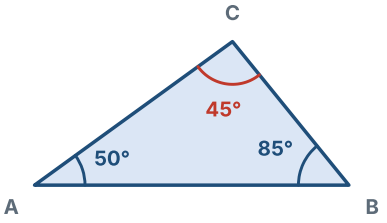
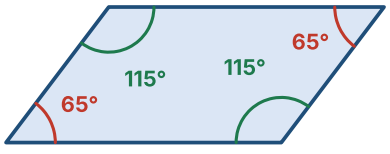
#	Question	Type into Desmos	Then	Answer
D1	Two complementary angles measure 3x° and (2x + 10)°. What is x?	`3x+2x+10=90`	Type the equation; Desmos solves it.	16
D2	Two angles on a straight line are 4x° and (x + 30)°. What is x?	`4x+x+30=180`	A straight line gives 180.	30
D3	A triangle has angles 50° and 60°. What is the third angle?	`180-50-60`	The three must total 180.	70
D4	A triangle's angles are in the ratio 2 : 3 : 4. What is the largest angle?	`180/9*4`	Nine parts in total.	80
D5	An isosceles triangle has apex angle 40°. What is each base angle?	`(180-40)/2`	The two base angles are equal.	70
D6	What is the sum of the interior angles of a hexagon?	`(6-2)*180`	The formula works for any polygon.	720
D7	What is each interior angle of a regular hexagon?	`180-360/6`	Exterior first, then subtract from 180.	120
D8	A regular polygon has interior angles of 150°. How many sides does it have?	`360/(180-150)`	Exterior angles always total 360.	12
D9	Two sides of a triangle are 7 and 10. What is the largest whole-number third side?	`7+10-1`	Strictly less than the sum.	16
D10	Two parallel lines are cut by a transversal and one angle is 72°. What is the co-interior angle?	`180-72`	Co-interior angles are supplementary.	108

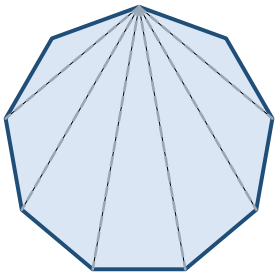
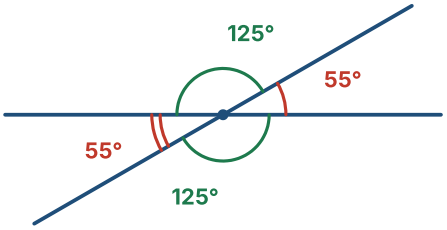
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Mark every angle you can deduce on the figure before you write an equation. Most of these fall out in two steps once the diagram is labelled.

#	Level	Question	Answer	Solution
P1	Hard	Two angles are complementary. One angle measures 3x° and the other measures (2x + 10)°. What is the value of x? 	16	Complementary means they total 90°: 3x + 2x + 10 = 90, so 5x = 80. x = **16**. Using 180 instead of 90 gives 34, which is not an option — a useful check.

#	Level	Question	Answer	Solution
P2	Hard	In the figure, two parallel lines are cut by a transversal. If angle 1 measures 72°, what is the measure of angle 3, in degrees? 	108	Angles 1 and 3 sit on a straight line, so they are supplementary. 180 − 72 = **108°**.
P3	Hard	In a triangle, two of the angles measure 50° and 60°. What is the measure of the exterior angle at the third vertex, in degrees? 	110	The third interior angle is 180 − 110 = 70°, and its exterior angle is 180 − 70 = **110°**. Equivalently, the exterior angle equals the two remote interior angles: 50 + 60 = 110.
P4	Hard	An isosceles triangle has an apex angle of 40°. What is the measure of each base angle? 	70°	The two base angles are equal and the three total 180°. (180 − 40)/2 = **70°**. A common wrong answer is the sum of the two base angles, not each one.
P5	Hard	The angles of a triangle are in the ratio 2 : 3 : 4. What is the measure of the largest angle, in degrees? 	80	The ratio has 2 + 3 + 4 = 9 parts, and 180/9 = 20° per part. 4 × 20 = **80°**.
P6	Hard	Two sides of a triangle measure 7 and 10. State the range of possible lengths for the third side. 	3 < x < 17	The third side x must satisfy **3 < x < 17**. Both boundaries are excluded, and each would give a flat, degenerate triangle.

#	Level	Question	Answer	Solution
P7	Very hard	In a regular polygon, each interior angle measures 150°. How many sides does the polygon have?  interior + exterior = 180°	12	Each exterior angle is 180 − 150 = 30°, and the exterior angles always total 360°. 360/30 = **12 sides**.
P8	Very hard	Two parallel lines are cut by a transversal. One of the eight angles formed measures (3x + 10)° and the co-interior angle on the same side measures (2x − 5)°. What is the value of x?  	35	Co-interior angles are supplementary: 3x + 10 + 2x − 5 = 180, so 5x + 5 = 180. x = **35**. Setting them equal instead gives x = −15, which is impossible — a sign you chose the wrong angle relationship.
P9	Very hard	In triangle ABC, angle A measures (2x)°, angle B measures (3x + 10)°, and angle C measures (x + 20)°. What is the measure of the largest angle, in degrees?  	85°	2x + 3x + 10 + x + 20 = 180 gives 6x + 30 = 180, so x = 25. The angles are 50°, 85° and 45°, so the largest is **85°**. A common wrong answer is x itself, which is not an angle of the triangle.
P10	Very hard	A parallelogram has one angle measuring 65°. What is the measure of the angle adjacent to it, in degrees?  	115	Adjacent angles of a parallelogram are supplementary. 180 − 65 = **115°**. The opposite angle, by contrast, is also 65°.

#	Level	Question	Answer	Solution
P11	Very hard	The sum of the interior angles of a polygon is 1,260°. How many sides does it have?  **7 triangles × 180° = 1260°**	9	(n − 2) × 180 = 1260, so n − 2 = 7. n = **9**. A common wrong answer is n − 2, stopped one step early.
P12	Very hard	In the figure, two lines intersect and one of the four angles formed measures (5x − 20)°. Its vertical angle measures (3x + 10)°. What is the measure of each of these angles, in degrees? 	55°	Vertical angles are equal: 5x − 20 = 3x + 10, so 2x = 30 and x = 15. The angle is 5(15) − 20 = **55°**. A common wrong answer is x itself, not the angle — read what the question asks for.

The one habit to build. Read what the question asks for. Several of these solve to a value of x that is *not* the answer — the angle is 5x − 20, not x.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

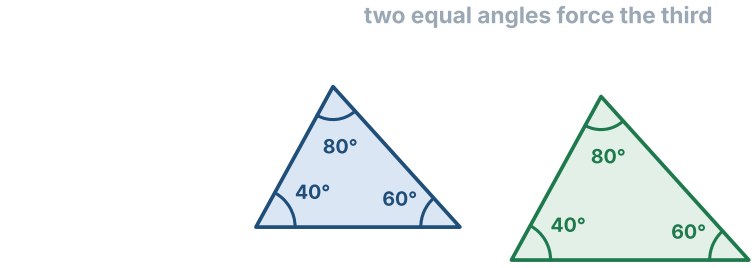
SAT Math 4.2 — Similarity & Congruence

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 When are two triangles similar?

AA — two pairs of equal angles is enough | similar means same shape, different size

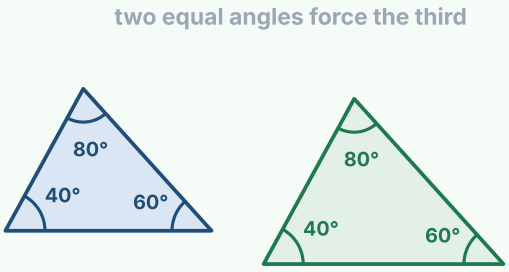
You never need all three angles: if two match, the third must. Similar triangles have equal angles and proportional sides, and the SAT almost always gives you the angles and asks for a side.



Same three angles, different sizes. Nothing here fixes the size, which is exactly why AA proves similarity but never congruence.

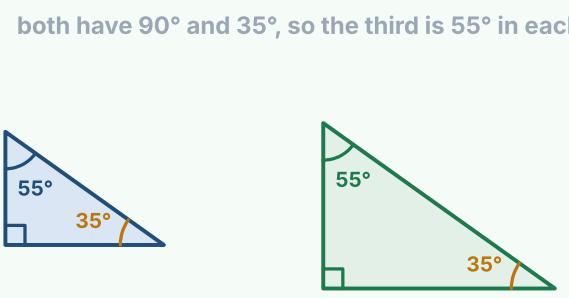
Example 1. **Two triangles have angles 40°, 60° and 40°, 80°. Are they similar?**

Solution.
The first has third angle 80°, the second 60°. Both are 40, 60, 80.
Yes, by AA.



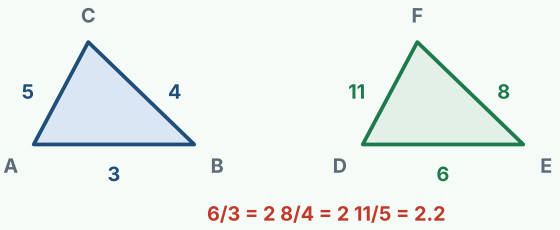
Example 2. **Two right triangles each have an angle of 35°. Are they similar?**

Solution.
Both have 90° and 35°, so the third is 55° in each.
Yes, by AA.



Example 3. **Two triangles have sides 3, 4, 5 and 6, 8, 11. Are they similar?**

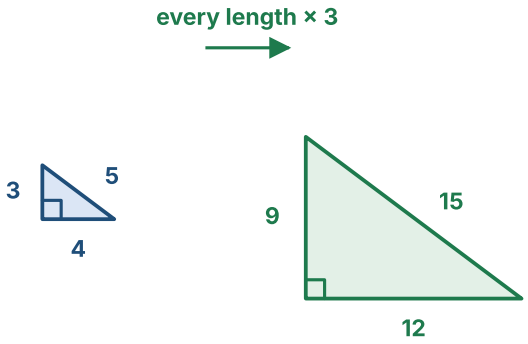
Solution.
6/3 = 2 and 8/4 = 2, but 11/5 = 2.2.
No — the ratios must all match.



2 The scale factor

scale factor k = new ÷ old | every matching length multiplies by k

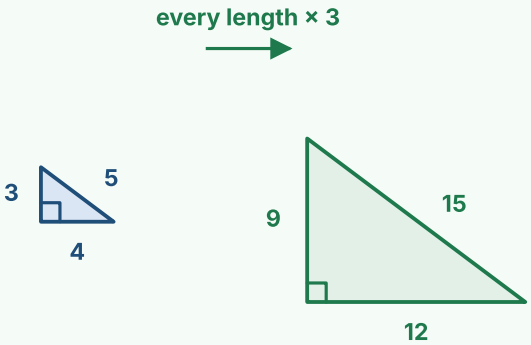
Find k from the one pair of sides you are given, then apply it to the side you want. Pair the sides correctly: the side opposite the 30° angle in one triangle matches the side opposite the 30° angle in the other.



One pair of matching sides gives k = 9/3 = 3. From there every other length follows: 4 → 12 and 5 → 15.

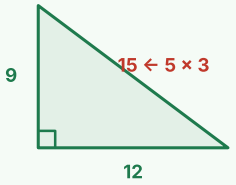
Example 1. **A triangle with sides 3, 4, 5 is enlarged so the shortest side becomes 9. Find the scale factor.**

Solution.
k = 9/3 = **3**.



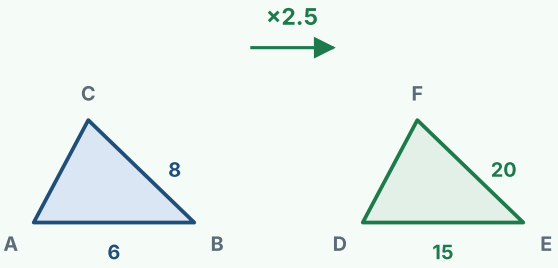
Example 2. **For that enlargement, what is the new longest side?**

Solution.
5 × 3 = **15**.



Example 3. **Triangle ABC has AB = 6; similar triangle DEF has DE = 15. If BC = 8, find EF.**

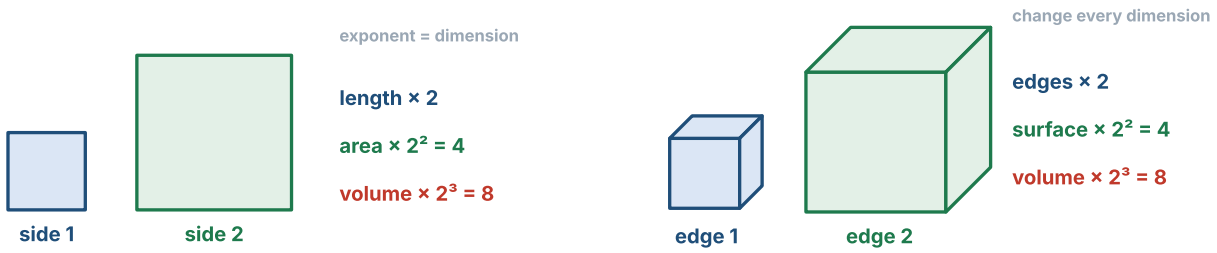
Solution.
k = 15/6 = 2.5.
EF = 8 × 2.5 = **20**.



3 Area and volume scale differently

lengths × k | areas × k^2 | volumes × k^3

This is the trap the SAT returns to. Doubling every length multiplies the area by 4 and the volume by 8. Going the other way, if areas are in the ratio 9 : 4, the lengths are in the ratio 3 : 2 — take the square root.



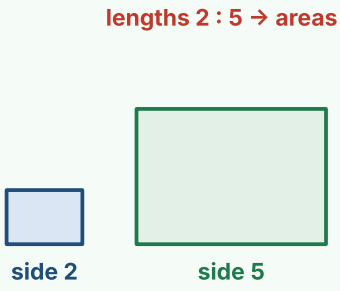
Double the side and four copies of the small square fit inside the big one; eight copies of the small cube fit inside the big cube. The exponent matches the dimension: 1 for length, 2 for area, 3 for volume.

Example 1. **Two similar rectangles have sides in the ratio 2 : 5. What is the ratio of their areas?**

Solution.

Square the factor.

4 : 25.

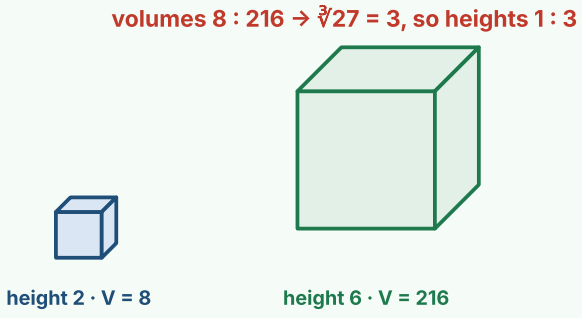


Example 2. **Two similar solids have volumes 8 cm³ and 216 cm³. Find the ratio of their heights.**

Solution.

216/8 = 27, and the cube root of 27 is 3.

1 : 3.

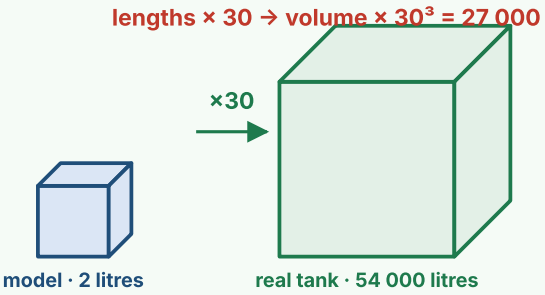


Example 3. **A model is built to scale 1 : 30. The model holds 2 litres. How much does the real tank hold?**

Solution.

Volume scales by $30^3 = 27{,}000$.

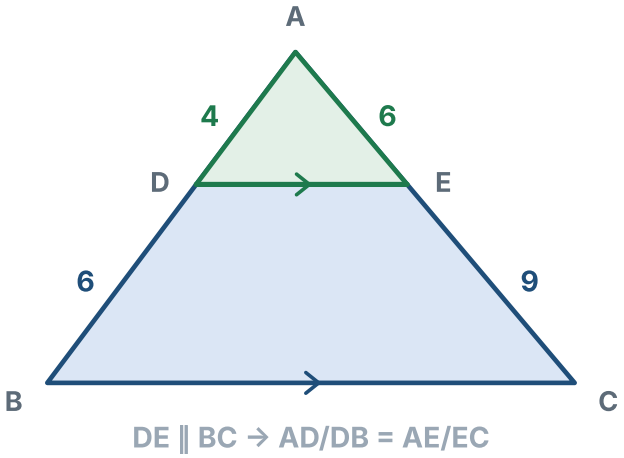
$2 \times 27{,}000 =$ **54,000 litres**.



4 Similar triangles hidden in one figure

a line parallel to one side cuts off a similar triangle

When a figure contains a line parallel to a side, two similar triangles are sitting inside it. Redraw them separately, the same way round, before writing any ratio.



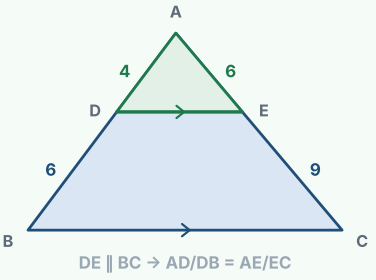
DE ∥ BC makes the angles at D and B equal, and the angle at A is shared, so triangle ADE is similar to triangle ABC. The matching sides give AD/DB = AE/EC.

Example 1. **In triangle ABC, DE is parallel to BC with AD = 4, DB = 6 and AE = 6. Find EC.**

Solution.

AD/DB = AE/EC, so 4/6 = 6/EC.

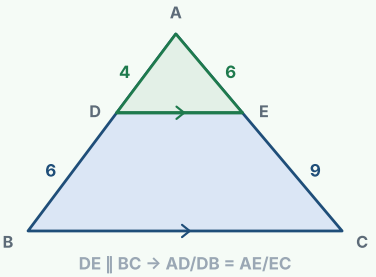
EC = **9**.



Example 2. **Why are the two triangles similar?**

Solution.

DE parallel to BC makes the corresponding angles equal, so **AA** applies.

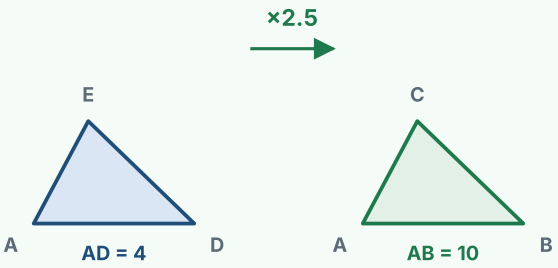


Example 3. **In that figure, AB = 10 and AC = 15. What is the scale factor from ADE to ABC?**

Solution.

AD/AB = 4/10.

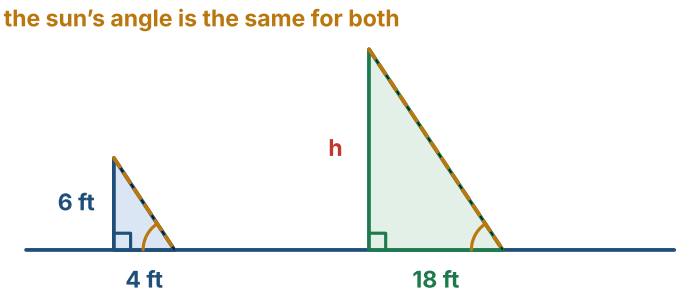
k = **2.5** going from the small triangle to the large one.



5 Shadows, mirrors and indirect measurement

same sun angle → similar triangles → height/shadow is the same ratio for both

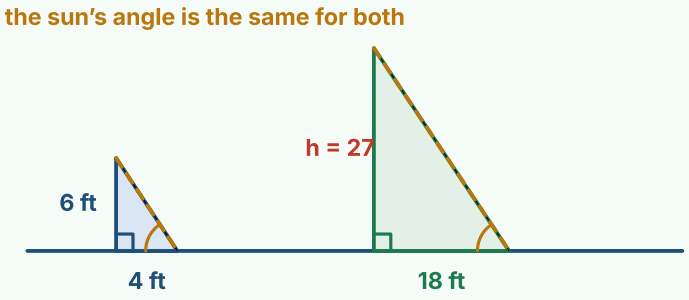
The classic SAT word problem. Write the ratio the same way round for both objects: height ÷ shadow for one equals height ÷ shadow for the other.



Both objects stand upright and both shadows run along the same ground, so the two triangles have the same three angles. 6/4 = h/18 gives h = 27.

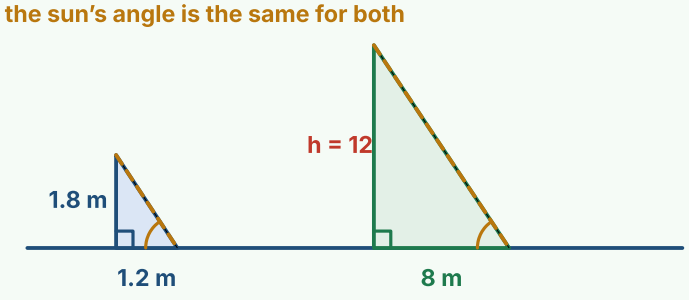
Example 1. **A 6-foot pole casts a 4-foot shadow. A signpost casts an 18-foot shadow. Find its height.**

Solution.
6/4 = h/18, so h = 18 × 1.5 = **27 feet**.



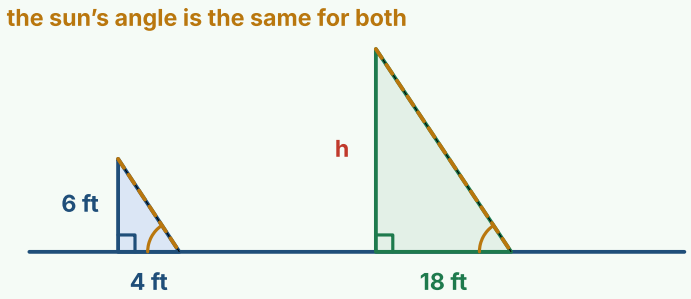
Example 2. **A 1.8 m person casts a 1.2 m shadow. A tree's shadow is 8 m. Find the tree's height.**

Solution.
1.8/1.2 = 1.5, and 8 × 1.5 = **12 m**.



Example 3. **Why must the two shadows be measured at the same time?**

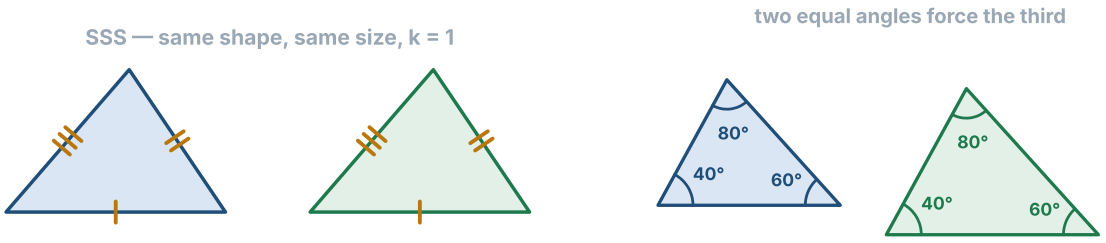
Solution.
The sun's angle changes through the day. Only at the same moment are the two triangles **similar**.



6 Congruence

congruent = same shape and same size, so the scale factor is 1

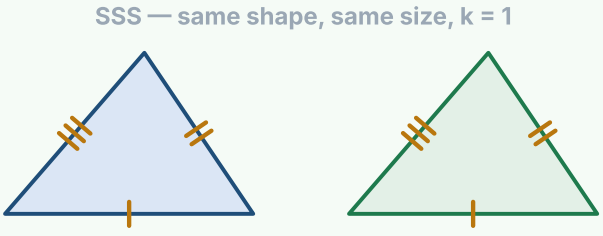
Congruent triangles match exactly: SSS, SAS, ASA or AAS. Note that AAA gives similarity only, never congruence, because the sizes can differ.



On the left the three pairs of sides match, so the triangles are congruent. On the right only the angles match, so they are similar — one is an enlargement of the other.

Example 1. **Two triangles have all three pairs of sides equal. Are they congruent?**

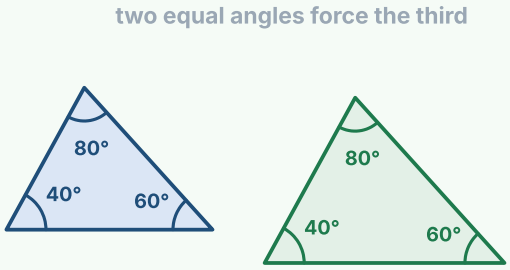
Solution.
Yes, by SSS.



Example 2. **Two triangles have all three pairs of angles equal. Are they congruent?**

Solution.

Not necessarily — they are similar, but one could be an enlargement of the other.

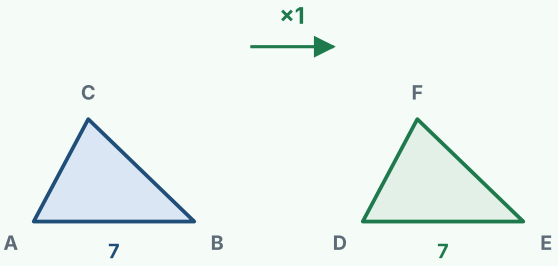


Example 3. **Triangles ABC and DEF are congruent with AB = 7. What is DE?**

Solution.

Congruent means the scale factor is 1.

DE = **7**.



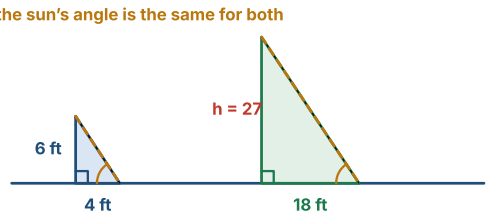
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

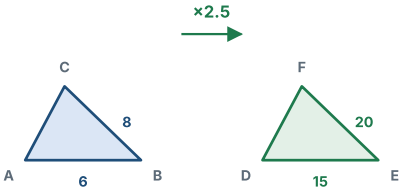
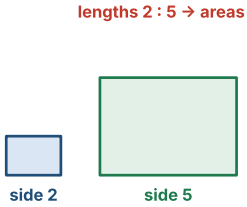
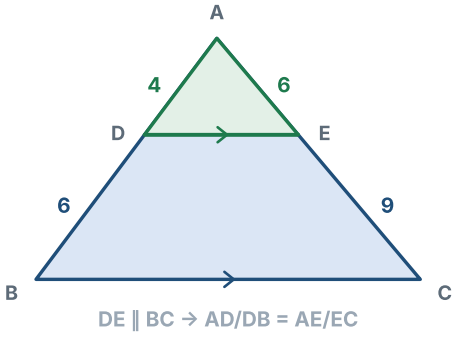
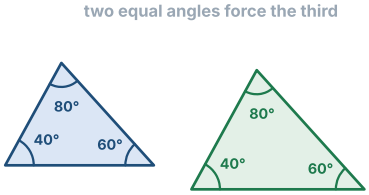
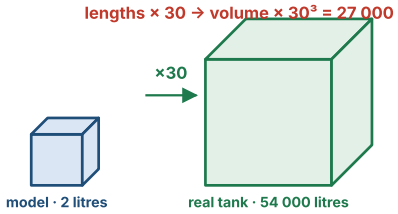
The SAT gives you Desmos on every maths question. Here it is **a proportion solver**. Type the ratio as an equation with the unknown in it and read the answer straight off.

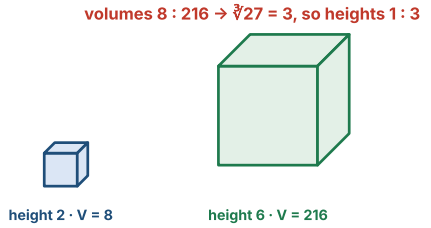
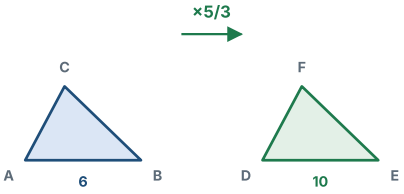
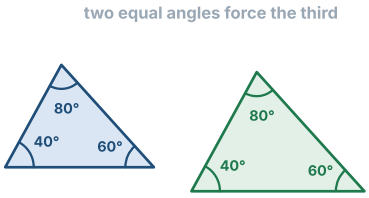
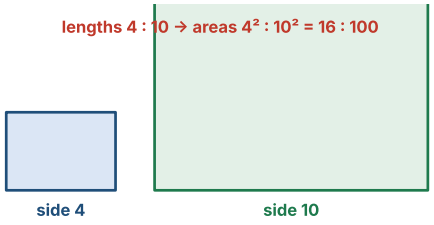
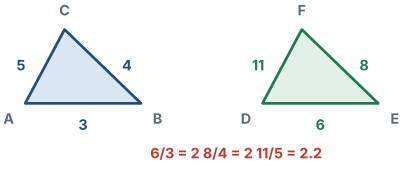
#	Question	Type into Desmos	Then	Answer
D1	A triangle with sides 3, 4, 5 is enlarged so the shortest side becomes 9. What is the new longest side?	`5*(9/3)`	Find the factor, then scale.	15
D2	Triangle ABC has AB = 6; similar triangle DEF has DE = 15. If BC = 8, what is EF?	`8*(15/6)`	The same factor applies to every side.	20
D3	Two similar rectangles have sides in the ratio 2 : 5. What is the ratio of their areas as a decimal?	`(2/5)^2`	Square the length ratio.	0.16
D4	Two similar solids have volumes 8 and 216. What is the ratio of their heights?	`(216/8)^(1/3)`	Cube root of the volume ratio.	3
D5	A model is built to scale 1 : 30 and holds 2 litres. How much does the real tank hold?	`2*30^3`	Volume scales by the cube.	54000
D6	A 6-foot pole casts a 4-foot shadow. A signpost casts an 18-foot shadow. How tall is it?	`18*(6/4)`	Same ratio for both objects.	27
D7	A 1.8 m person casts a 1.2 m shadow. A tree's shadow is 8 m. How tall is the tree?	`8*(1.8/1.2)`	Height over shadow is constant.	12
D8	In triangle ABC, DE is parallel to BC with AD = 4, DB = 6 and AE = 6. What is EC?	`6*(6/4)`	Set up AD/DB = AE/EC.	9
D9	Two similar triangles have areas 9 and 25. What is the ratio of their corresponding sides?	`sqrt(9/25)`	Take the square root of the area ratio.	0.6
D10	A photo 4 cm by 6 cm is enlarged by a factor of 2.5. What is the new area in square centimetres?	`4*6*2.5^2`	Area scales by the square of the factor.	150

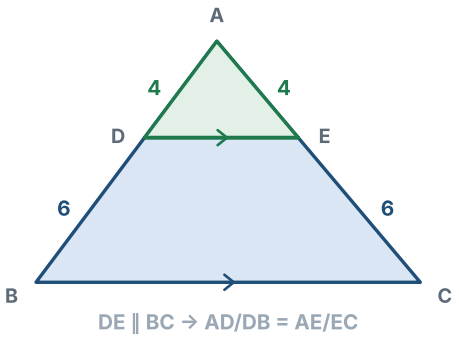
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. For every question, first ask whether the quantity being scaled is a **length**, an **area** or a **volume**. That one decision picks the exponent.

#	Level	Question	Answer	Solution
P1	Hard	Beside a market stall, a vertical pole 6 feet tall casts a shadow 4 feet long. At the same time of day, a signpost casts a shadow 18 feet long. What is the height of the signpost, in feet? 	27	The sun's angle is the same, so the triangles are similar: 6/4 = h/18. h = 18 × 1.5 = **27 feet**. A common wrong answer comes from 18 − 6, not a ratio at all.

#	Level	Question	Answer	Solution
P2	Hard	Triangle ABC is similar to triangle DEF, with AB = 6 and DE = 15. If BC = 8, what is the length of EF? 	20	The scale factor is 15/6 = 2.5, and it applies to every pair of matching sides. EF = 8 × 2.5 = **20**.
P3	Hard	Two similar rectangles have corresponding sides in the ratio 2 : 5. What is the ratio of their areas? 	4 : 25	Area scales by the **square** of the length ratio. $2^2 : 5^2$ = **4 : 25**. A common wrong answer forgets to square; D cubes it, which is the volume rule for solids.
P4	Hard	In triangle ABC, point D lies on AB and point E lies on AC so that DE is parallel to BC. If AD = 4, DB = 6 and AE = 6, what is the length of EC? 	9	DE parallel to BC makes the two triangles similar, so AD/DB = AE/EC. 4/6 = 6/EC gives EC = 36/4 = **9**.
P5	Hard	Two triangles have angles of 40° and 60° in the first, and 40° and 80° in the second. State what is true here. 	They are similar, because two angles of each match.	The first triangle's third angle is 80° and the second's is 60°, so both have angles 40, 60 and 80. **Similar by AA.** Congruence would need the sizes to match too, which nothing here establishes.
P6	Hard	A model of a water tank is built to a scale of 1 : 30. The model holds 2 litres. How many litres does the real tank hold? 	54,000	Volume scales by the **cube** of the length factor: 30^3 = 27,000. 2 × 27,000 = **54,000 litres**. A common wrong answer multiplies by 30 once; B squares it.

#	Level	Question	Answer	Solution
P7	Very hard	Two similar solid figures have volumes of 8 cubic centimetres and 216 cubic centimetres. What is the ratio of their corresponding heights? 	1 : 3	216/8 = 27, and the height ratio is the **cube root** of the volume ratio. ∛27 = 3, so the ratio is **1 : 3**. A common wrong answer uses the volume ratio itself.
P8	Very hard	Two similar triangles have areas of 9 and 25 square units. The shorter triangle has a side of length 6. What is the length of the corresponding side in the larger triangle? 	10	The side ratio is the square root of the area ratio: √(25/9) = 5/3. 6 × 5/3 = **10**.
P9	Very hard	Triangles ABC and DEF have all three pairs of angles equal. Which conclusion is guaranteed? 	The triangles are similar.	Equal angles guarantee the same **shape** but say nothing about size. **Similar**, and possibly congruent, but AAA alone never proves congruence — one could be an enlargement of the other.
P10	Very hard	A rectangular photograph measuring 4 cm by 6 cm is enlarged by a scale factor of 2.5. What is the area of the enlarged photograph, in square centimetres? 	150	Original area 24 cm²; area scales by 2.5² = 6.25. 24 × 6.25 = **150**. A common wrong answer multiplies the area by 2.5 once, the commonest error here.
P11	Very hard	Two triangles have sides 3, 4, 5 and 6, 8, 11 respectively. Are they similar, and why? 	No, because the three ratios are not all equal.	6/3 = 2 and 8/4 = 2, but 11/5 = 2.2. Similarity needs **all** matching ratios to agree, so they are not similar. A common wrong answer is true but irrelevant — similar triangles normally do have different perimeters.

#	Level	Question	Answer	Solution
P12	Very hard	In a figure, triangle ADE sits inside triangle ABC with DE parallel to BC. AD = 4, AB = 10 and the area of triangle ADE is 12 square units. What is the area of triangle ABC, in square units?  DE ∥ BC → AD/DB = AE/EC	75	The length factor is AB/AD = 10/4 = 2.5, so the area factor is $2.5^2 = 6.25$. 12 × 6.25 = **75**. A common wrong answer applies the length factor 2.5 to the area directly instead of squaring it.

The one habit to build. Areas scale by k squared and volumes by k cubed. Your class scored 47% here, and almost every lost mark was a length factor applied to an area or a volume.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

SAT Math 4.3 — Right Triangles & Trigonometry

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

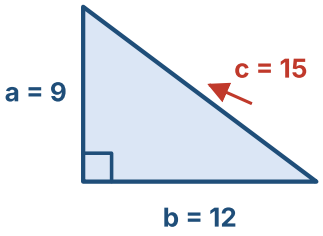
1 Pythagoras

$a^2 + b^2 = c^2$, where c is the hypotenuse — the side opposite the right angle

Identify the hypotenuse before substituting. It is always the longest side and always opposite the right angle. Putting a leg where c belongs is the error that costs this topic its marks.

$$a^2 + b^2 = c^2$$

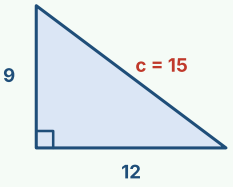
$$81 + 144 = 225$$



The arrow points from the right angle to the side facing it: that side is c. Here 81 + 144 = 225, so c = 15.

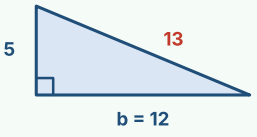
Example 1. **A right triangle has legs 9 and 12. Find the hypotenuse.**

Solution.
81 + 144 = 225.
c = √225 = **15**.



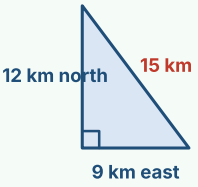
Example 2. **A right triangle has hypotenuse 13 and one leg 5. Find the other leg.**

Solution.
Here 13 is c, so 25 + b^2 = 169.
b = √144 = **12**.



Example 3. **A drone flies 9 km east then 12 km north. How far is it from the start?**

Solution.
The path makes a right angle.
√(81 + 144) = **15 km**.

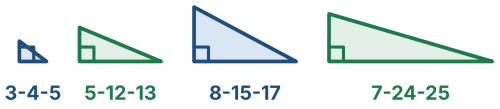


2 The triples worth memorising

3-4-5 | 5-12-13 | 8-15-17 | 7-24-25 — and all their multiples

Recognising a triple turns a calculation into a glance. 9-12-15 is 3-4-5 tripled; 10-24-26 is 5-12-13 doubled. If two sides fit a triple, the third is immediate.

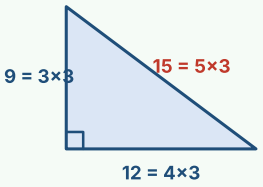
and every multiple: 6-8-10, 9-12-15, 10-24-26 ...



Four right triangles whose sides are whole numbers. Spotting one saves squaring, adding and taking a root.

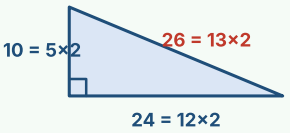
Example 1. **A right triangle has legs 9 and 12. Which triple is this?**

Solution.
3-4-5 multiplied by 3.
The hypotenuse is 5 × 3 = **15**.



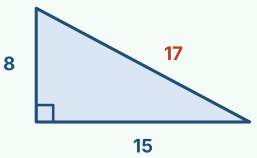
Example 2. **A right triangle has hypotenuse 26 and one leg 10. Find the other leg.**

Solution.
5-12-13 doubled.
The other leg is 12 × 2 = **24**.



Example 3. **Is a triangle with sides 8, 15, 17 right-angled?**

Solution.
$64 + 225 = 289 = 17^2$.
Yes — the converse of Pythagoras.

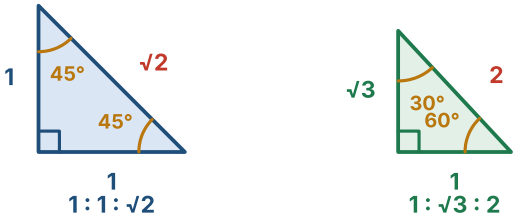


3 Special right triangles

45-45-90 → sides 1 : 1 : √2 | 30-60-90 → sides 1 : √3 : 2

These two appear constantly and are on the reference sheet. In a 30-60-90 the shortest side faces the 30° angle and is half the hypotenuse; the √3 side faces the 60°.

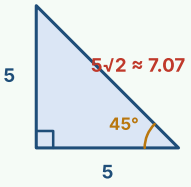
the short side faces 30° — half the hypotenuse



Memorise which side faces which angle. The hypotenuse is always the largest number in the ratio — √2 on the left, 2 on the right.

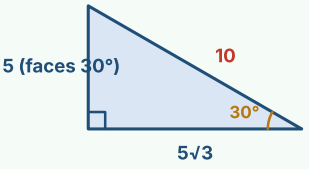
Example 1. **A 45-45-90 triangle has legs of 5. Find the hypotenuse.**

Solution.
5 × √2 = **5√2** ≈ 7.07.



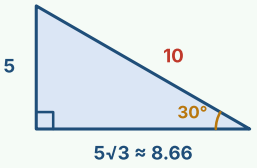
Example 2. **A 30-60-90 triangle has hypotenuse 10. Find the shortest side.**

Solution.
The side opposite 30° is half the hypotenuse.
5.



Example 3. **For that triangle, find the remaining side.**

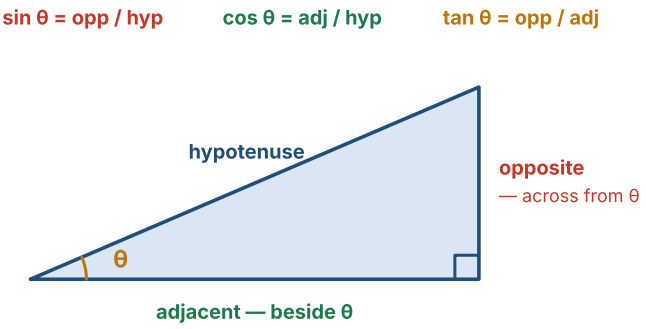
Solution.
It faces the 60° angle: 5√3 ≈ **8.66**.



4 SOH-CAH-TOA

sin = opp/hyp | cos = adj/hyp | tan = opp/adj

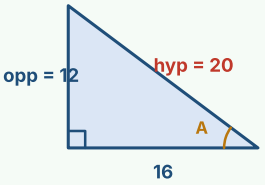
Label the three sides relative to the angle you are working with before choosing a ratio. Opposite and adjacent swap when you switch angles; the hypotenuse never moves.



Stand at θ: the side across the triangle is opposite, the side you are standing on is adjacent, and the long side facing the right angle is the hypotenuse.

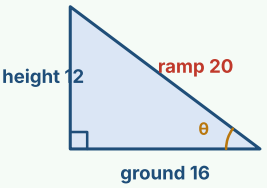
Example 1. **In a right triangle, sin A = 0.6 and the hypotenuse is 20. Find the side opposite A.**

Solution.
sin A = opp/hyp, so opp = 0.6 × 20 = **12**.



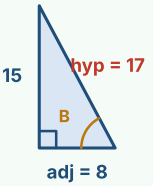
Example 2. **A ramp makes an angle whose sine is 0.6 and is 20 feet long. How high is the top?**

Solution.
The ramp is the hypotenuse.
0.6 × 20 = **12 feet**.



Example 3. **In a right triangle, the side adjacent to angle B is 8 and the hypotenuse is 17. Find cos B.**

Solution.
cos B = adj/hyp = **8/17** ≈ 0.471.

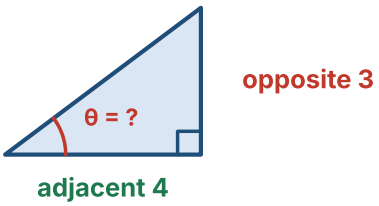


5 Finding an angle

use inverse trig: if $\tan\theta = 3/4$ then $\theta = \tan^{-1}(3/4)$

When the sides are known and the angle is not, the inverse functions give it. In Desmos type arctan, arcsin or arccos and make sure the calculator is in degrees.

$$\tan\theta = 3/4 \rightarrow \theta = \tan^{-1}(0.75) \approx 37°$$

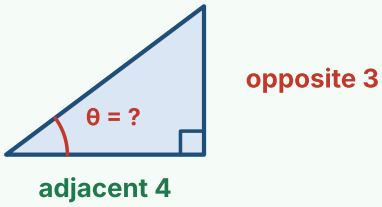


Two sides are known, the angle is not — so run the ratio backwards with the inverse function, and check the calculator says degrees.

Example 1. **A right triangle has opposite 3 and adjacent 4. Find the angle, to the nearest degree.**

Solution.
$\tan\theta = 0.75$, so $\theta = \tan^{-1}(0.75) \approx$ **37°**.

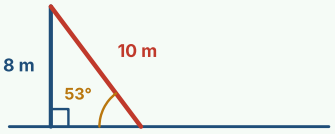
$$\tan\theta = 3/4 \rightarrow \theta = \tan^{-1}(0.75) \approx 37°$$



Example 2. **A ladder 10 m long reaches 8 m up a wall. What angle does it make with the ground?**

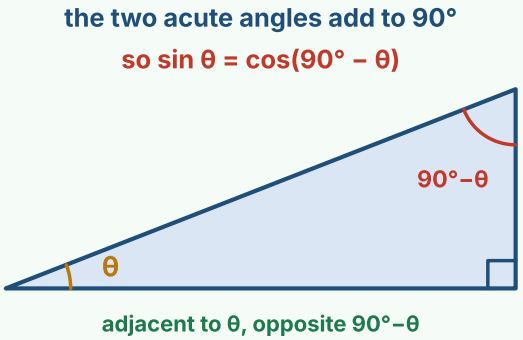
Solution.
$\sin\theta = 8/10 = 0.8$.
$\theta = \sin^{-1}(0.8) \approx$ **53°**.

$\sin\theta = 8/10 \rightarrow \theta \approx 53°$



Example 3. **Why do those two answers add to about 90°?**

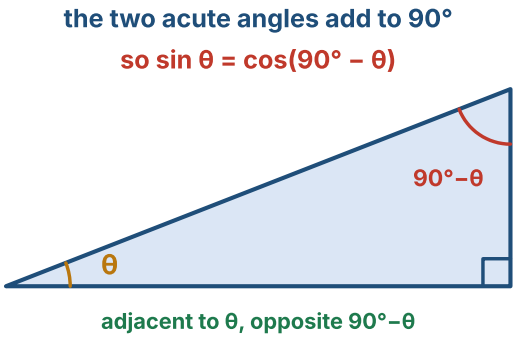
Solution.
They are the **two acute angles** of similar right triangles — 3-4-5 in both cases.



6 sin θ = cos(90° − θ)

the sine of an angle equals the cosine of its complement

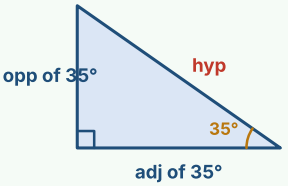
In a right triangle the two acute angles add to 90°, and what is opposite one is adjacent to the other. So sin and cos swap. Questions set sin(expression) = cos(expression) and ask you to make the two angles add to 90.



The side marked opposite for θ is the side adjacent for 90° − θ. Same two lengths, same fraction — only the name of the ratio changes.

Example 1. **If sin 35° = cos x° and x is acute, find x.**

Solution.
x = 90 − 35 = **55**.

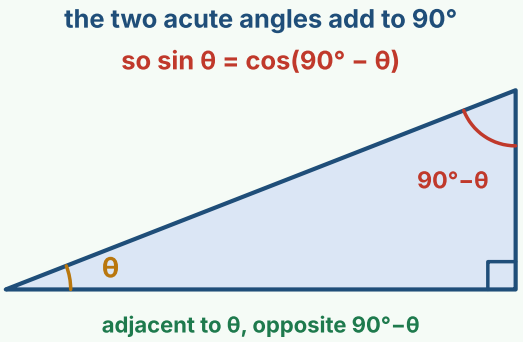


Example 2. **In a right triangle the acute angles are (2x + 10)° and (3x − 20)°. Find x.**

Solution.

They are complementary: 2x + 10 + 3x − 20 = 90, so 5x = 100.

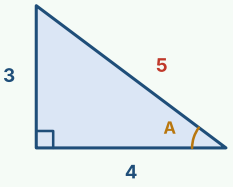
x = **20**.



Example 3. **If sin A = 3/5 for an acute angle A, what is cos(90° − A)?**

Solution.

3/5 — it is the same quantity, by the identity.



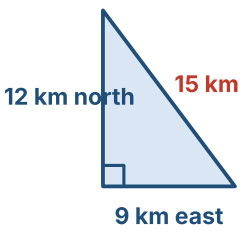
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

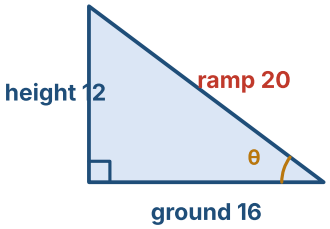
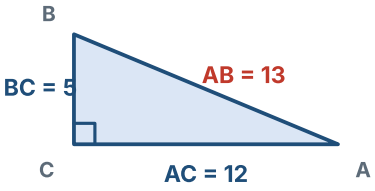
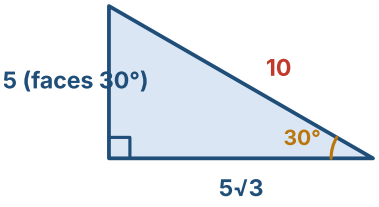
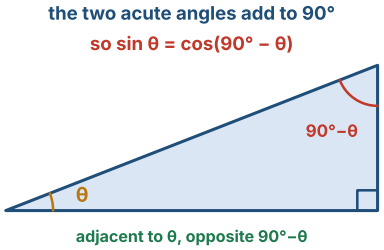
The SAT gives you Desmos on every maths question. Here it is **a trig calculator**. Use arcsin, arccos and arctan for angles, and check the setting is **degrees** before you trust anything.

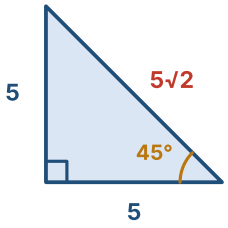
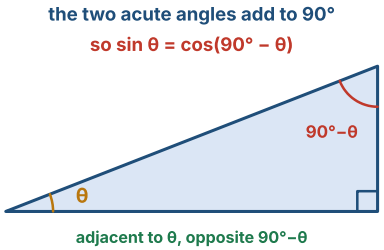
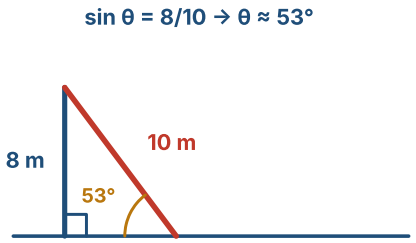
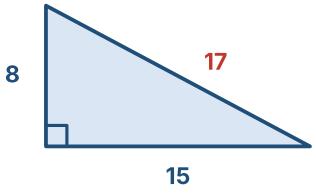
#	Question	Type into Desmos	Then	Answer
D1	A right triangle has legs 9 and 12. What is the hypotenuse?	`sqrt(9^2+12^2)`	Square, add, root — in one line.	15
D2	A right triangle has hypotenuse 13 and one leg 5. What is the other leg?	`sqrt(13^2-5^2)`	Subtract when you are given the hypotenuse.	12
D3	A 45-45-90 triangle has legs of 5. What is the hypotenuse, to two decimal places?	`5*sqrt(2)`	Desmos keeps the exact form too.	7.07
D4	A 30-60-90 triangle has hypotenuse 10. What is the side opposite 60°, to two decimals?	`5*sqrt(3)`	Half the hypotenuse, times √3.	8.66
D5	If sin A = 0.6 and the hypotenuse is 20, what is the opposite side?	`0.6*20`	sin = opp/hyp rearranged.	12
D6	A right triangle has opposite 3 and adjacent 4. What is the angle in degrees, to the nearest degree?	`arctan(3/4)`	Check Desmos is set to degrees.	37
D7	A ladder 10 m long reaches 8 m up a wall. What angle does it make with the ground?	`arcsin(8/10)`	Inverse sine.	53
D8	If sin 35° = cos x°, what is x?	`90-35`	Complementary angles.	55
D9	A right triangle has sides 8 and 17 as leg and hypotenuse. What is cos of that angle, to three decimals?	`8/17`	Adjacent over hypotenuse.	0.471
D10	In a right triangle the acute angles are (2x + 10)° and (3x − 20)°. What is x?	`2x+10+3x-20=90`	The two acute angles total 90.	20

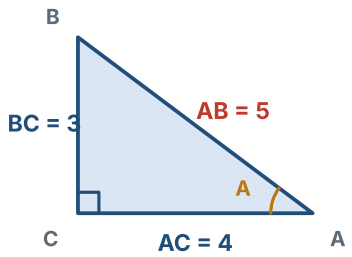
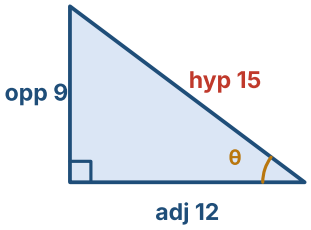
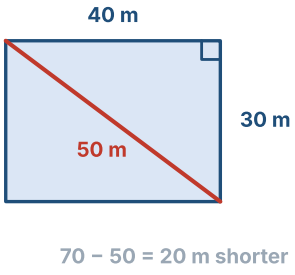
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Label opposite, adjacent and hypotenuse relative to the angle in the question *before* choosing a ratio. Opposite and adjacent swap when the angle does.

#	Level	Question	Answer	Solution
P1	Hard	A drone takes off from a stadium, flies 9 kilometres due east, then 12 kilometres due north. What is the straight-line distance, in kilometres, between the drone and the stadium?  	15	East and north are perpendicular, so this is a right triangle. √(81 + 144) = √225 = **15 km**. It is 3-4-5 tripled.

#	Level	Question	Answer	Solution
P2	Hard	A straight loading ramp forms a right triangle with the level ground. The sine of the angle between the ramp and the ground is 0.6, and the ramp is 20 feet long. How many feet above the ground is the top of the ramp? 	12	The ramp is the hypotenuse, and sin = opp/hyp. 0.6 × 20 = **12 feet**. A common wrong answer divides 20 by 0.6 × 2 or similar — the clue is that the opposite side must be shorter than the hypotenuse.
P3	Hard	In right triangle ABC, the right angle is at C and the hypotenuse AB = 13. If BC = 5, what is the length of AC? 	12	AB is the hypotenuse, so $5^2 + AC^2 = 13^2$. AC = √144 = **12**. This is the 5-12-13 triple.
P4	Hard	A 30-60-90 triangle has a hypotenuse of length 10. What is the length of the side opposite the 30° angle? 	5	In a 30-60-90 triangle the side opposite 30° is **half the hypotenuse**. 10/2 = **5**. A common wrong answer is the side opposite 60°.
P5	Hard	If sin 35° = cos x° and 0 < x < 90, what is the value of x? 	55	sin θ = cos(90° − θ). x = 90 − 35 = **55**.

#	Level	Question	Answer	Solution
P6	Hard	A 45-45-90 triangle has legs of length 5. What is the length of the hypotenuse? 	5√2	In a 45-45-90 triangle the sides are in the ratio 1 : 1 : √2. The hypotenuse is **5√2** ≈ 7.07. A common wrong answer doubles the leg instead.
P7	Very hard	In a right-triangle design on a banner, the two acute angles measure (2x + 10)° and (3x − 20)°, so sin((2x + 10)°) = cos((3x − 20)°). What is the value of x? 	20	The identity holds exactly when the two angles are complementary. 2x + 10 + 3x − 20 = 90 gives 5x = 100, so x = **20**. The angles are 50° and 40°.
P8	Very hard	A ladder 10 metres long leans against a wall and reaches 8 metres up it. What angle, to the nearest degree, does the ladder make with the ground? 	53°	The wall height is opposite the angle and the ladder is the hypotenuse: sin θ = 0.8. θ = $\sin^{-1}(0.8)$ ≈ **53°**. A common wrong answer is the angle at the *top* of the ladder, the complement.
P9	Very hard	A right triangle has legs of length 8 and 15. What is the length of the hypotenuse? 	17	64 + 225 = 289, and √289 = **17** — the 8-15-17 triple. A common wrong answer comes from subtracting instead of adding the squares.

#	Level	Question	Answer	Solution
P10	Very hard	In right triangle ABC the right angle is at C, and sin A = 3/5. What is the value of cos B? 	3/5	Angles A and B are complementary, and sin A = cos(90° − A) = cos B. cos B = **3/5**. A common wrong answer is cos A, which answers a different question.
P11	Very hard	A right triangle has an angle whose tangent is 3/4. If the side adjacent to that angle is 12, what is the length of the hypotenuse? 	15	tan = opp/adj = 3/4, so with adj = 12 the opposite is 9. This is 9-12-15, which is 3-4-5 tripled, so the hypotenuse is **15**. A common wrong answer is the opposite side, not the hypotenuse.
P12	Very hard	A rectangular field measures 40 metres by 30 metres. A path runs diagonally from one corner to the opposite corner. How much shorter, in metres, is the diagonal path than walking along two sides? 	20	The diagonal is √(1600 + 900) = √2500 = 50 m, and two sides total 70 m. 70 − 50 = **20 metres shorter**.

The one habit to build. Find the hypotenuse first and write it down. Your class scored 20% on Pythagoras, and the usual cause is putting a leg where c belongs — the hypotenuse is always opposite the right angle and always the longest side.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

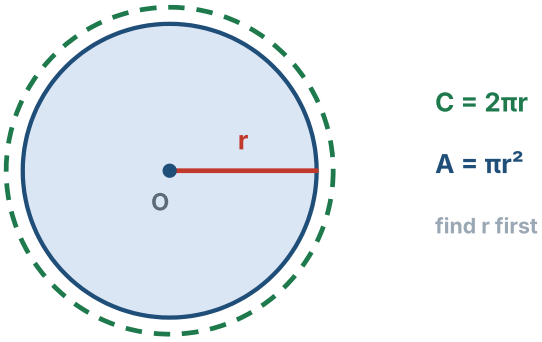
SAT Math 4.4 — Circles

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Circumference and area

$C = 2\pi r$ | $A = \pi r^2$ — both are on the reference sheet

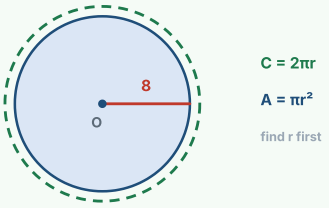
Almost every circle question starts by finding r. If you are given the area, take the square root; if the circumference, divide by 2π. Then answer what was asked.



The radius is the one number that unlocks both formulas: the green ring is the circumference and the shaded disc is the area.

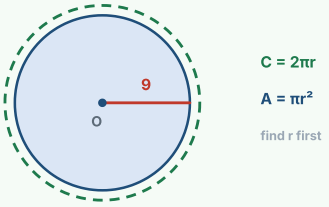
Example 1. **A circle has area 64π. Find its circumference.**

Solution.
$\pi r^2 = 64\pi$ gives $r = 8$.
$C = 2\pi(8) =$ **16π**.



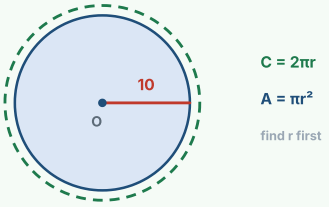
Example 2. **A circle has circumference 18π. Find its area.**

Solution.
$2\pi r = 18\pi$ gives $r = 9$.
$A = \pi(81) =$ **81π**.



Example 3. **A circular path encloses 100π square metres. How long is the path?**

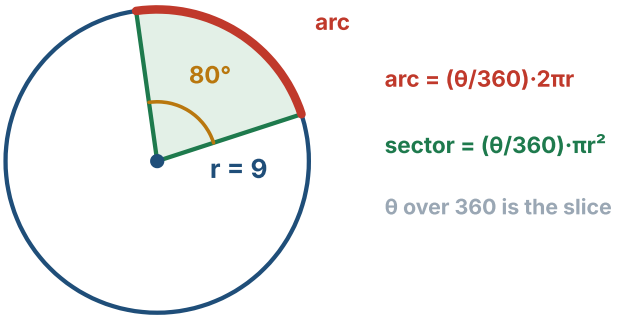
Solution.
$r = 10$, so the path is the circumference.
$2\pi(10) =$ **20π** metres.



2 Arcs and sectors

arc = (θ/360) × 2πr | sector area = (θ/360) × πr²

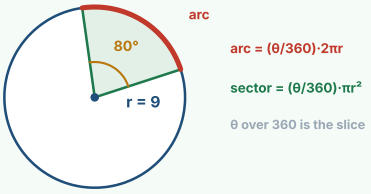
A sector is a fraction of the whole circle, and the fraction is the angle over 360. Work out the whole first, then take the slice.



The central angle picks the fraction: 80/360 = 2/9. Multiply the whole circumference by it for the arc, or the whole area for the sector.

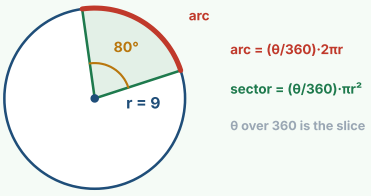
Example 1. **A circle of radius 9 has a sector with central angle 80°. Find its area.**

Solution.
Whole area = 81π, and 80/360 = 2/9.
(2/9)(81π) = **18π**.



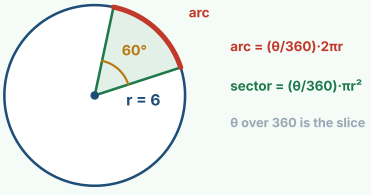
Example 2. **For that same circle, find the arc length of the sector.**

Solution.
Whole circumference = 18π.
(2/9)(18π) = **4π**.



Example 3. **A sector of a circle of radius 6 has area 6π. Find its central angle.**

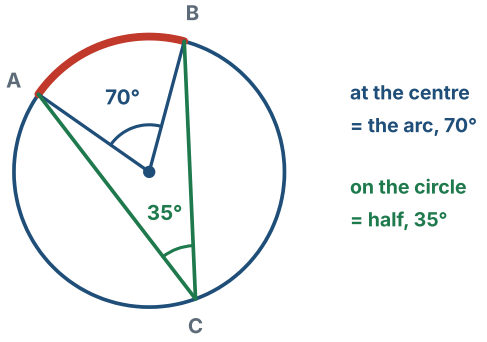
Solution.
Whole area 36π, and 6π/36π = 1/6.
(1/6)(360) = **60°**.

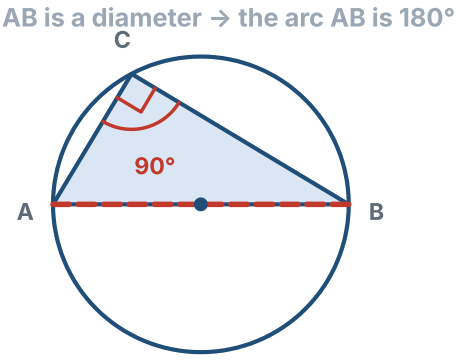


3 Central and inscribed angles

a central angle equals its arc | an inscribed angle is half its arc

The vertex position decides the rule: at the centre, the angle equals the arc; on the circle, it is half. The special case worth knowing cold is that an angle inscribed in a semicircle is a right angle.

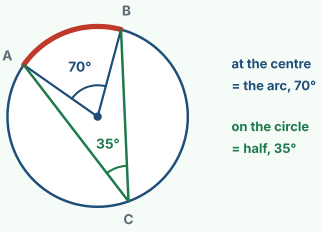




Both angles stand on the same red arc. The one at the centre is 70°; the one on the circle is 35°. When the arc is a semicircle (180°), half of it is a right angle.

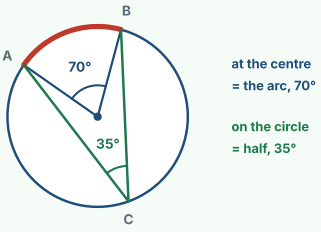
Example 1. **A central angle cuts an arc of 70°. What is the central angle?**

Solution.
70° — they are equal.



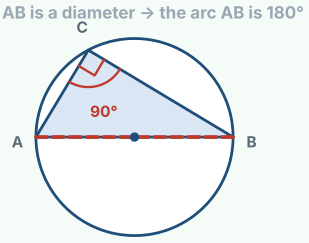
Example 2. **An inscribed angle cuts that same 70° arc. What is the inscribed angle?**

Solution.
Half the arc.
35°.



Example 3. **Triangle ABC is inscribed in a circle with AB a diameter. What is angle ACB?**

Solution.
The arc AB is 180°, and the inscribed angle is half.
90°.



4 Tangents and chords

a tangent is perpendicular to the radius at the point of contact

A tangent gives you a right angle for free, which usually means Pythagoras is next. A radius drawn perpendicular to a chord also bisects it.



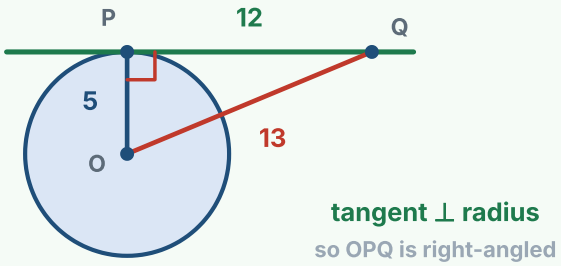
Left: the radius OP meets the tangent at 90°, so OPQ is a right triangle with legs 5 and 12. Right: the perpendicular from the centre cuts the chord in half, giving legs 8 and d with hypotenuse 10.

Example 1. **A tangent touches a circle of radius 5 at P. From an external point Q, PQ = 12. Find OQ.**

Solution.

Angle OPQ is 90°, so $OQ^2 = 25 + 144$.

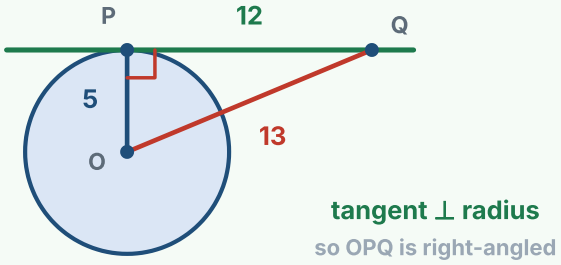
OQ = **13**.



Example 2. **Why is triangle OPQ right-angled?**

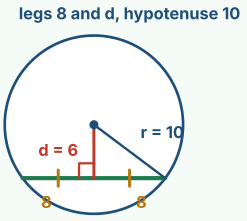
Solution.

OP is a radius to the point of contact, and a **tangent is perpendicular to the radius** there.



Example 3. **A chord of length 16 sits in a circle of radius 10. How far is it from the centre?**

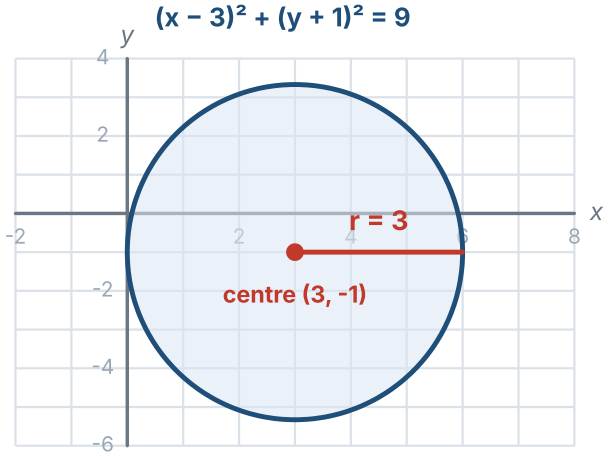
Solution.
The perpendicular from the centre bisects the chord, giving legs 8 and d with hypotenuse 10.
d = √(100 − 64) = **6**.



5 The equation of a circle

$(x - h)^2 + (y - k)^2 = r^2$, centre (h, k), radius r

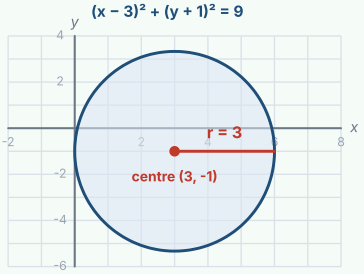
Read the centre with the signs flipped and remember the right-hand side is r squared, not r. This formula is *not* on the reference sheet.



$(x - 3)^2 + (y + 1)^2 = 9$ sits at (3, −1): the x term reads −3 so h = +3, the y term reads +1 so k = −1. And 9 is r^2, so r = 3.

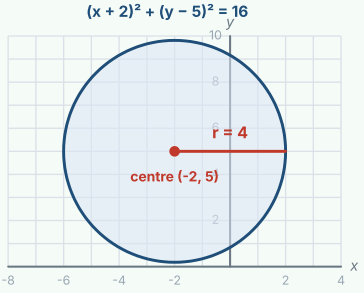
Example 1. **Find the centre and radius of $(x - 3)^2 + (y + 1)^2 = 9$.**

Solution.
Centre **(3, −1)**; radius **3**, because $9 = 3^2$.



Example 2. **Write the equation of a circle with centre (−2, 5) and radius 4.**

Solution.
$(x + 2)^2 + (y - 5)^2 = 16$.

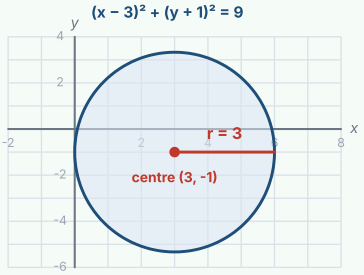


Example 3. **Does the point (6, −1) lie on the circle in the first example?**

Solution.

$(6 - 3)^2 + (-1 + 1)^2 = 9 + 0 = 9$.

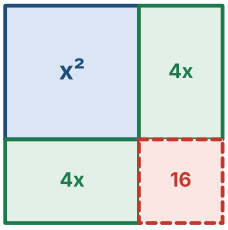
Yes, it is exactly on the circle.



6 Completing the square for a circle

group the x terms and the y terms, then halve and square each coefficient

An equation given as $x^2 + y^2 + Dx + Ey = F$ has to be rearranged before you can read anything off. Complete the square twice, once for x and once for y.



$x^2 + 8x$

$= (x + 4)^2 - 16$

halve 8, square it, then subtract it back

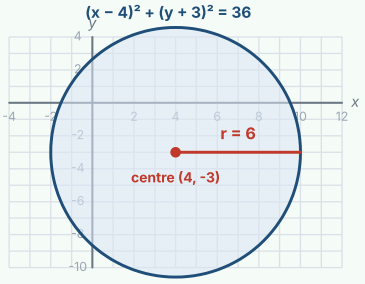
The square of side x plus the two strips of width 4 almost make a square of side x + 4. The dashed 16 is the missing corner — add it to both sides and the left-hand side becomes $(x + 4)^2$.

Example 1. **Find the radius of $x^2 + y^2 - 8x + 6y = 11$.**

Solution.

$(x - 4)^2 - 16 + (y + 3)^2 - 9 = 11$.

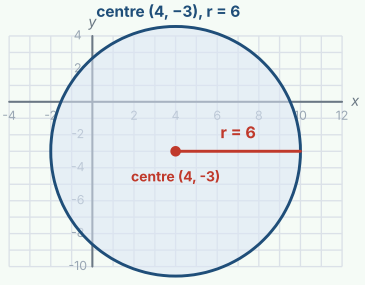
$(x - 4)^2 + (y + 3)^2 = 36$, so r = **6**.



Example 2. **Find its centre.**

Solution.

(4, −3) — the signs flip.

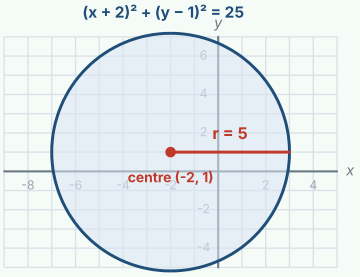


Example 3. **Find the radius of $x^2 + y^2 + 4x - 2y = 20$.**

Solution.

$(x + 2)^2 + (y - 1)^2 = 20 + 4 + 1 = 25$.

$r = \mathbf{5}$.



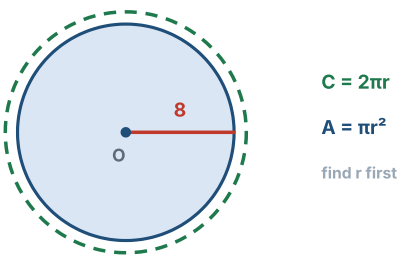
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

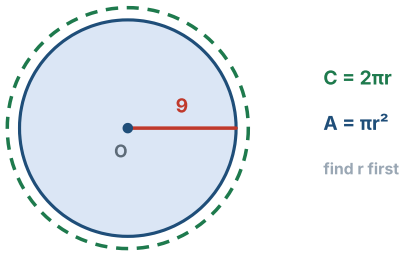
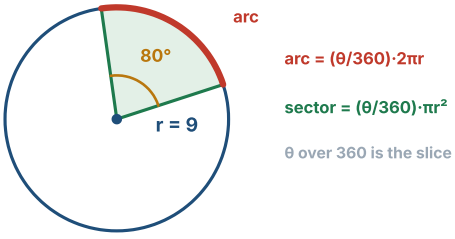
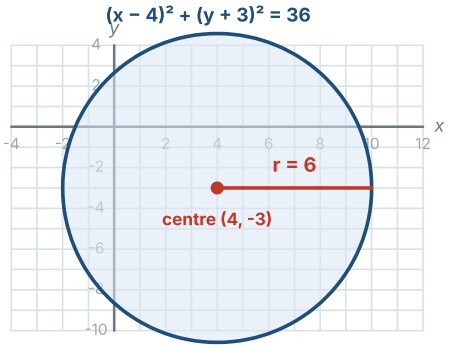
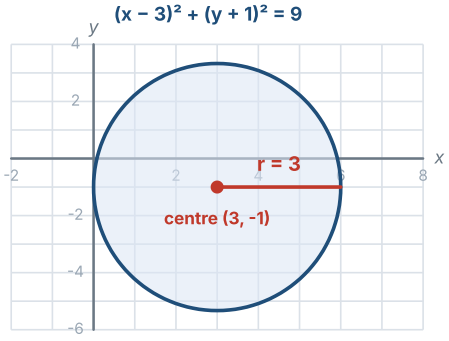
The SAT gives you Desmos on every maths question. Here it is **a circle grapher**. Type the equation in any form and Desmos draws it, centre and radius included — a fast check on your completing the square.

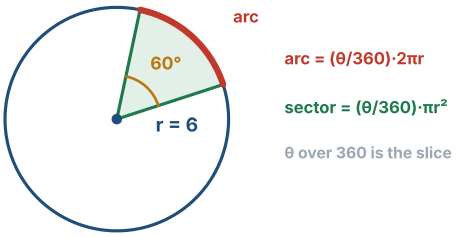
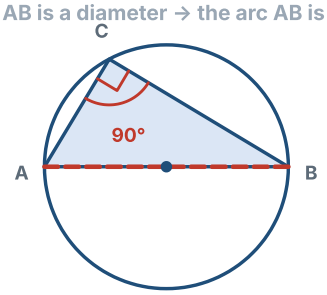
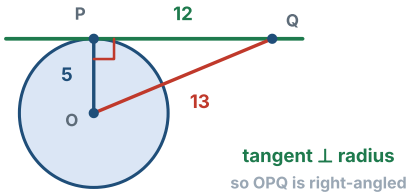
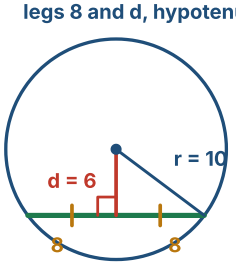
#	Question	Type into Desmos	Then	Answer
D1	A circle has area 64π. What is its circumference as a multiple of π?	`2*sqrt(64)`	Radius first, then the circumference.	16
D2	A circle has circumference 18π. What is its area as a multiple of π?	`(18/2)^2`	r = 9, then r squared.	81
D3	A sector of a circle of radius 9 has central angle 80°. What is its area as a multiple of π?	`80/360*81`	Fraction of the whole.	18
D4	For that same sector, what is the arc length as a multiple of π?	`80/360*18`	Same fraction, applied to the circumference.	4
D5	A sector of a circle of radius 6 has area 6π. What is its central angle in degrees?	`6/36*360`	The ratio of areas gives the ratio of angles.	60
D6	A tangent meets a circle of radius 5 at P, and PQ = 12. What is OQ?	`sqrt(5^2+12^2)`	The tangent makes a right angle with the radius.	13
D7	A chord of length 16 is in a circle of radius 10. How far is it from the centre?	`sqrt(10^2-8^2)`	Half the chord is one leg.	6
D8	What is the radius of $x^2 + y^2 - 8x + 6y = 11$?	`sqrt(11+16+9)`	Add the completing-the-square constants to both sides.	6
D9	What is the radius of $x^2 + y^2 + 4x - 2y = 20$?	`sqrt(20+4+1)`	Same method, different numbers.	5
D10	An inscribed angle cuts an arc of 70°. What is the inscribed angle?	`70/2`	Half the arc.	35

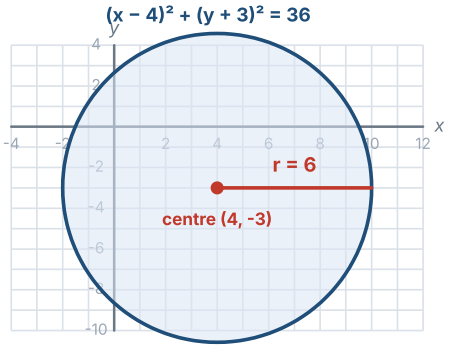
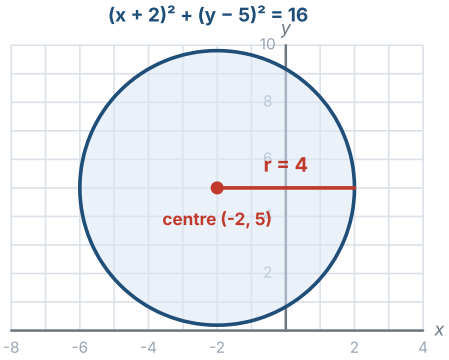
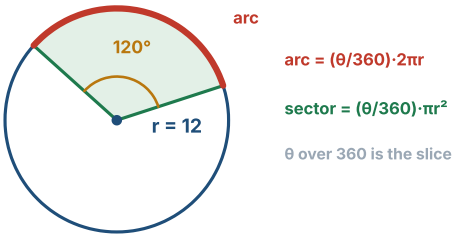
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Find the radius first in almost every question, then answer what was actually asked: circumference, area, arc or sector.

#	Level	Question	Answer	Solution
P1	Hard	The area enclosed by a circular jogging path is 64π square metres. What is the length of the path, in metres? 	16π	$\pi r^2 = 64\pi$ gives r = 8, and the path is the circumference. $2\pi(8) = \mathbf{16\pi}$. A common wrong answer is the radius times π, a common slip.

#	Level	Question	Answer	Solution
P2	Hard	A round cheese wheel has a circumference of 18π centimetres. What is the area, in square centimetres, of its circular top face? 	81π	2πr = 18π gives r = 9. A = π(9)² = **81π**. A common wrong answer squares the diameter instead of the radius.
P3	Hard	A wedge cut from a circular cheese of radius 9 inches has a central angle of 80°. What is the area, in square inches, of the top face of the wedge? 	18π	The whole area is 81π and the wedge is 80/360 = 2/9 of it. (2/9)(81π) = **18π**.
P4	Hard	What is the radius of the circle given by $x^2 + y^2 - 8x + 6y = 11$? 	6	Complete the square twice: $(x - 4)^2 + (y + 3)^2 = 11 + 16 + 9 = 36$. r = √36 = **6**. Answering 36 gives r^2, not r.
P5	Hard	A circle in the xy-plane has equation $(x - 3)^2 + (y + 1)^2 = 9$. What are the coordinates of its centre? 	(3, −1)	The form is $(x - h)^2 + (y - k)^2 = r^2$, so h = 3 and k = −1. **(3, −1)** — the signs flip. A common wrong answer flips them the wrong way.

#	Level	Question	Answer	Solution
P6	Hard	A sector of a circle of radius 6 has an area of 6π. What is its central angle, in degrees? 	60	The whole circle has area 36π, and 6π/36π = 1/6. (1/6)(360) = **60°**.
P7	Very hard	In a circle with centre O, AB is a diameter and C is a point on the circle distinct from A and B. What is the measure of angle ACB, in degrees? 	90	Angle ACB is inscribed in a semicircle, so it cuts an arc of 180° and is half of it. **90°** wherever C sits — which is exactly why any other answer is wrong.
P8	Very hard	A tangent touches a circle with centre O and radius 5 at point P. An external point Q lies on the tangent with PQ = 12. What is the length of OQ? 	13	The tangent is perpendicular to the radius OP, so triangle OPQ has a right angle at P. OQ = √(25 + 144) = **13**, the 5-12-13 triple.
P9	Very hard	A chord of length 16 is drawn in a circle of radius 10. What is the distance from the centre of the circle to the chord? 	6	The perpendicular from the centre bisects the chord, giving a right triangle with hypotenuse 10 and one leg 8. d = √(100 − 64) = **6**. A common wrong answer is half the chord itself.

#	Level	Question	Answer	Solution
P10	Very hard	The circular fountain plaza at the centre of a market square is modelled by $x^2 + y^2 - 8x + 6y = 11$, with units in metres. What is the radius, in metres? 	6	Completing the square gives $(x - 4)^2 + (y + 3)^2 = 36$. r = **6**. A common wrong answer is r^2, and C is the number on the right of the original equation — both are answers to steps along the way, not the question.
P11	Very hard	A circle has centre (−2, 5) and radius 4. Write the equation of this circle. 	$(x + 2)^2 + (y - 5)^2 = 16$	Signs flip inside the brackets and the right-hand side is r^2. **$(x + 2)^2 + (y - 5)^2 = 16$.** A common wrong answer uses r instead of r^2; another gets both signs wrong.
P12	Very hard	A circle of radius 12 centimetres has a sector whose arc length is 8π centimetres. What is the area of that sector, in square centimetres? 	48π	The full circumference is 24π, so the sector is 8π/24π = 1/3 of the circle. The full area is 144π, and (1/3)(144π) = **48π**.

The one habit to build. The right-hand side of a circle equation is r^2, not r. Your class scored 47% on the circle equation, and answering 36 instead of 6 was the single most common slip.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

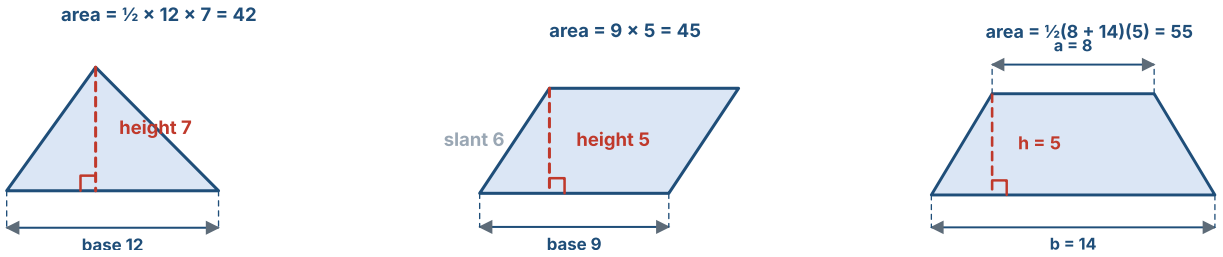
SAT Math 4.5 — Area & Volume

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Areas of plane figures

triangle ½bh | parallelogram bh | trapezoid ½(a + b)h | circle πr²

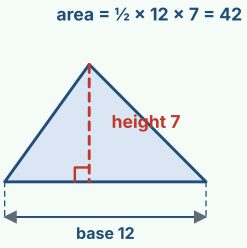
The height is always the *perpendicular* distance, not the slanted side. A triangle drawn leaning over still uses the vertical height.



In all three the red dashed line is the height: it meets the base at a right angle. The slanted side of 6 in the parallelogram is a distraction — the height is 5.

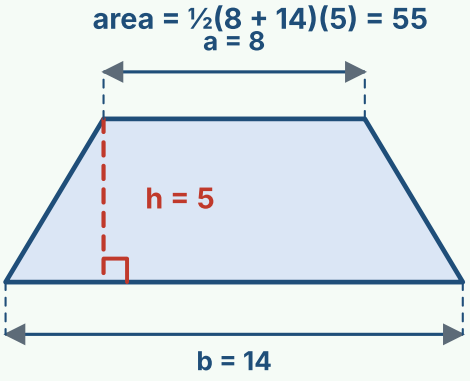
Example 1. **A triangle has base 12 and perpendicular height 7. Find its area.**

Solution.
½(12)(7) = **42**.



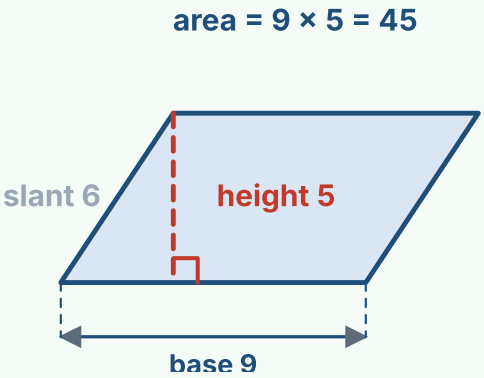
Example 2. **A trapezoid has parallel sides 8 and 14 and height 5. Find its area.**

Solution.
½(8 + 14)(5) = ½(22)(5) = **55**.



Example 3. **A parallelogram has base 9 and slant side 6, with perpendicular height 5. Find its area.**

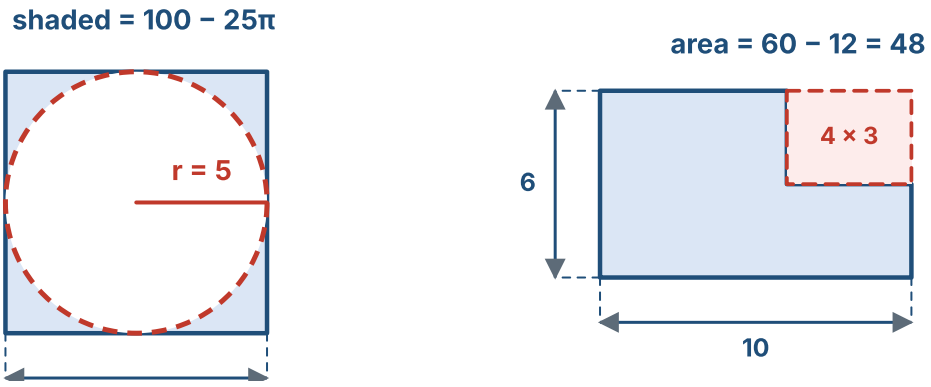
Solution.
Use the perpendicular height, not the slant.
9 × 5 = **45**.



2 Composite figures

split the shape into pieces you know, or subtract a hole from a whole

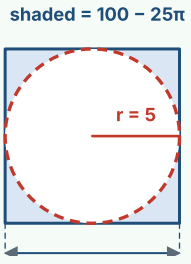
Decide whether adding or subtracting is quicker, then label every piece before calculating. A shaded region is nearly always one area minus another.



Left: the shaded part is the square minus the circular hole. Right: the L-shape is the full rectangle minus the dashed corner that was removed.

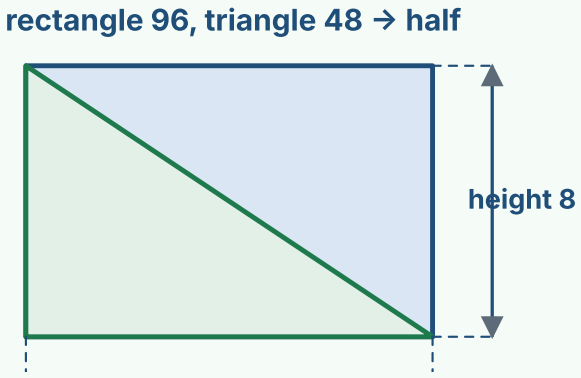
Example 1. **A square of side 10 has a circle of radius 5 cut from it. Find the shaded area left over.**

Solution.
100 − 25π ≈ 100 − 78.54 = **21.46**.



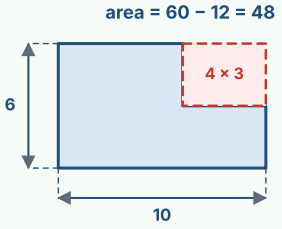
Example 2. **A rectangle 12 by 8 has a triangle of base 12 and height 8 removed. What fraction remains?**

Solution.
Rectangle 96, triangle 48.
Half remains.



Example 3. **An L-shape is a 10 by 6 rectangle with a 4 by 3 rectangle removed from one corner. Find its area.**

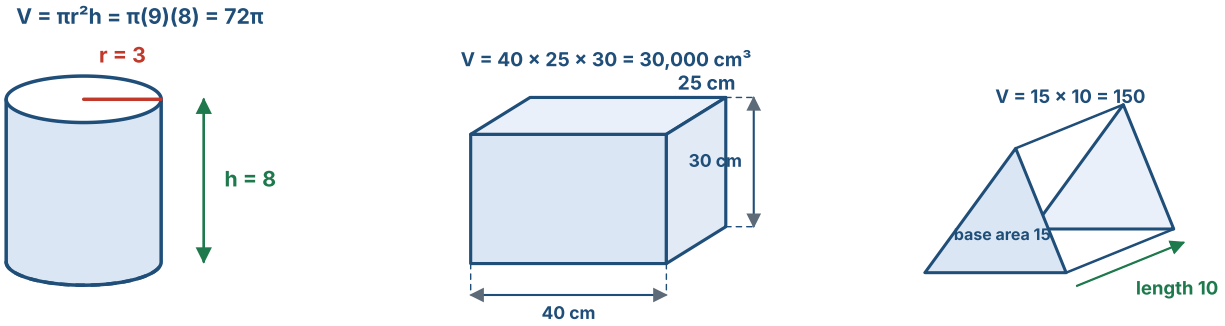
Solution.
60 − 12 = **48**.



3 Volume of prisms and cylinders

V = (area of the base) × height | cylinder V = πr²h

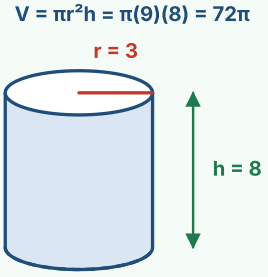
Any solid with the same cross-section all the way up follows one rule: find the base area, multiply by the height. The reference sheet gives these, so the work is choosing the base.



Each solid has one cross-section repeated along its length. Find that face's area — a circle, a rectangle, a triangle — then multiply by how far it travels.

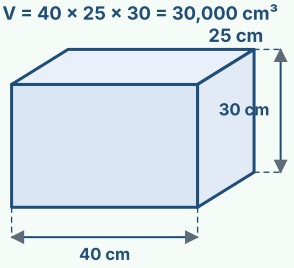
Example 1. **A cylinder has radius 3 and height 8. Find its volume.**

Solution.
π(9)(8) = **72π** ≈ 226.19.



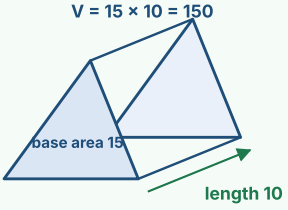
Example 2. **A rectangular tank measures 40 by 25 by 30 centimetres. Find its volume.**

Solution.
40 × 25 × 30 = **30,000 cm³**, which is 30 litres.



Example 3. **A triangular prism has a base triangle of area 15 and length 10. Find its volume.**

Solution.
15 × 10 = **150**.

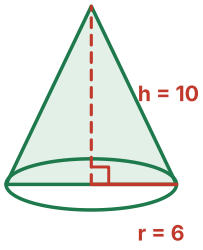


4 Cones, pyramids and spheres

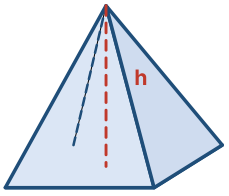
cone $\frac{1}{3}\pi r^2 h$ | pyramid $\frac{1}{3}Bh$ | sphere $\frac{4}{3}\pi r^3$

The one-third is the whole difficulty: a cone is a third of the cylinder that would contain it. All three formulas are on the reference sheet, so check you have copied the fraction.

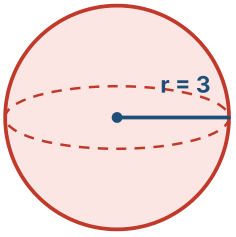
$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$



$V = \frac{1}{3} \times$ base area $\times h$



$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$

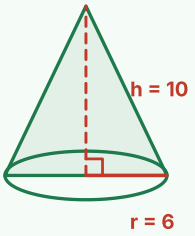


The cone and the pyramid both carry the one third; the sphere carries four thirds and cubes the radius. Copy the fraction before you substitute.

Example 1. **A cone has radius 6 and height 10. Find its volume.**

Solution.
$(1/3)\pi(36)(10) = \mathbf{120\pi}$.

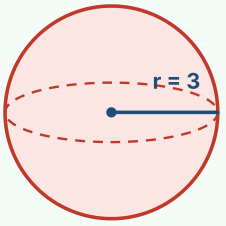
$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$



Example 2. **A sphere has radius 3. Find its volume.**

Solution.
$(4/3)\pi(27) = \mathbf{36\pi}$.

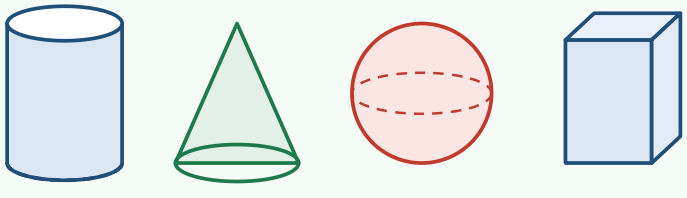
$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$



Example 3. **A cylinder and a cone share the same radius and height. How do their volumes compare?**

Solution.
The cone is exactly **one third** of the cylinder.

all four are on the reference sheet

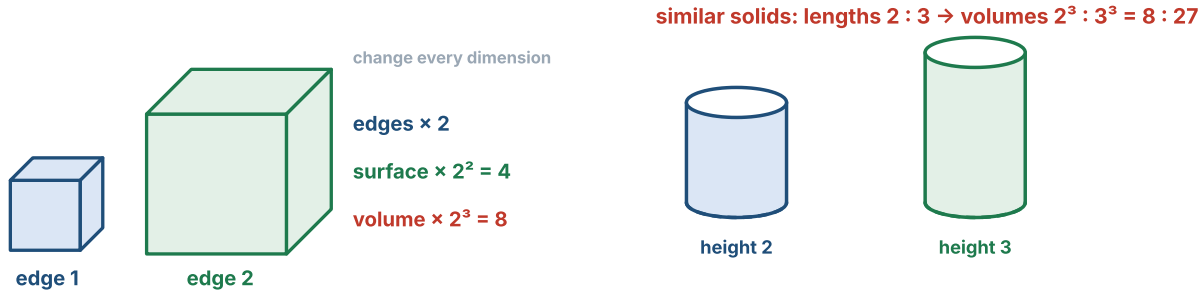


$V = \pi r^2 h$ $V = \frac{1}{3}\pi r^2 h$ $V = \frac{4}{3}\pi r^3$ $V = Bh$

5 Scaling a solid

lengths × k | surface area × k^2 | volume × k^3

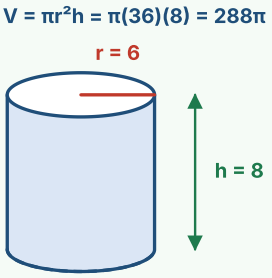
Changing one dimension multiplies the volume by that factor alone. Changing every dimension by k multiplies the volume by k cubed. Read carefully which the question means.



Left: every edge doubles, so the volume is eight times as large. Right: two similar cans in the ratio 2 : 3 have volumes in the ratio 8 : 27.

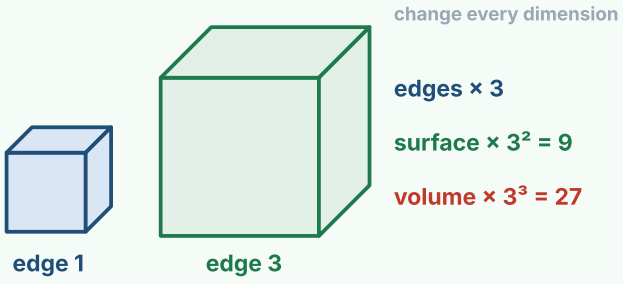
Example 1. **A cylinder's radius is doubled and its height is unchanged. What happens to the volume?**

Solution.
V depends on r^2.
The volume is multiplied by **4**.



Example 2. **A cube's edges are all tripled. What happens to its volume?**

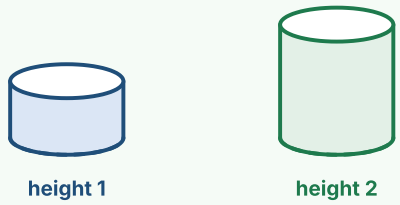
Solution.
Every dimension triples.
Volume × 3^3 = **27**.



Example 3. **A cylinder's height alone is doubled. What happens to the volume?**

Solution.
Only one dimension changes.
Volume **doubles**.

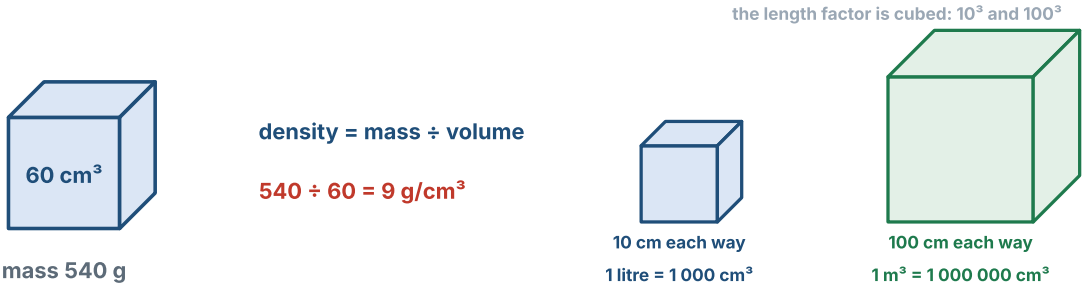
similar solids: lengths $1:2 \rightarrow$ volumes $1^3:2^3 = 1:8$



6 Density and unit conversion

density = mass ÷ volume | for volume the conversion factor is cubed

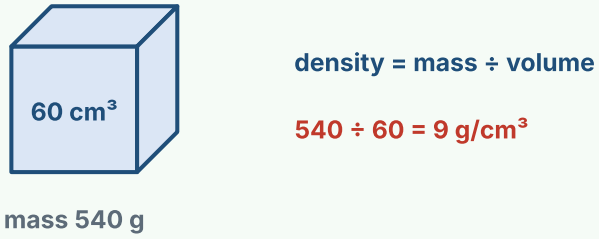
Density questions are a rate in disguise. For units, remember 1 litre = 1000 cm^3 while 1 m^3 = 1,000,000 cm^3 — the two are not the same factor.



A litre is a 10 cm cube, so 10^3 = 1 000 cm^3. A cubic metre is a 100 cm cube, so 100^3 = 1 000 000 cm^3. Using one factor for the other is the classic slip.

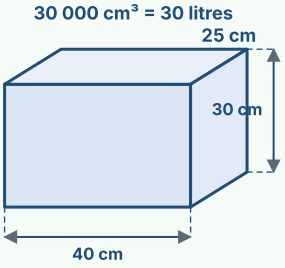
Example 1. **A block has mass 540 g and volume 60 cm^3. Find its density.**

Solution.
540/60 = **9 g/cm^3**.



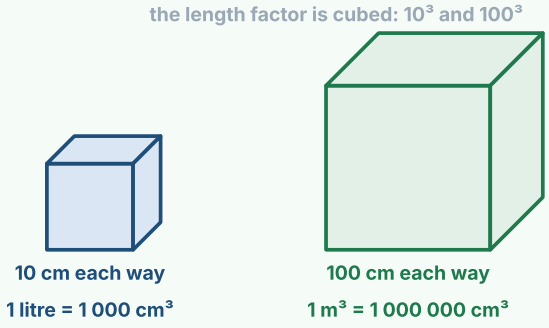
Example 2. **A tank holds 30,000 cm³. How many litres is that?**

Solution.
30,000 ÷ 1000 = **30 litres**.



Example 3. **How many cubic centimetres are in 1 cubic metre?**

Solution.
100^3 = **1,000,000**. The length factor 100 gets cubed.



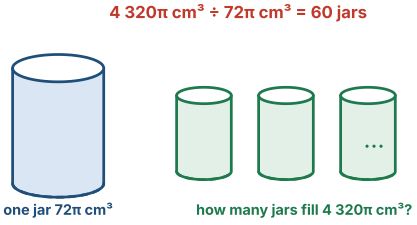
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

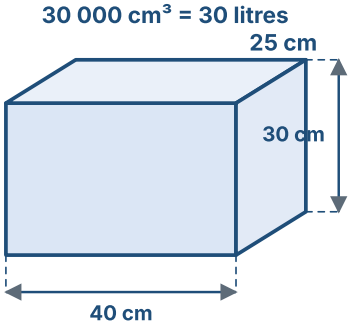
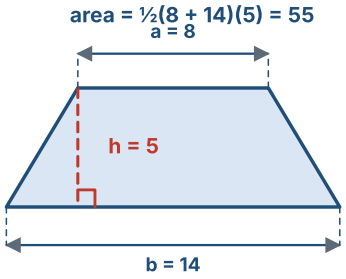
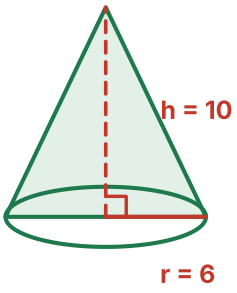
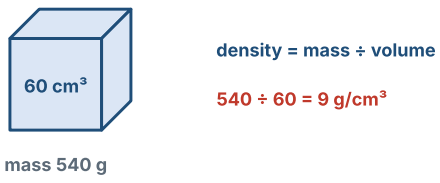
The SAT gives you Desmos on every maths question. Here it is **an arithmetic engine for formulas**. Type the whole formula with its numbers in one line so the 1/3 or the ½ cannot go missing.

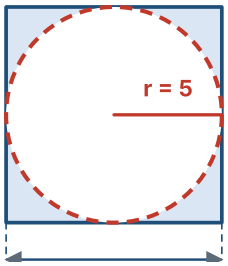
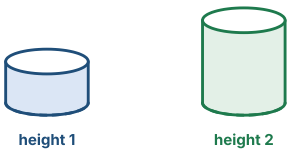
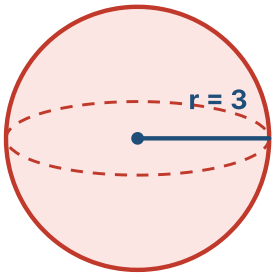
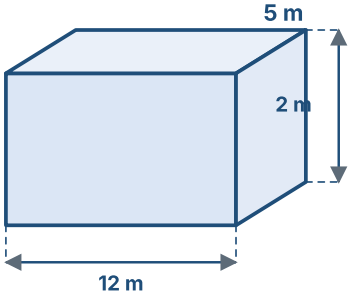
#	Question	Type into Desmos	Then	Answer
D1	A triangle has base 12 and height 7. What is its area?	`0.5*12*7`	Half base times height.	42
D2	A trapezoid has parallel sides 8 and 14 and height 5. What is its area?	`0.5*(8+14)*5`	Average the parallel sides, then multiply by the height.	55
D3	A cylinder has radius 3 and height 8. What is its volume as a multiple of π?	`3^2*8`	r squared times h.	72
D4	A cone has radius 6 and height 10. What is its volume as a multiple of π?	`(1/3)*6^2*10`	Do not forget the one third.	120
D5	A sphere has radius 3. What is its volume as a multiple of π?	`(4/3)*3^3`	Four thirds times r cubed.	36
D6	A tank measures 40 by 25 by 30 cm. How many litres does it hold?	`40*25*30/1000`	Divide by 1000, not by a million.	30
D7	A block has mass 540 g and volume 60 cm³. What is its density in g/cm³?	`540/60`	Mass over volume.	9
D8	A square of side 10 has a circle of radius 5 removed. What area remains, to two decimals?	`100-pi*25`	Subtract the hole from the whole.	21.46
D9	A cylinder's radius is doubled and its height is unchanged. By what factor does the volume grow?	`2^2`	Volume depends on r squared.	4
D10	How many cubic centimetres are in one cubic metre?	`100^3`	The length factor is cubed.	1000000

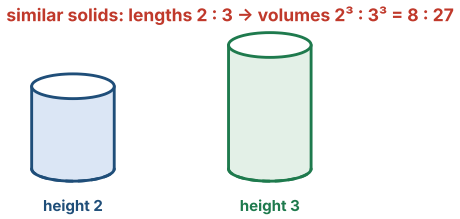
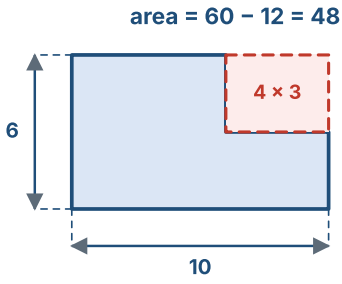
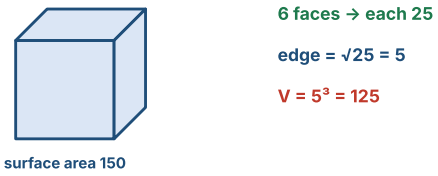
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Write the formula down before substituting. Most lost marks in this topic are a correct calculation of the wrong formula.

#	Level	Question	Answer	Solution
P1	Hard	A market supplier pours 4,320π cubic centimetres of olive oil into identical cylindrical jars. Each jar has an inside radius of 3 centimetres and is filled to a height of 8 centimetres. How many jars does the supplier fill? 	60	Each jar holds π(9)(8) = 72π cm³. 4320π ÷ 72π = **60 jars**. A common wrong answer is the volume of one jar in multiples of π, not a count.

#	Level	Question	Answer	Solution
P2	Hard	An oil tin is a rectangular prism measuring 40 cm by 25 cm by 30 cm and is completely full. How many litres of oil does it hold? (1 litre = 1000 cubic centimetres) 	30	40 × 25 × 30 = 30,000 cm³. 30,000 ÷ 1000 = **30 litres**. Dividing by 1,000,000 gives 0.03 — that factor converts to cubic *metres*, and eight of your class used it on Test 5.
P3	Hard	A trapezoid has parallel sides of 8 and 14 and a perpendicular height of 5. What is its area? 	55	Area = ½(a + b)h = ½(22)(5) = **55**. A common wrong answer forgets the one half; another averages the sides but omits the height.
P4	Hard	A cone has a radius of 6 and a height of 10. What is its volume? $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$ 	120π	V = (1/3)πr²h = (1/3)π(36)(10). = **120π**. A common wrong answer is the cylinder's volume — the one third was dropped.
P5	Hard	A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre? 	9	Density = mass ÷ volume = 540/60 = **9 g/cm³**. A common wrong answer inverts the fraction; another multiplies instead of dividing.

#	Level	Question	Answer	Solution
P6	Hard	A square of side 10 centimetres has a circle of radius 5 centimetres cut out of it. What is the area of the remaining region, to the nearest hundredth of a square centimetre? **shaded = 100 − 25π** 	21.46	100 − π(25) = 100 − 78.5398… = **21.46 cm²**.
P7	Very hard	A cylinder's radius is doubled while its height stays the same. By what factor is its volume multiplied? **similar solids: lengths 1 : 2 → volumes $1^3 : 2^3 = 1 : 8$** 	4	V = πr²h, and only r changes, so the factor is 2^2. The volume is **4 times** as large. A common wrong answer would be right if *every* dimension doubled.
P8	Very hard	A sphere and a cylinder both have radius 3. The cylinder has height 3. What is the ratio of the sphere's volume to the cylinder's? **$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$** 	4 : 3	Sphere = (4/3)π(27) = 36π; cylinder = π(9)(3) = 27π. 36 : 27 = **4 : 3**. A common wrong answer inverts the ratio.
P9	Very hard	A rectangular swimming pool is 12 metres long, 5 metres wide and 2 metres deep. Water is pumped in at 40 litres per minute. How many minutes does it take to fill? **120 m³ = 120 000 litres** 	3000	Volume = 120 m³, and 1 m³ = 1000 litres, so 120,000 litres. 120,000 ÷ 40 = **3,000 minutes**. Using 1 m³ = 1,000,000 cm³ and then forgetting to convert to litres is the usual slip here.

#	Level	Question	Answer	Solution
P10	Very hard	Two similar cylindrical cans have heights in the ratio 2 : 3. What is the ratio of their volumes? 	8 : 27	Similar solids scale by k in length, so volumes scale by k³. $2^3 : 3^3$ = **8 : 27**. A common wrong answer squares instead of cubing, which is the surface-area rule.
P11	Very hard	An L-shaped floor is formed by removing a 4 m by 3 m rectangle from one corner of a 10 m by 6 m rectangle. Tiles cost $12 per square metre. What is the total cost of tiling the floor? 	$576	Area = 60 − 12 = 48 m². 48 × 12 = **$576**. A common wrong answer tiles only the 12 m² corner that was removed.
P12	Very hard	A cube has a surface area of 150 square centimetres. What is its volume, in cubic centimetres? 	125	A cube has 6 faces, so each face has area 150/6 = 25 and the edge is 5. $V = 5^3$ = **125**. A common wrong answer is the area of one face, not the volume.

The one habit to build. Litres and cubic metres use different conversion factors: 1 litre = 1,000 cm³ but 1 m³ = 1,000,000 cm³. Eight of thirteen of your class used the wrong one on Test 5 and turned 30 litres into 0.03.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.

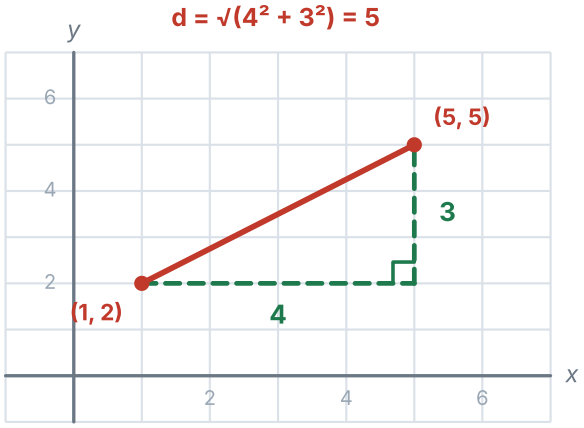
SAT Math 4.6 — Coordinate Geometry

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Distance between two points

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} \text{ — it is Pythagoras in disguise}$$

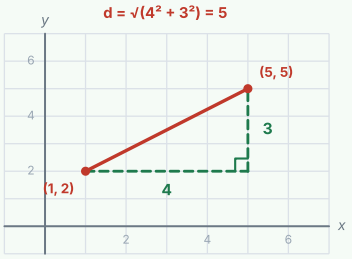
Draw the horizontal and vertical legs and the formula becomes obvious. Because both differences are squared, the order of subtraction does not matter.



The green legs are just the differences in x and in y: 4 across and 3 up. The red segment is the hypotenuse of that right triangle, so d = 5.

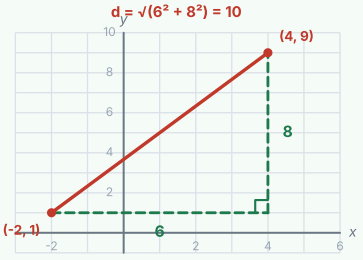
Example 1. **Find the distance between (1, 2) and (5, 5).**

Solution.
Legs 4 and 3.
√(16 + 9) = **5**.



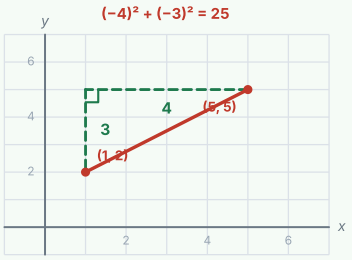
Example 2. **Find the distance between (−2, 1) and (4, 9).**

Solution.
Legs 6 and 8.
√(36 + 64) = **10**.



Example 3. **Why does subtracting the other way round give the same answer?**

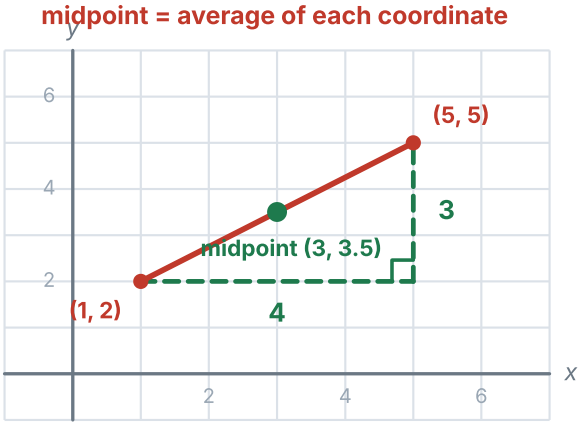
Solution.
Squaring removes the sign: $(-4)^2 = 4^2$.



2 Midpoint

midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ — average each coordinate

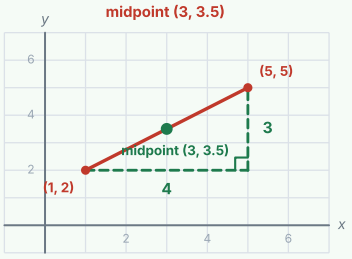
Average the x's and average the y's, separately. If you are given one endpoint and the midpoint, work backwards: the midpoint is as far past itself again.



The midpoint sits halfway along in both directions: x goes 1 → 3 → 5 and y goes 2 → 3.5 → 5.

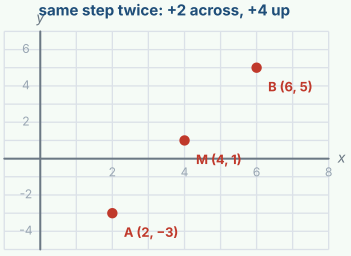
Example 1. **Find the midpoint of (1, 2) and (5, 5).**

Solution.
((1+5)/2, (2+5)/2) = **(3, 3.5)**.



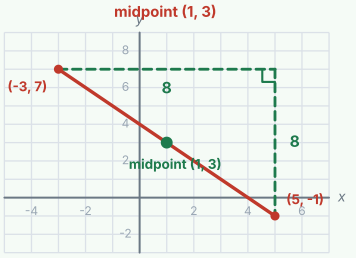
Example 2. **The midpoint of AB is (4, 1) and A is (2, −3). Find B.**

Solution.
B is the same step again: (4 + 2, 1 + 4) = **(6, 5)**.



Example 3. **Find the midpoint of (−3, 7) and (5, −1).**

Solution.
((−3+5)/2, (7−1)/2) = **(1, 3)**.

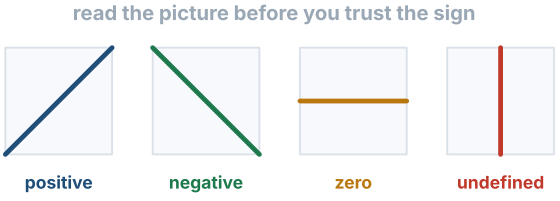


3 Slope

$m = (y_2 - y_1)/(x_2 - x_1)$ — rise over run

Keep the two points in the same order on top and bottom. A positive slope rises left to right; a horizontal line has slope 0 and a vertical line has no slope at all.

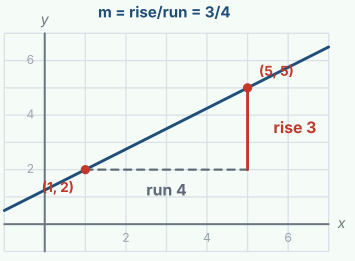




Rise 3 over run 4 gives m = 3/4. The four small panels show the only four possibilities: up, down, flat (0) and vertical (undefined).

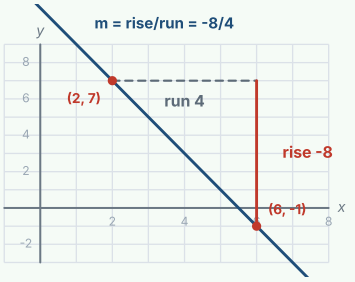
Example 1. **Find the slope through (1, 2) and (5, 5).**

Solution.
(5 − 2)/(5 − 1) = **3/4**.



Example 2. **Find the slope through (2, 7) and (6, −1).**

Solution.
(−1 − 7)/(6 − 2) = −8/4 = **−2**.



Example 3. **What is the slope of the line through (3, 4) and (3, 9)?**

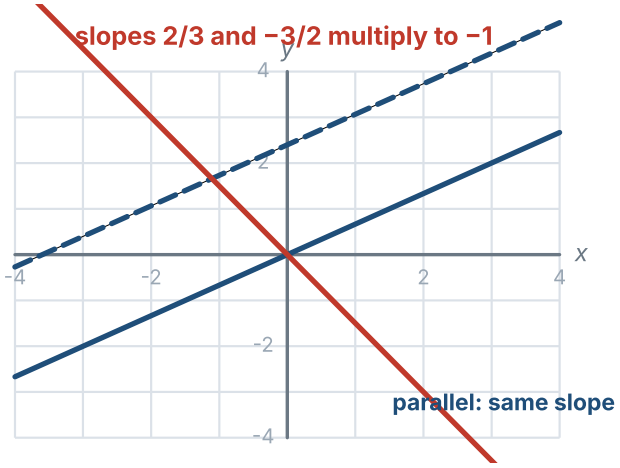
Solution.
The run is 0, so the slope is **undefined** — the line is vertical.



4 Parallel and perpendicular

parallel → equal slopes | perpendicular → slopes multiply to −1 (negative reciprocals)

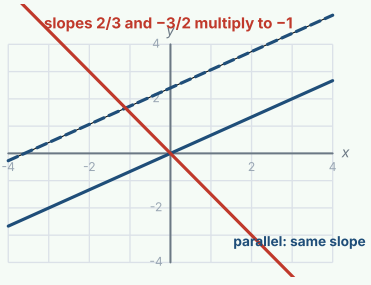
To get a perpendicular slope, flip the fraction and change the sign. The perpendicular to a slope of 2/3 is −3/2.



The two blue lines never meet — same slope 2/3. The red line crosses them at a right angle, and its slope −3/2 is the fraction flipped and negated.

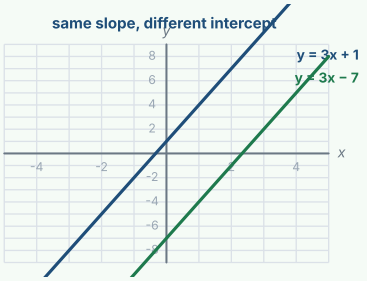
Example 1. **A line has slope 2/3. What is the slope of a line perpendicular to it?**

Solution.
Flip and negate.
−3/2.



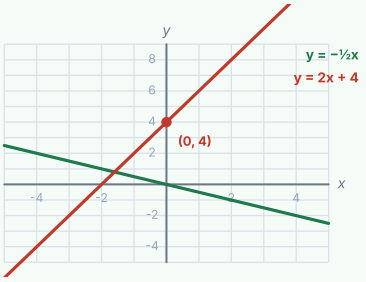
Example 2. **Is y = 3x + 1 parallel to y = 3x − 7?**

Solution.
Both have slope 3.
Yes — parallel, and never meeting.



Example 3. **A line through (0, 4) is perpendicular to y = −½x. Write its equation.**

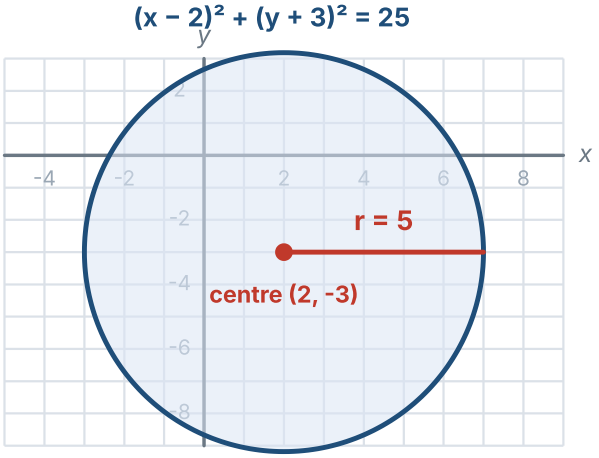
Solution.
The perpendicular slope is 2.
y = 2x + 4.



5 The equation of a circle in the plane

$$(x - h)^2 + (y - k)^2 = r^2$$

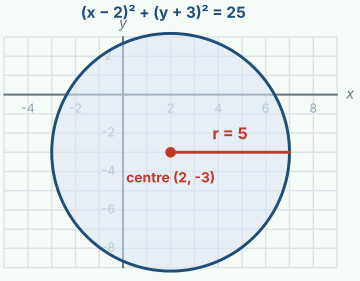
This is coordinate geometry's one memorised formula, and it is not on the reference sheet. The centre's coordinates appear with their signs flipped.



Centre (2, −3) with radius 5. In the equation the 2 appears as − 2 and the −3 appears as + 3, and the right-hand side is $5^2 = 25$.

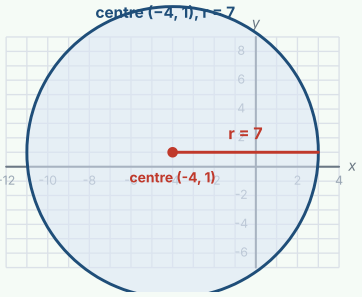
Example 1. **A circle has centre (2, −3) and radius 5. Write its equation.**

Solution.
$(x - 2)^2 + (y + 3)^2 = 25$.



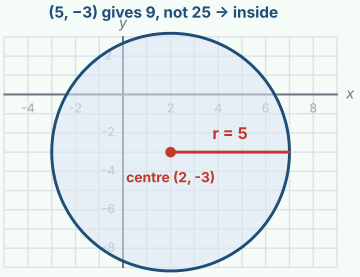
Example 2. **Find the centre and radius of $(x + 4)^2 + (y - 1)^2 = 49$.**

Solution.
Centre **(−4, 1)**, radius **7**.



Example 3. **Is (5, −3) on the circle with centre (2, −3) and radius 5?**

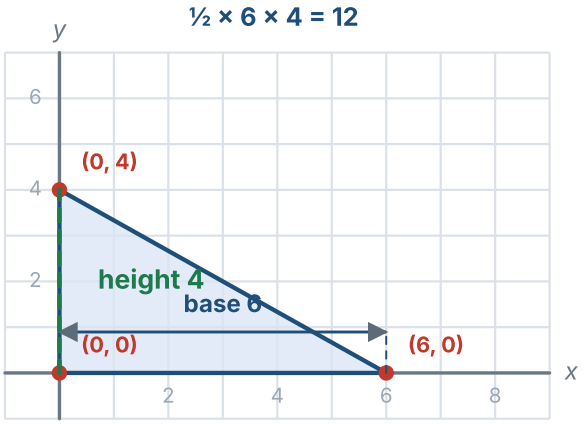
Solution.
$(5 - 2)^2 + 0 = 9$, not 25.
No — it is inside.



6 Area on a grid

for a right-angled or axis-aligned figure, read the legs straight off the grid

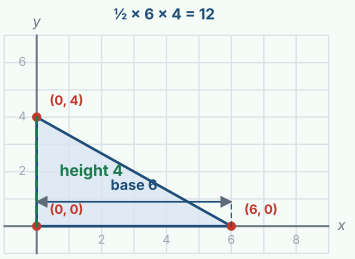
When a figure sits on the coordinate axes, you rarely need a formula beyond the basics: count the horizontal and vertical lengths and apply the ordinary area rule.



Both legs lie along the axes, so the base is 6 and the height is 4 — no distance formula needed.

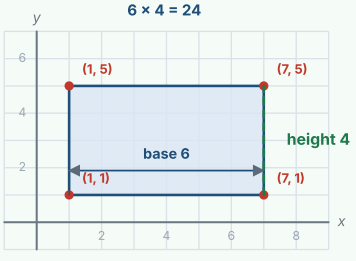
Example 1. **Find the area of a triangle with vertices (0, 0), (6, 0) and (0, 4).**

Solution.
Legs 6 and 4 along the axes.
½(6)(4) = **12**.



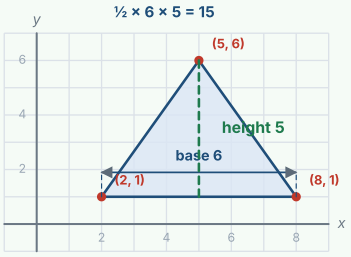
Example 2. **Find the area of a rectangle with vertices (1, 1), (7, 1), (7, 5) and (1, 5).**

Solution.
Width 6, height 4.
24.



Example 3. **Find the area of a triangle with vertices (2, 1), (8, 1) and (5, 6).**

Solution.
Base 6 along y = 1, height 5.
½(6)(5) = **15**.



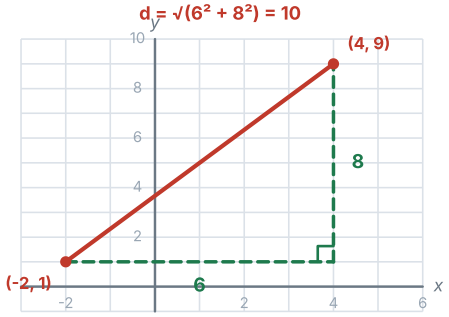
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

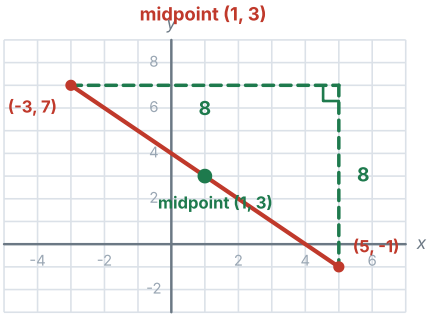
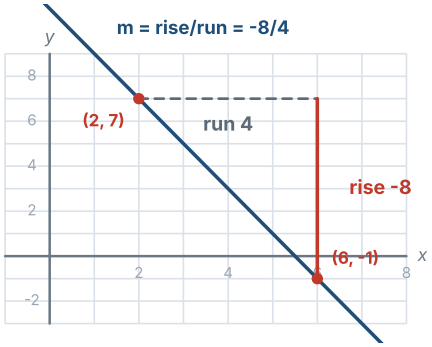
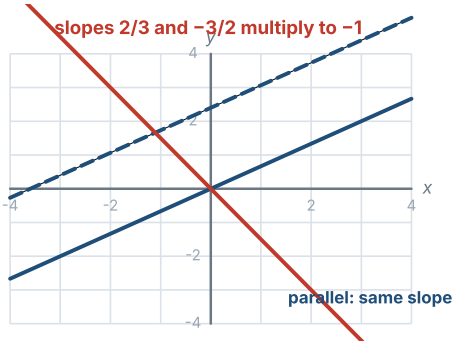
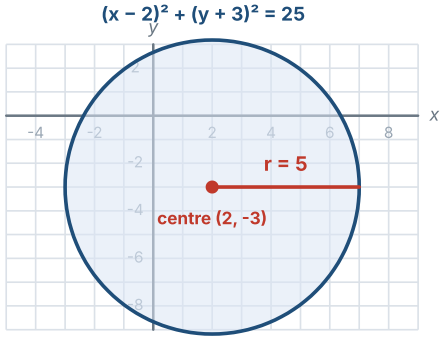
The SAT gives you Desmos on every maths question. Here it is **a plotter**. Enter the points and the line; seeing the picture settles slope signs and perpendicularity faster than any rule.

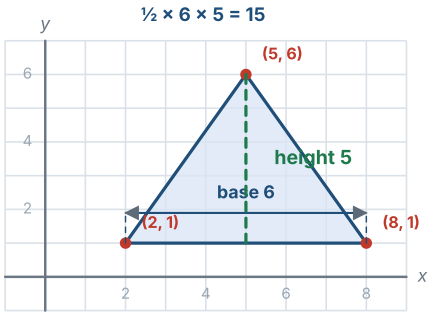
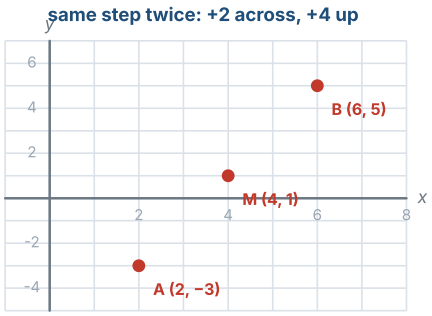
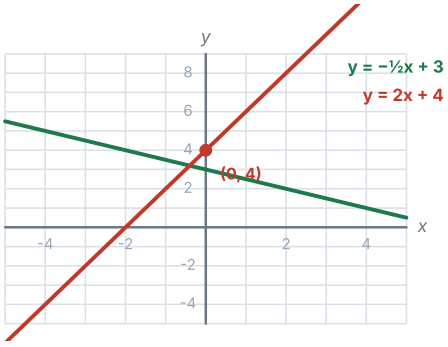
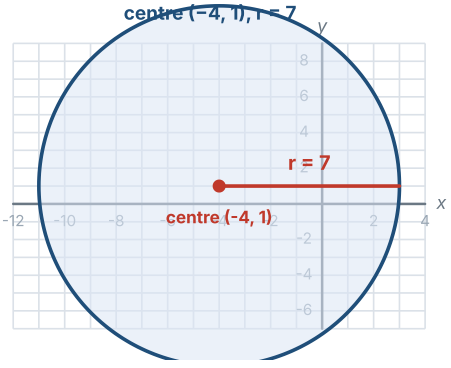
#	Question	Type into Desmos	Then	Answer
D1	What is the distance between (1, 2) and (5, 5)?	`sqrt((5-1)^2+(5-2)^2)`	Pythagoras on the two differences.	5
D2	What is the distance between (−2, 1) and (4, 9)?	`sqrt(6^2+8^2)`	A 6-8-10 triangle.	10
D3	What is the midpoint of (1, 2) and (5, 5)?	`((1+5)/2,(2+5)/2)`	Average each coordinate.	(3, 3.5)
D4	What is the slope through (2, 7) and (6, −1)?	`(-1-7)/(6-2)`	Keep the order consistent.	−2
D5	A line has slope 2/3. What is the perpendicular slope?	`-1/(2/3)`	Flip and negate.	−1.5
D6	The midpoint of AB is (4, 1) and A is (2, −3). What is the x-coordinate of B?	`2*4-2`	Twice the midpoint minus the known endpoint.	6
D7	What is the radius of $(x + 4)^2 + (y − 1)^2 = 49$?	`sqrt(49)`	The right-hand side is r squared.	7
D8	What is the area of a triangle with vertices (0, 0), (6, 0) and (0, 4)?	`0.5*6*4`	The legs lie along the axes.	12
D9	What is the area of a triangle with vertices (2, 1), (8, 1) and (5, 6)?	`0.5*6*5`	Base along y = 1, height from y = 1 to y = 6.	15
D10	Is (5, −3) on the circle centred (2, −3) with radius 5?	`(5-2)^2+(-3+3)^2`	Compare the result with r squared.	no, 9 is not 25

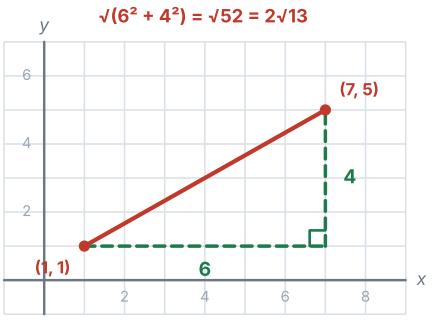
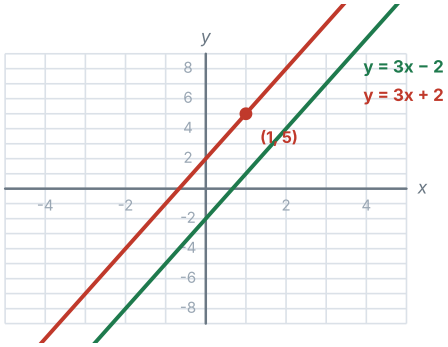
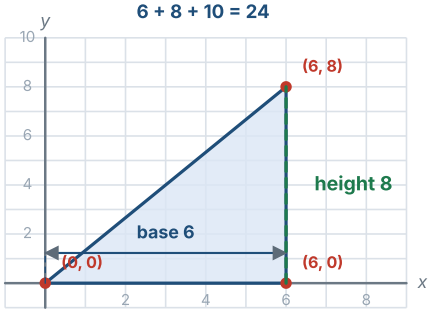
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Sketch the points on a quick set of axes first. Nearly every question in this topic is Pythagoras, an average, or a slope — and the sketch tells you which.

#	Level	Question	Answer	Solution
P1	Hard	What is the distance between the points (−2, 1) and (4, 9) in the xy-plane? 	10	The horizontal difference is 6 and the vertical difference is 8. √(36 + 64) = √100 = **10**.

#	Level	Question	Answer	Solution
P2	Hard	What is the midpoint of the segment joining (−3, 7) and (5, −1)? 	(1, 3)	Average each coordinate: ((−3 + 5)/2, (7 − 1)/2). **(1, 3)**. A common wrong answer subtracts the coordinates instead of averaging them.
P3	Hard	What is the slope of the line passing through (2, 7) and (6, −1)? 	−2	(−1 − 7)/(6 − 2) = −8/4 = **−2**. A common wrong answer inverts the fraction, giving run over rise.
P4	Hard	A line has a slope of 2/3. What is the slope of any line perpendicular to it? 	−3/2	Perpendicular slopes are negative reciprocals: flip the fraction and change the sign. **−3/2**. A common wrong answer flips without negating; another negates without flipping.
P5	Hard	A circle has centre (2, −3) and radius 5. Write the equation of that circle. 	(x − 2)² + (y + 3)² = 25	Signs flip inside the brackets and the right side is r². **(x − 2)² + (y + 3)² = 25.** A common wrong answer uses r instead of r².

#	Level	Question	Answer	Solution
P6	Hard	A triangle has vertices (2, 1), (8, 1) and (5, 6). What is its area, in square units? 	15	The base lies along y = 1 from x = 2 to x = 8, so it is 6 long, and the height is 6 − 1 = 5. ½(6)(5) = **15**.
P7	Very hard	The midpoint of segment AB is (4, 1). If A has coordinates (2, −3), what are the coordinates of B? 	(6, 5)	The midpoint is the average, so B is the same step again: (4 + 2, 1 + 4). **(6, 5)**. A common wrong answer is the midpoint of A and the given midpoint.
P8	Very hard	Line k passes through (0, 4) and is perpendicular to the line y = −½x + 3. What is the equation of line k? 	y = 2x + 4	The perpendicular slope to −½ is **2**, and the y-intercept is given as 4. **y = 2x + 4.** A common wrong answer flips the fraction without changing the sign.
P9	Very hard	In the xy-plane, a circle has equation $(x + 4)^2 + (y - 1)^2 = 49$. What is the sum of the coordinates of its centre and its radius? 	4	Centre (−4, 1) and radius 7. −4 + 1 + 7 = **4**.

#	Level	Question	Answer	Solution
P10	Very hard	A square has two opposite vertices at (1, 1) and (7, 5). What is the length of its diagonal? 	2√13	The diagonal joins the two given points: √(36 + 16) = √52. √52 = 2√13 ≈ 7.21, so **2√13**. A common wrong answer is the same value unsimplified — the SAT expects the simplified radical.
P11	Very hard	Line m has equation y = 3x − 2. Which line is parallel to m and passes through (1, 5)? 	y = 3x + 2	Parallel means the same slope, 3, and 5 = 3(1) + b gives b = 2. **y = 3x + 2.** A common wrong answer is the original line; another does not pass through (1, 5).
P12	Very hard	Points A(0, 0), B(6, 0) and C(6, 8) form a triangle. What is its perimeter? 	24	AB = 6 and BC = 8, and the right angle at B makes AC = 10 by the 6-8-10 triple. 6 + 8 + 10 = **24**. A common wrong answer omits the hypotenuse; another is twice the area.

The one habit to build. Distance is Pythagoras; midpoint is an average. Confusing the two is the commonest error here, and a thirty-second sketch prevents it.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by the checkers in the source folder; every figure is drawn to the numbers in the question.